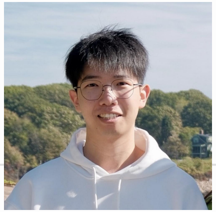
CIKM SEOUL 2025
November 10 - 14

large-genrec.github.io/

Towards Large Generative Recommendation: A Tokenization Perspective

Yupeng Hou[1], An Zhang[2], Leheng Sheng[3], Jiancan Wu[2], Xiang Wang[2], Tat-Seng Chua[3], Julian McAuley[1]

[1]UC San Diego [2]USTC [3]NUS

01

Background & Introduction

What are Generative Models?

A generative model learns the underlying distribution of data and can generate new samples from it.

Training data~$p_{data}(x)$

Learning →

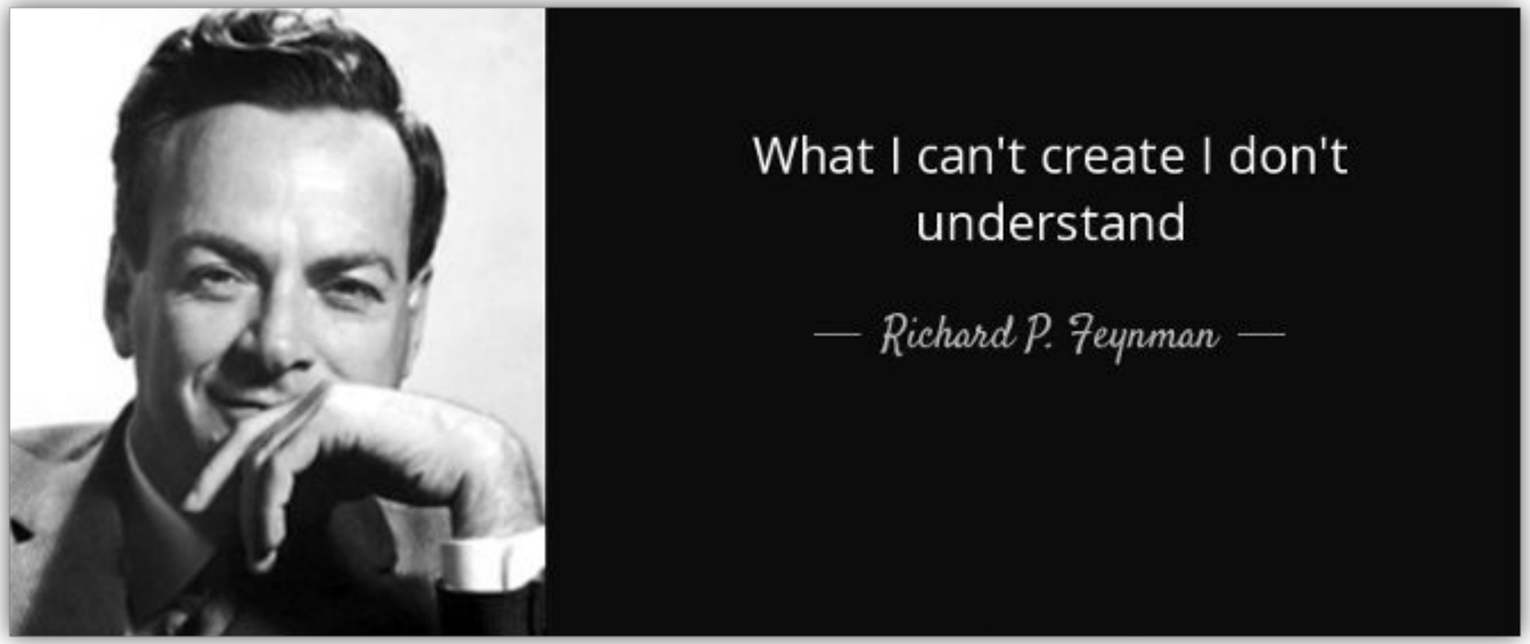
$p_\theta(x)$

Data generation distribution

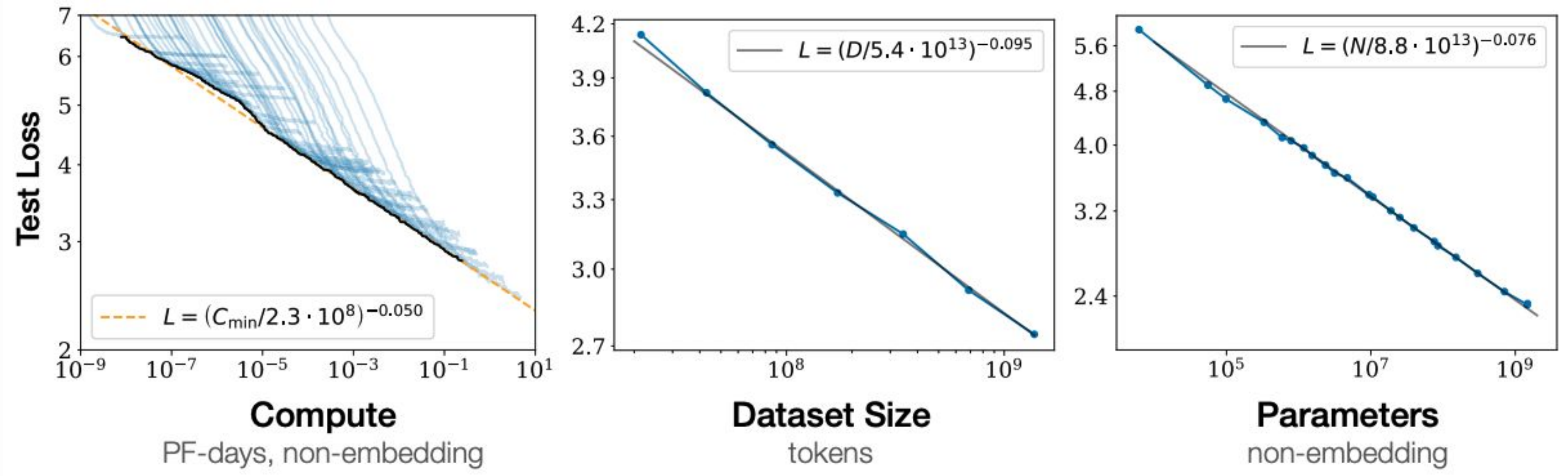
Generating →

New samples

Why “Generative” Models?



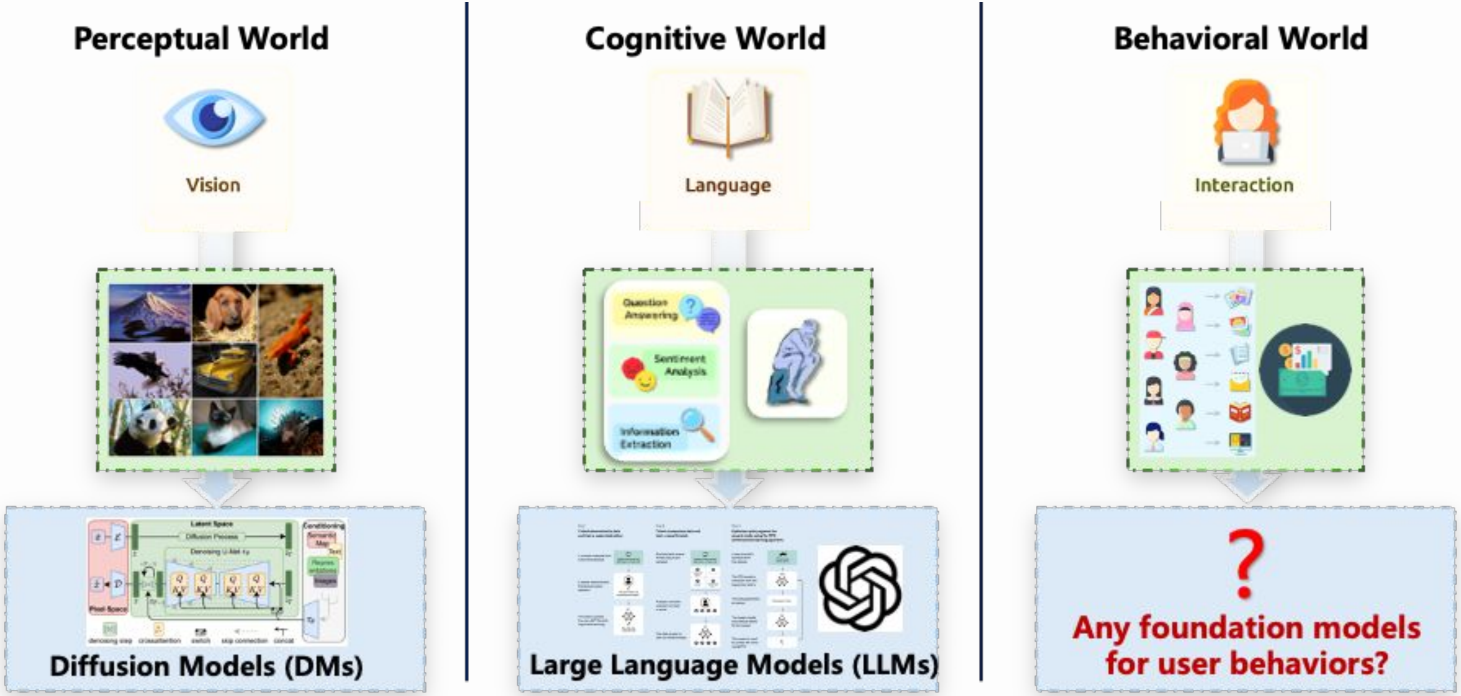
Scaling Law as a Pathway towards AGI



Scaling laws provide a framework for understanding how **model size**, **data volume**, and **test-time computing** might lead to advanced AI capabilities.

Understanding Scaling Laws for Recommendation Models. Arxiv 2022.

As the Reflection of Real World



Where are We Now?

In language and vision:

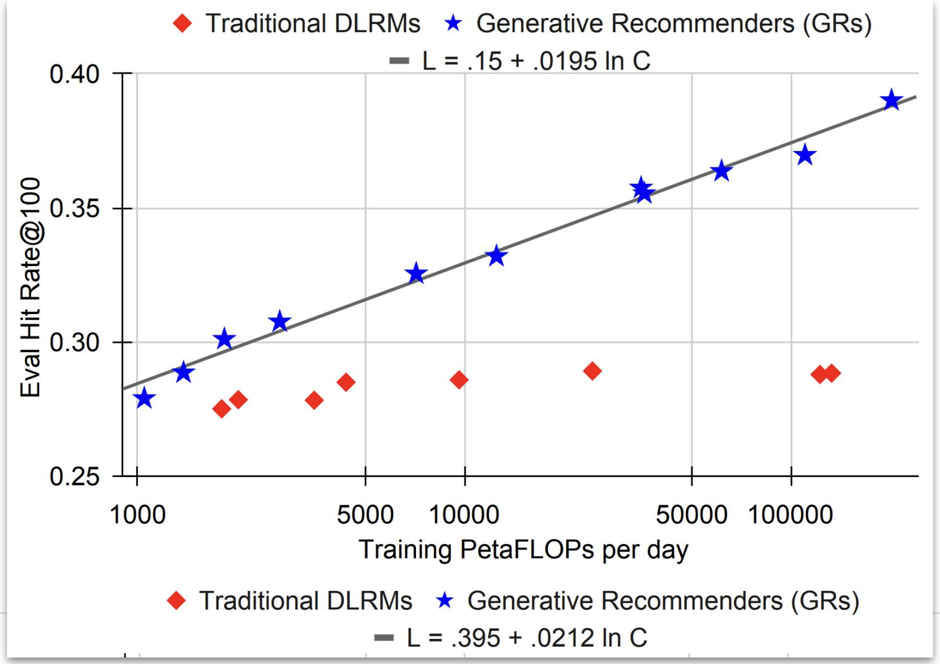
- Generative models are dominating.
- Scaling law has been witnessed.
- Large language/diffusion models have been established.

In recommendation:

- Initial attempts on building large generative recommendation models

Why "Generative" Recommendation?

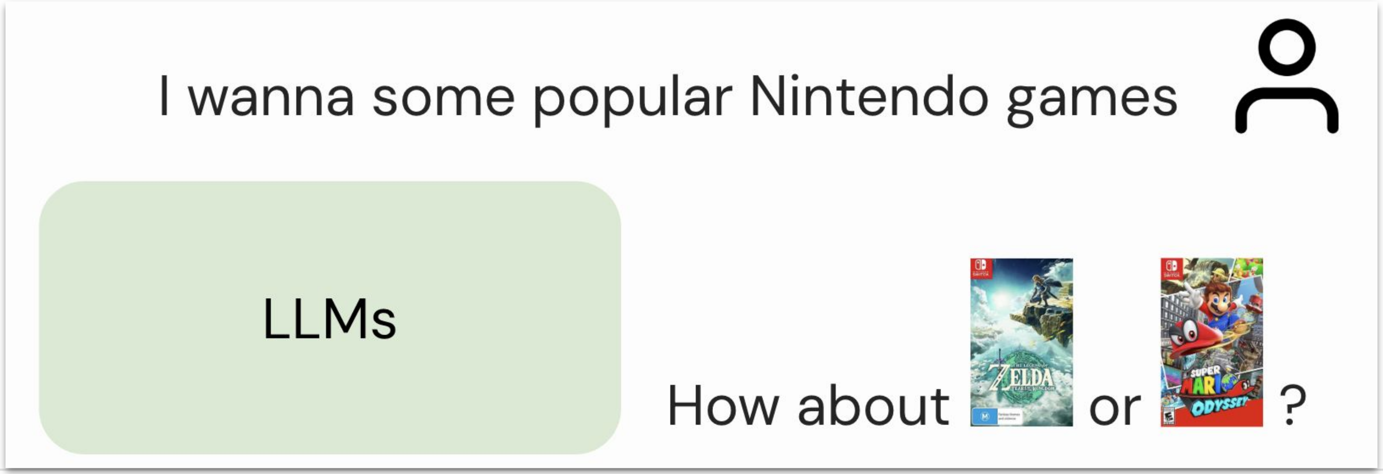
(1) Better scaling behavior



Zhai et al. Actions Speak Louder than Words: Trillion-Parameter Sequential Transducers for Generative Recommendations. ICML 2024.

Why “Generative” Recommendation?

(1) **Better scaling behavior**
(2) **Better alignment with other modalities (text, image, audio, …);**

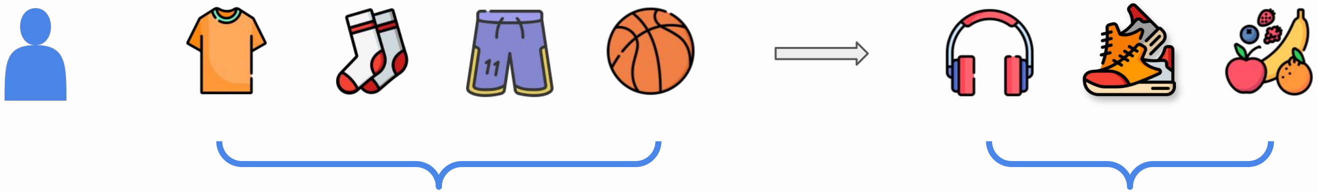


Language Modeling vs. Recommendation

Language Modeling

Will be the 46th film overall in the ⟶ Marvel Cinematic Universe

Recommendation



User Historical Actions ⟶ Future Actions

However …

Language Modeling

- Dense world knowledge
- Text tokens (10^5)

Recommendation

- Sparse user-item interactions
- Items (10^9)

Scaling laws rarely apply to traditional recommendation models.

What is “Tokenization”?

Convert

human-readable data — Text, Image, Action, …

Into

machine-readbble formats — Sequence of Tokens

Language Tokenization

Human-readable Data 👨👩

Will be the 46th film overall in the

❓Machine cannot directly read

Language Tokenization

Human-readable Data 👨👩

Will be the 46th film overall in the

Tokenize

Machine-readable Data 🤖

8743, 307, 262, 6337, 400, 2646, 4045, 287, 262

a token

Language Tokenization

Human-readable Data 👨 👩

Will be the 46th film overall in the

Tokenize

Machine-readable Data 🤖

8743, 307, 262, 6337, 400, 2646, 4045, 287, 262 9923, 21932, 1512, 11950

token-by-token generation

Language Tokenization

Human-readable Data 👨👩

Will be the 46th film overall in the

Tokenize

Machine-readable Data 🤖

8743, 307, 262, 6337, 400, 2646, 4045, 287, 262 9923, 21932, 1512, 11950

Detokenize

Marvel Cinematic Universe

Action Tokenization

Human-readable Data 👨 👩

Machine-readable Data 🤖

Tokenize

?

Action Tokenization

Human-readable Data 👨 👩

Machine-readable Data 🤖

Tokenize

i3487, i124, i19240, i772

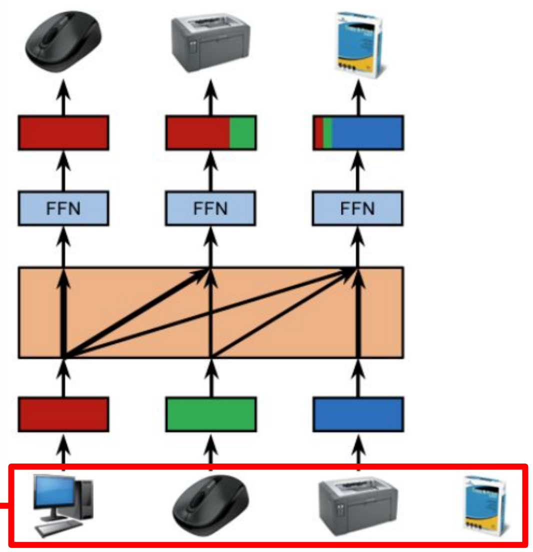
One token per action? 🤔

Action Tokenization

One token per action

Example: SASRec [ICDM'18]

Each item is indexed by a unique **item ID**, corresponding to a **learnable embedding**



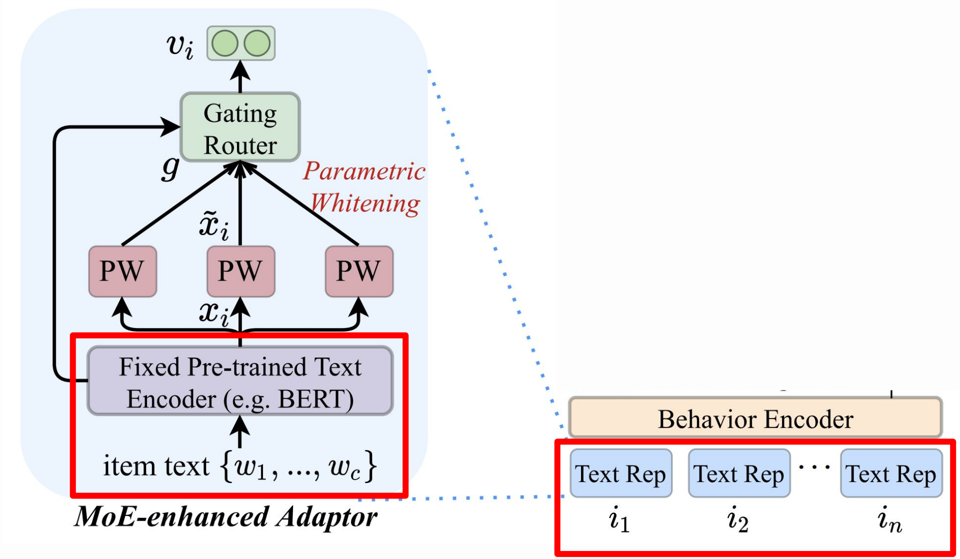
Kang and McAuley. Self-Attentive Sequential Recommendation. ICDM 2018.

Action Tokenization

One token per action

Example: UniSRec [KDD'22]

Each item is indexed by a unique **item ID**, corresponding to a **fixed representation**

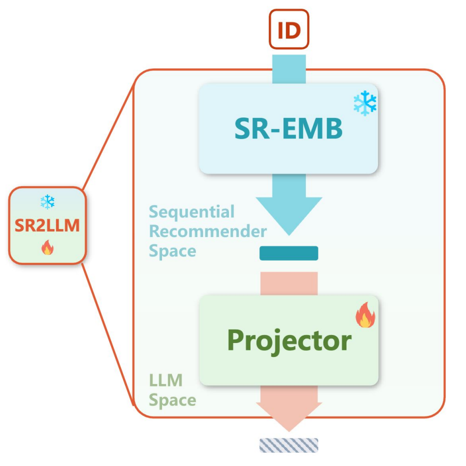


Hou et al. Towards Universal Sequence Representation Learning for Recommender Systems. KDD 2022.

Action Tokenization

One token per action

Example: LLaRA [SIGIR'24]



Align **item representations** with text tokens in LLMs

(b) Hybrid prompting method.

Input: This user has watched Titanic [emb_s^{14}], Roman Holiday [emb_s^{20}], ... Gone with the wind [emb_s^{37}] in the previous. Please predict the next movie this user will watch. The movie title candidates are The Wizard of Oz [emb_s^{5}], Braveheart [emb_s^{42}],..., Waterloo Bridge [emb_s^{20}],... Batman & Robin [emb_s^{19}]. Choose only one movie from the candidates. The answer is

Output: Waterloo Bridge.

Liao et al. LLaRA: Large Language-Recommendation Assistant. SIGIR 2024.

Action Tokenization

One token per action

How many tokens in LLMs?

Meta	Llama 3	~128,000
OpenAI	GPT-5	~200,000
Google DeepMind	Gemma 2	~256,000

Action Tokenization

One token per action

How many tokens in LLMs?

Meta	Llama 3	~128,000
OpenAI	GPT-5	~200,000
Google DeepMind	Gemma 2	~256,000

How many item IDs?

Amazon-Reviews-2023	~48,200,000

https://amazon-reviews-2023.github.io/

Action Tokenization

One token per action

How many tokens/item IDs in LLMs/RecSys?

Having too many tokens may make the data too sparse to effectively train a large generative model

~128,000

~200,000

~256,000

~48,200,000

Action Tokenization

Human-readable Data

Machine-readable Data

Tokenize

Can we tokenize the action data into a distribution that generative models can easily fit?

Action Tokenization

Human-readable Data

Machine-readable Data

Tokenize

- **Text Descriptions** (LLM-based Generative Rec)
- **Semantic IDs** (SemID-based Generative Rec)

Schedule Overview

Time	Session
13:45 - 14:15	Part 1: Background and Introduction
14:15 - 15:00	Part 2: LLM-based Generative Recommendation
15:00 - 15:30	Part 3: Semantic IDs
15:30 - 16:00	Coffee Break
16:00 - 17:15	Part 4: Semantic ID-based Generative Recommendation
17:15 - 17:45	Part 5: Open Challenges and Beyond
17:45 - 18:00	Q&A Discussion

Schedule Overview

Time	Session
13:45 - 14:15	Part 1: Background and Introduction
14:15 - 15:00	Part 2: LLM-based Generative Recommendation
15:00 - 15:30	Part 3: Semantic IDs
15:30 - 16:00	Coffee Break
16:00 - 17:15	Part 4: Semantic ID-based Generative Recommendation
17:15 - 17:45	Part 5: Open Challenges and Beyond
17:45 - 18:00	Q&A Discussion

Schedule Overview

Time	Session
13:45 - 14:15	Part 1: Background and Introduction
14:15 - 15:00	Part 2: LLM-based Generative Recommendation
15:00 - 15:30	Part 3: Semantic IDs
15:30 - 16:00	Coffee Break
16:00 - 17:15	Part 4: Semantic ID-based Generative Recommendation
17:15 - 17:45	Part 5: Open Challenges and Beyond
17:45 - 18:00	Q&A Discussion

Schedule Overview

Time	Session
13:45 - 14:15	Part 1: Background and Introduction
14:15 - 15:00	Part 2: LLM-based Generative Recommendation
15:00 - 15:30	Part 3: Semantic IDs
15:30 - 16:00	Coffee Break
16:00 - 17:15	Part 4: Semantic ID-based Generative Recommendation
17:15 - 17:45	Part 5: Open Challenges and Beyond
17:45 - 18:00	Q&A Discussion

Takeaway of Intro

(1) Why generative rec?

Scaling law; Alignment with other modalities

Takeaway of Intro

(1) Why generative rec?

Scaling law; Alignment with other modalities

(2) Tokenization

Human-readable data -> Machine-readable data

Takeaway of Intro

(1) Why generative rec?

Scaling law; Alignment with other modalities;

(2) Tokenization

Human-readable data -> machine-readable data

(3) Action Tokenization

One token per action -> too sparse for generative models;

Text or semantic IDs?

02

Large Language Model

-based Generative Recommendation

Action Tokenization

Human-readable Data 👨👩

Tokenize

Machine-readable Data 🤖

Premium Men’s Short Sleeve Athletic Training T-Shirt Made of Lightweight Breathable Fabric, Ideal for Running, Gym Workouts, and Casual Sportswear in All Seasons; High-Performance Breathable Cotton Crew Socks for Men with Arch Support, Cushioned Heel and Toe, and Moisture Control, Perfect for Sports, Walking, and Everyday Comfort; Men’s Loose-Fit Basketball Shorts with Elastic Drawstring Waistband, Quick-Dry Mesh Fabric, and Printed Number 11 for Professional and Recreational Play; Official Size 7 Composite Leather Basketball Designed for Indoor and Outdoor Use, Deep Channel Design for Enhanced Grip and Ball Control, Ideal for Training and Competitive Matches;

Text description of each action

Action Tokenization

Cons: *Inefficient;*

Human-readable Data 👨 👩

Tokenize

Machine-readable Data 🤖

Premium Men’s Short Sleeve Athletic Training T-Shirt Made of Lightweight Breathable Fabric, Ideal for Running, Gym Workouts, and Casual Sportswear in All Seasons; High-Performance Breathable Cotton Crew Socks for Men with Arch Support, Cushioned Heel and Toe, and Moisture Control, Perfect for Sports, Walking, and Everyday Comfort; Men’s Loose-Fit Basketball Shorts with Elastic Drawstring Waistband, Quick-Dry Mesh Fabric, and Printed Number 11 for Professional and Recreational Play; Official Size 7 Composite Leather Basketball Designed for Indoor and Outdoor Use, Deep Channel Design for Enhanced Grip and Ball Control, Ideal for Training and Competitive Matches;

Text description of each action

Action Tokenization

Cons: *Inefficient;*

Pros: *The underlying distribution aligns with that of LLMs*

Human-readable Data 👨 👩

Tokenize

Machine-readable Data 🤖

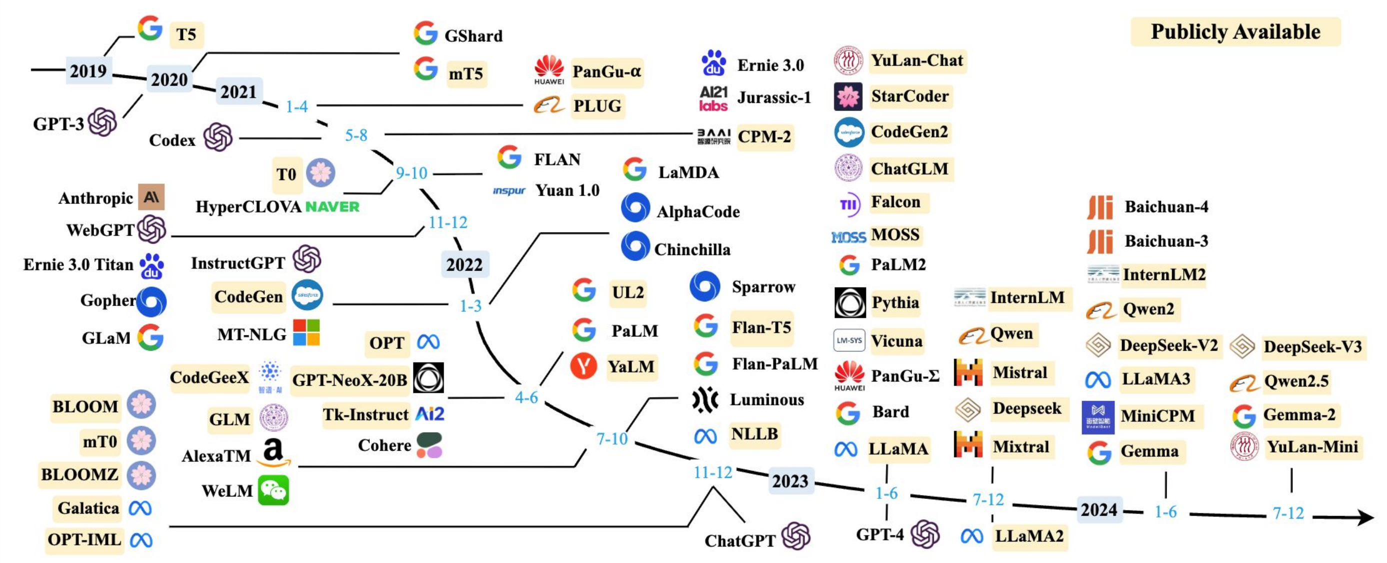
Premium Men's Short Sleeve Athletic Training T-Shirt Made of Lightweight Breathable Fabric, Ideal for Running, Gym Workouts, and Casual Sportswear in All Seasons; High-Performance Breathable Cotton Crew Socks for Men with Arch Support, Cushioned Heel and Toe, and Moisture Control, Perfect for Sports, Walking, and Everyday Comfort; Men's Loose-Fit Basketball Shorts with Elastic Drawstring Waistband, Quick-Dry Mesh Fabric, and Printed Number 11 for Professional and Recreational Play; Official Size 7 Composite Leather Basketball Designed for Indoor and Outdoor Use, Deep Channel Design for Enhanced Grip and Ball Control, Ideal for Training and Competitive Matches;

Text description of each action

The Rise of Large Language Models

Transformer

2017



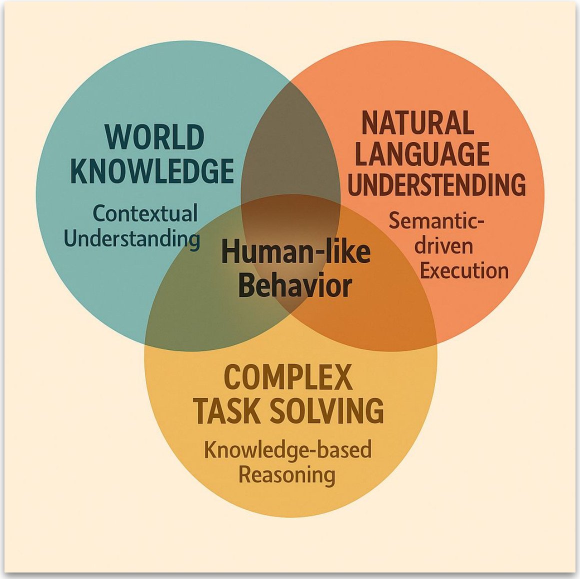
O3, R1...

2025

LLMs are developing so fast recently...

Zhao et al. A Survey of Large Language Models. arXiv:2303.18223

Large Language Models

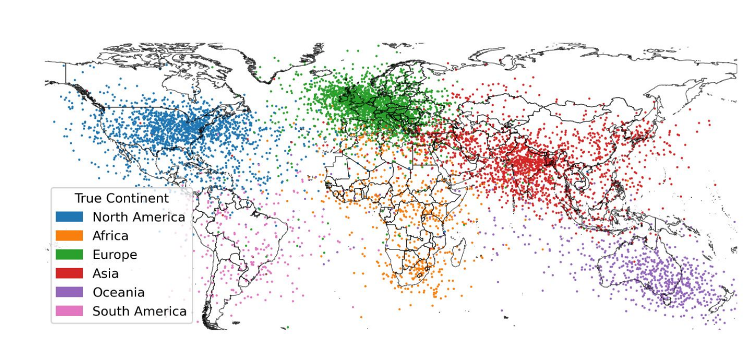


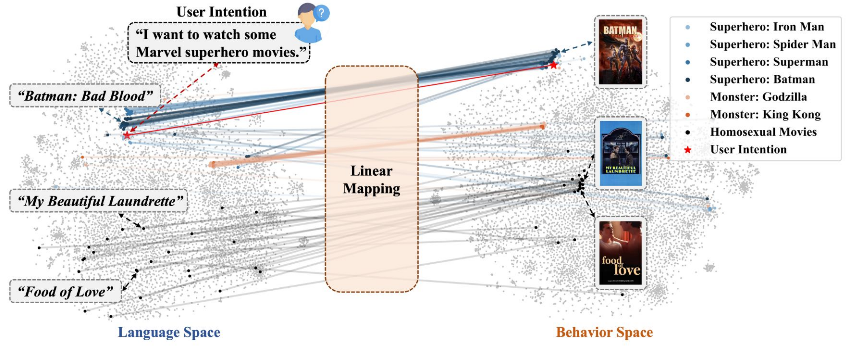
Key features:

- World knowledge.
- Natural language understanding.
- Human-like behavior.

Benefits of LLMs for Recommendation

(1) **World knowledge** – from pretraining





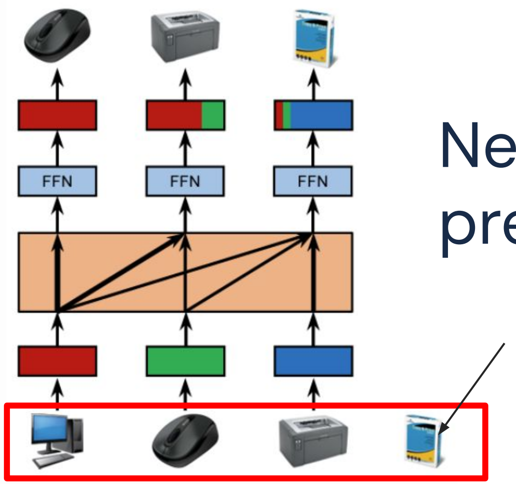
In space

In recommendation

Gurnee et al. Language Models Represent Space and Time. ICLR 2024.
Sheng et al. Language Representations Can be What Recommenders Need: Findings and Potentials. ICLR 2025.

Benefits of LLMs for Recommendation

(1) World knowledge



Next ID prediction

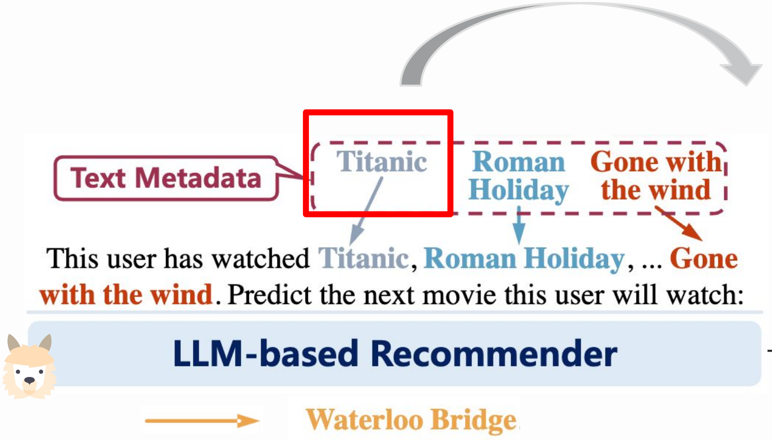
Item IDs

ID-based item modeling lack semantic meanings

Example: SASRec [*ICDM'18*]

Benefits of LLMs for Recommendation

(1) World knowledge



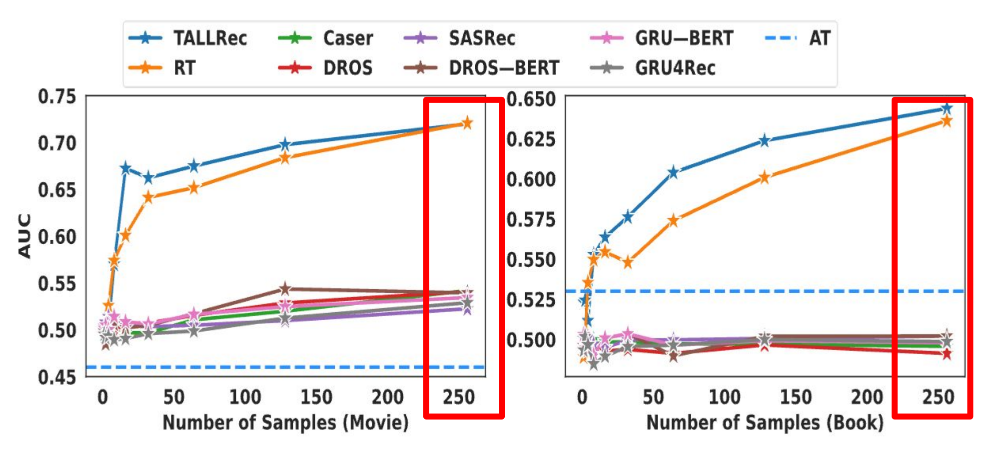
Titanic is a 1997 epic romance and disaster film directed by James Cameron, telling the tragic love story between Jack and Rose aboard the ill-fated RMS Titanic. It blends historical events with fictional drama, becoming one of the most iconic and emotionally powerful films of all time.

→ Abundant prior knowledge about items

Liao et al. LLaRA: Large Language-Recommendation Assistant. SIGIR 2024.

Benefits of LLMs for Recommendation

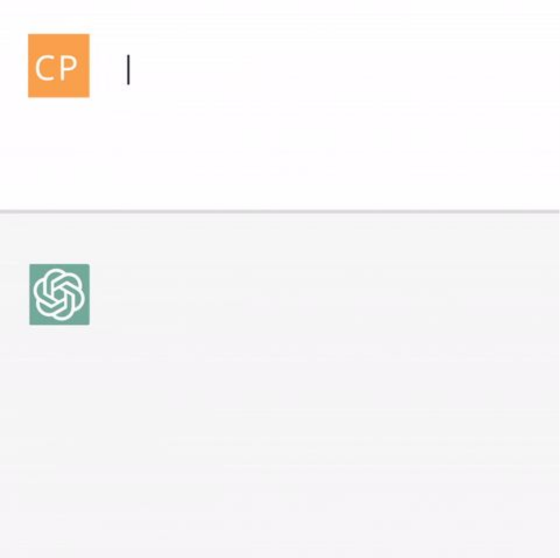
(1) World knowledge



Few data -> a good recommender

Benefits of LLMs for Recommendation

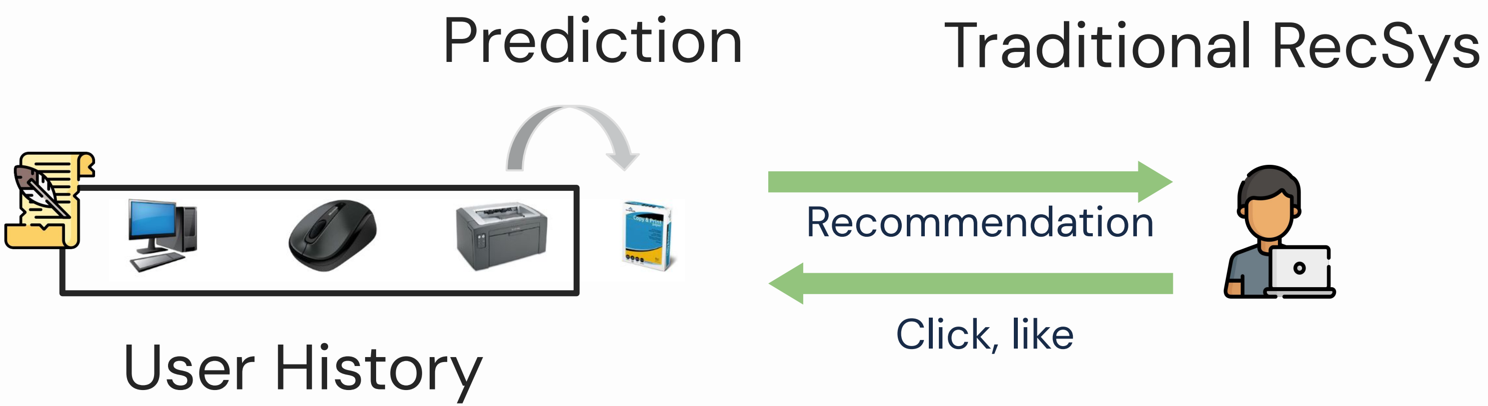
(2) Natural language understanding & generation



LLMs can interact with users fluently

Benefits of LLMs for Recommendation

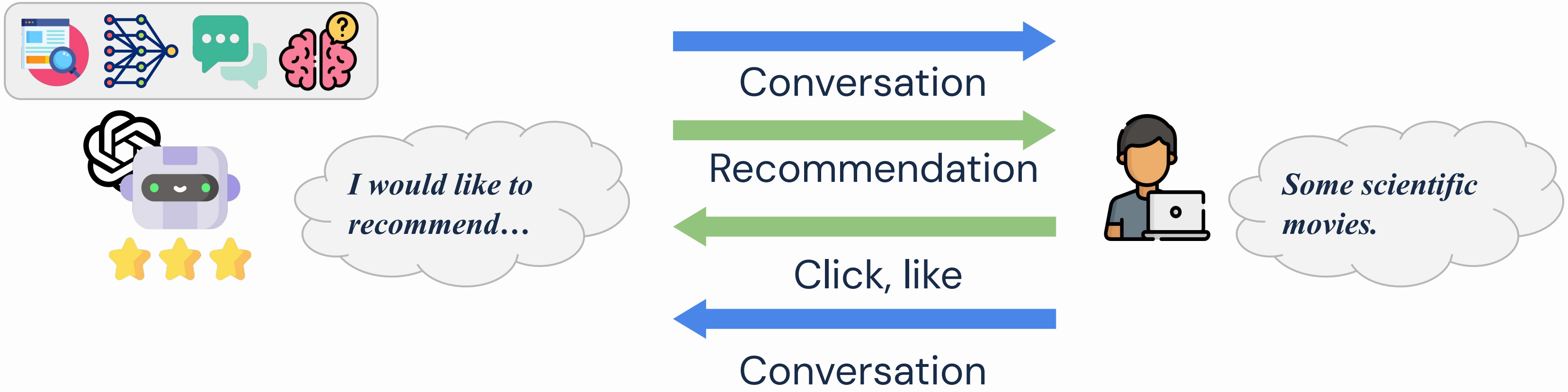
(2) Natural language understanding & generation



Benefits of LLMs for Recommendation

(2) Natural language understanding & generation

Conversational RecSys



Benefits of LLMs for Recommendation

(3) Human-like behavior

Park et al. Generative Agents: Interactive Simulacra of Human Behavior. UIST 2023

Benefits of LLMs for Recommendation

(3) Human-like behavior

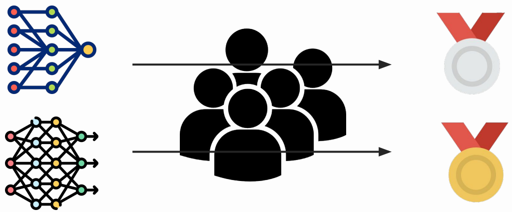
Offline recommender evaluation

Inaccurate, but
affordable

Benefits of LLMs for Recommendation

(3) Human-like behavior

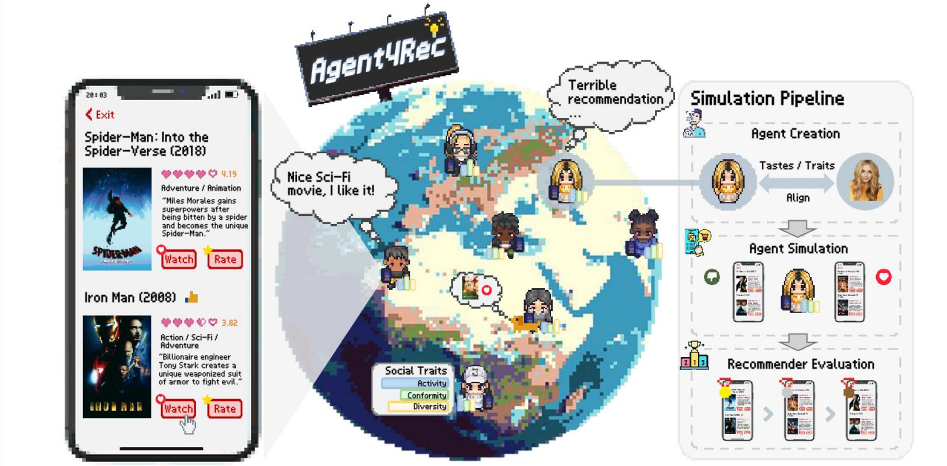
Online recommender evaluation



Accurate, but costly

Benefits of LLMs for Recommendation

(3) Human-like behavior



LLM as user simulator

Faithful
Affordable
Controllable
...

Part 1: LLM as Sequential Recommender

(i) **Early efforts**: Pretrained LLMs for recommendation;

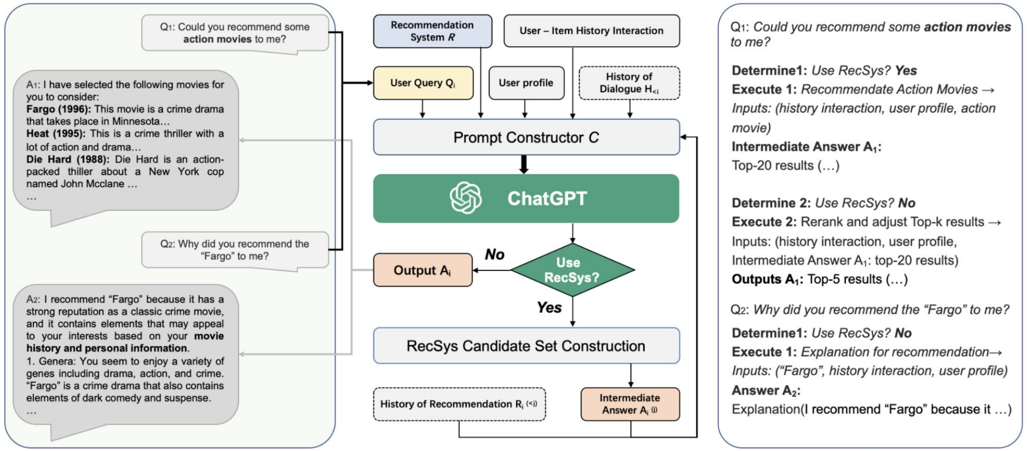
Early efforts

- Directly use freezed LLMs (e.g., GPT 4) for recommendation.

Early efforts

Prompt Engineering + In-Context Learning (ChatRec)

Key idea: LLMs as the recsys controller

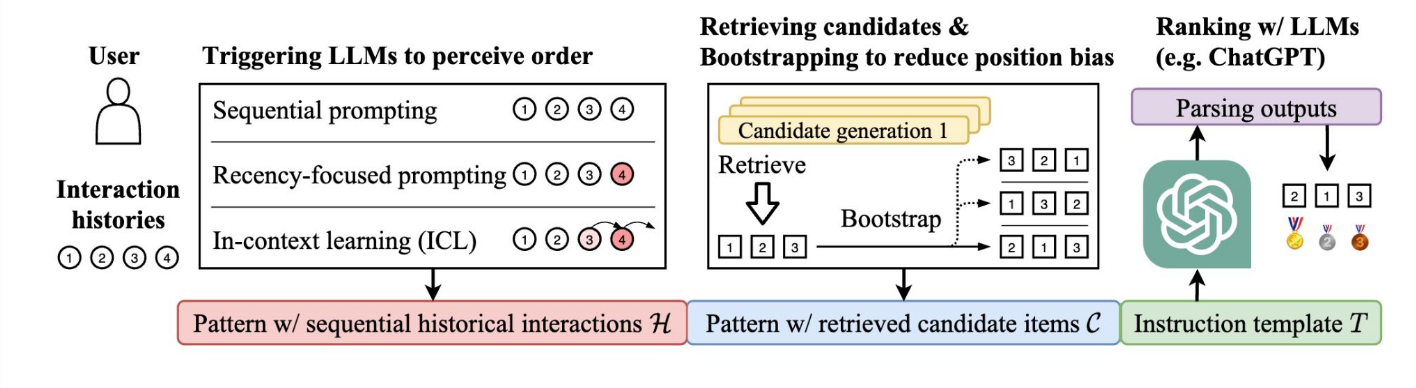


Gao et al. Chat-REC: Towards Interactive and Explainable LLMs-Augmented Recommender System. arXiv 2303.14524.

Early efforts

Prompt Engineering + In-Context Learning (LLMRank)

Key idea: LLMs as the reranker



Hou et al. Large Language Models are Zero-Shot Rankers for Recommender Systems. ECIR 2024.

Early efforts

- Directly use freezed LLMs (e.g., GPT 4) for recommendation.
- A performance gap compared to traditional recommenders exists.

Early efforts

Sub-optimal performance

	Method	ML-1M N@1	ML-1M N@5	ML-1M N@10	ML-1M N@20	Games N@1	Games N@5	Games N@10	Games N@20
full	Pop	0.08	1.20	4.13	5.79	0.13	1.00	2.27	2.62
full	BPRMF [49]	0.26	1.69	4.41	6.04	0.55	1.98	**2.96**	**3.19**
full	SASRec [33]	**3.76**	**9.79**	**10.45**	**10.56**	**1.33**	**3.55**	**4.02**	**4.11**
zero-shot	BM25 [50]	0.26	0.87	2.32	5.28	0.18	1.07	1.80	2.55
zero-shot	UniSRec [30]	0.88	3.46	5.30	6.92	0.00	1.86	2.03	2.31
zero-shot	VQ-Rec [29]	0.20	1.60	3.29	5.73	0.20	1.21	1.91	2.64
zero-shot	Ours	**1.74**	**5.22**	**6.91**	**7.90**	**0.90**	**2.26**	2.80	3.08

Hou et al. Large Language Models are Zero-Shot Rankers for Recommender Systems. ECIR 2024.

Part 1: LLM as Sequential Recommender

(i) Early efforts: Pretrained LLMs for recommendation;
(ii) **Aligning** LLMs for recommendation;

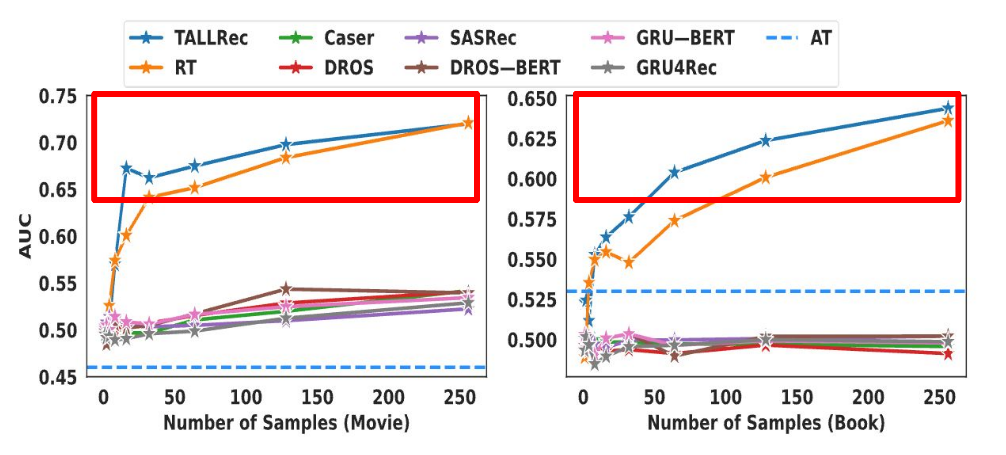
Aligning LLMs for recommendation

TALLRec

General task alignment -> Recommendation alignment

Aligning LLMs for recommendation

TALLRec



Good recommender with few training instances

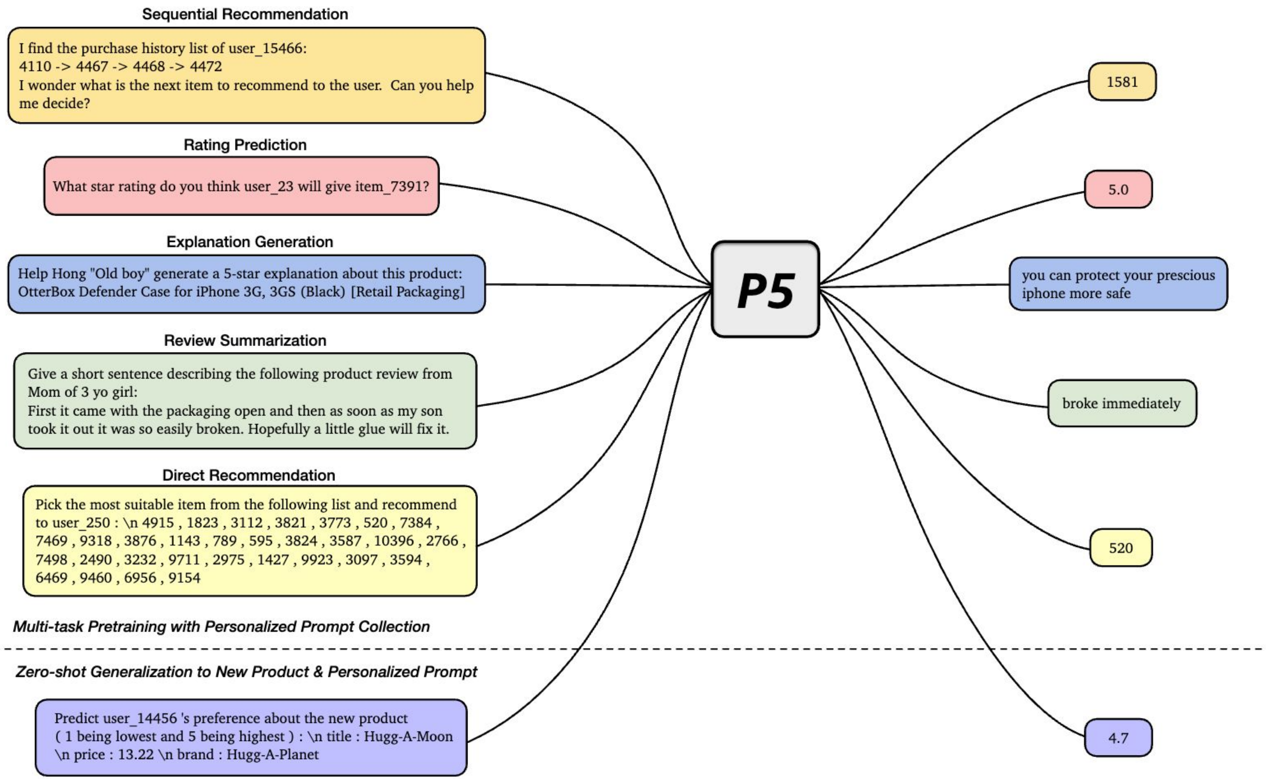
Bao et al. TALLRec: An Effective and Efficient Tuning Framework to Align Large Language Model with Recommendation. RecSys 2023.

Aligning LLMs for recommendation

P5

Multi-task

Cross-task generalization



Geng et al. Recommendation as Language Processing (RLP): A Unified Pretrain, Personalized Prompt & Predict Paradigm (P5). RecSys 2022.

Aligning LLMs for recommendation

InstructRec

Table 1: Example instructions with various types of user preferences, intentions, and task forms. To enhance the readability, we make some modifications to the original instructions that are used in our experiments.

Instantiation	Model Instructions
$\langle P_1, I_0, T_0 \rangle$	The user has purchased these items: <historical interactions>. Based on this information, is it likely that the user will interact with <target item> next?
$\langle P_2, I_0, T_3 \rangle$	You are a search engine and you meet a user's query: <explicit preference>. Please respond to this user by selecting items from the candidates: <candidate items>.
$\langle P_0, I_1, T_2 \rangle$	As a recommender system, your task is to recommend an item that is related to the user's <vague intention>. Please provide your recommendation.
$\langle P_0, I_2, T_2 \rangle$	Suppose you are a search engine, now the user search that <specific Intention>, can you generate the item to respond to user's query?
$\langle P_1, P_2, T_2 \rangle$	Here is the historical interactions of a user: <historical interactions>. His preferences are as follows: <explicit preference>. Please provide recommendations.
$\langle P_1, I_1, T_2 \rangle$	The user has interacted with the following <historical interactions>. Now the user search for <vague intention>, please generate products that match his intent.
$\langle P_1, I_2, T_2 \rangle$	The user has recently purchased the following <historical items>. The user has expressed a desire for <specific intention>. Please provide recommendations.

Unify recommendation & search via instruction tuning

Zhang et al. Recommendation as Instruction Following: A Large Language Model Empowered Recommendation Approach. TOIS.

Part 1: LLM as Sequential Recommender

(i) Early efforts: Pretrained LLMs for recommendation;
(ii) Aligning LLMs for recommendation;
(iii) **Training** objective & **inference**

Training objective

(1) Supervised finetuning (SFT)

I have watched Titanic, Roman Holiday, … Gone with the wind. Predict the next movie I will watch:

Training objective

(1) Supervised finetuning (SFT)

I have watched Titanic, Roman Holiday, … Gone with the wind. Predict the next movie I will watch:

Waterloo Bridge.

Prediction

Training objective

(1) Supervised finetuning (SFT)

I have watched Titanic, Roman Holiday, … Gone with the wind. Predict the next movie I will watch: Waterloo Bridge.

Prediction

Training objective

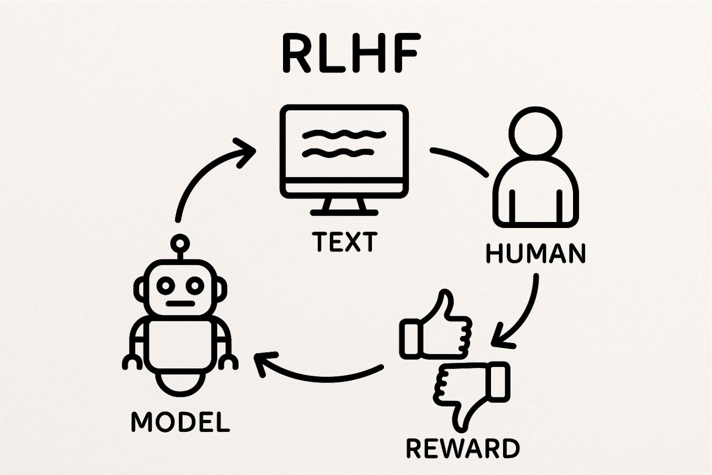
(1) Supervised finetuning (SFT)

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\mathbb{E}_{(x,y)\sim\mathcal{D}}\left[\sum_{t=1}^{T} \log \boxed{P_\theta(y_t \mid y_{<t})}\right]$$

Always predict the next token

Training objective

(2) Preference learning



LLMs are trained to align **human preferences**

Recommendation is about **user preferences**

Training objective

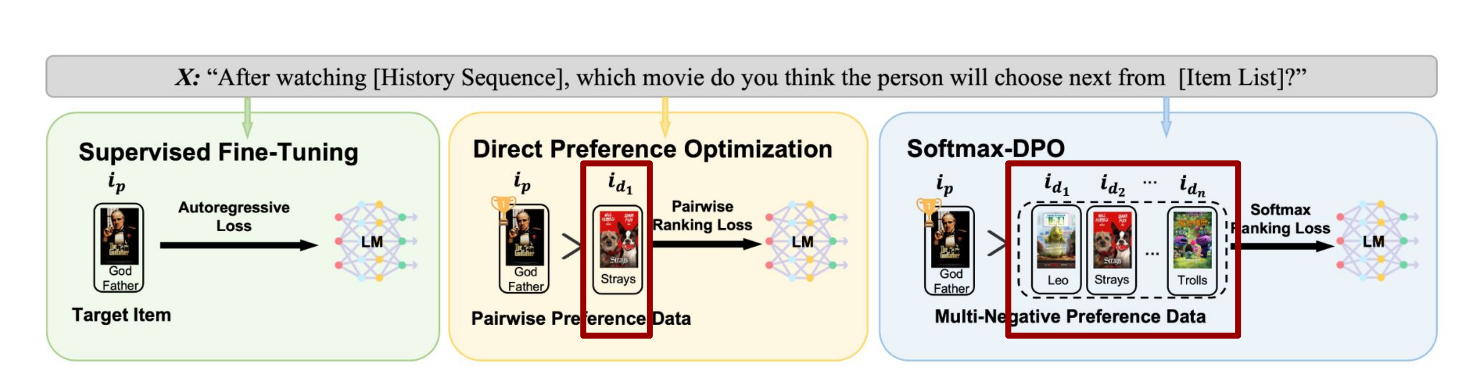
(2) Preference learning

I have watched Titanic, Roman Holiday, … Gone with the wind. Predict the next movie I will watch:

Waterloo Bridge > Harry Potter

Training objective

(2) Preference learning (S-DPO [NeurIPS'24])

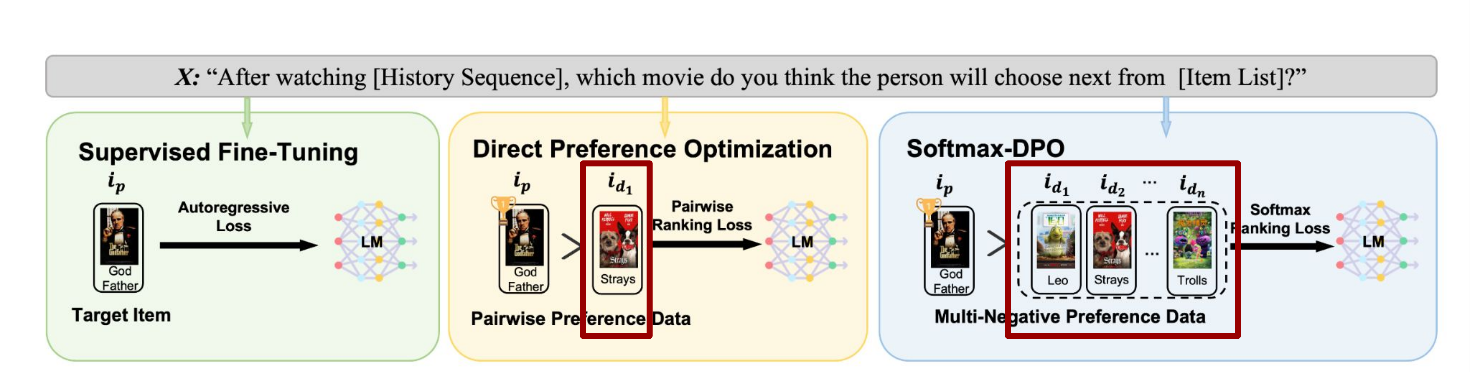


Single negative → Multiple negatives

Chen et al. On Softmax Direct Preference Optimization for Recommendation. NeurIPS 2024.

Training objective

(2) Preference learning (S-DPO [NeurIPS'24])



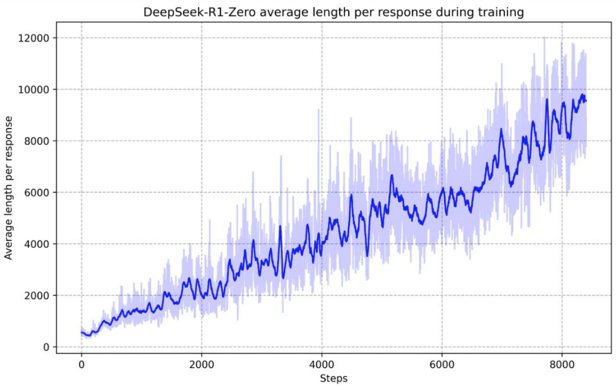
$$\mathcal{L}_{\mathrm{S-DPO}}(\pi_\theta; \pi_{\mathrm{ref}}) = -\mathbb{E}_{(x_u, e_p, \mathcal{E}_d) \sim \mathcal{D}} \left[\log \sigma \left(-\log \sum_{e_d \in \mathcal{E}_d} \exp \left(\beta \log \frac{\pi_\theta(e_d|x_u)}{\pi_{\mathrm{ref}}(e_d|x_u)} - \beta \log \frac{\pi_\theta(e_p|x_u)}{\pi_{\mathrm{ref}}(e_p|x_u)} \right) \right) \right].$$

Chen et al. On Softmax Direct Preference Optimization for Recommendation. NeurIPS 2024.

Training objective

(3) Reinforce learning

Emergent reasoning capabilities through RL



$$
\begin{aligned}
\mathcal{J}_{GRPO}(\theta) &= \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{old}}(O|q)] \\
&\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}A_i, \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon\right)A_i\right) - \beta\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right)\right), \quad (1)
\end{aligned}
$$

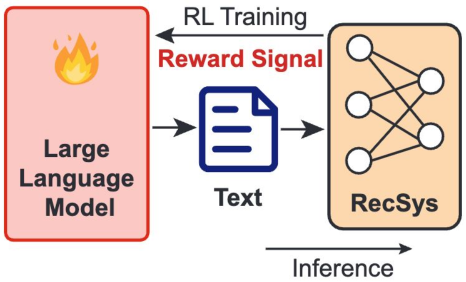
$$
\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right) = \frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - 1, \quad (2)
$$

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>
To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$
$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$
Rearrange to isolate the inner square root term:
$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$
$\ldots$
Wait, wait. Wait. That's an aha moment I can flag here.
Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$
We started with the equation:
$\sqrt{a - \sqrt{a + x}} = x$
First, let's square both sides:
$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$
Next, I could square both sides again, treating the equation: $\cdots$
$\ldots$

Training objective

(3) Reinforce learning (Rec-R1)



$$\max_{\theta} \mathbb{E}_{s\sim p(s),\, a\sim \pi_\theta(a|s)} \boxed{[f(a|s)]}.$$

Ranking metrics as rewards

Prompt Template for Rec-R1 + Dense Retriever (Product Search)

```
You are an expert in generating queries for dense retrieval. Given a customer
query, your task is to retain the original query while expanding it with
additional semantically relevant information, retrieve the most relevant
products, ensuring they best meet customer needs. If no useful expansion is
needed, return the original query as is.

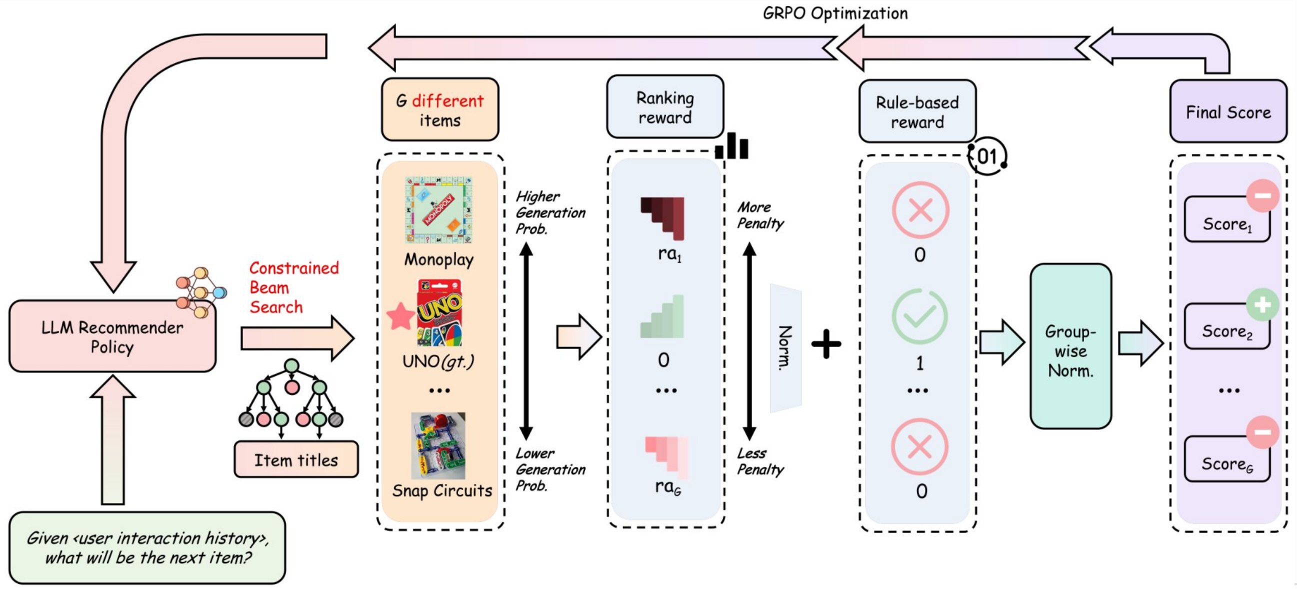
Below is the query:
``` {user_query} ```

<|im_start|>system
You are a helpful AI assistant. You first think about the reasoning process
in the mind and then provide the user with the answer.
<|im_end|>
<|im_start|>user
[PROMPT as above]
Show your work in <think>\think> tags. Your final response must be in JSON
format within <answer>\answer> tags. For example,
<answer>
{ "query": xxx }
</answer>.
<|im_end|>
<|im_start|>assistant
Let me solve this step by step.
<think>
```
```

Training objective

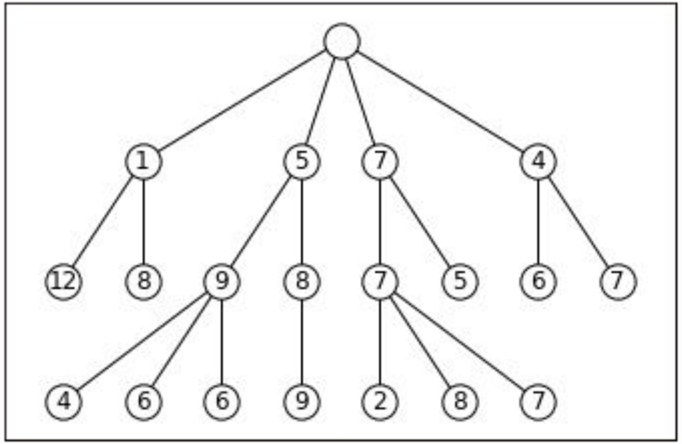
(3) Reinforce learning (ReRe)

Ranking rewards + rule-based rewards



Inference

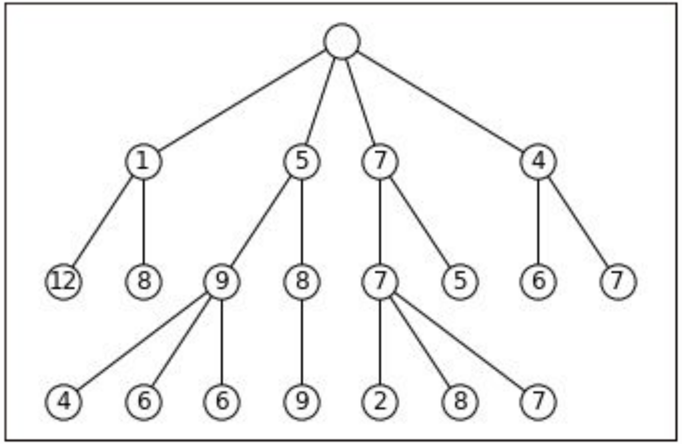
(1) Beam Search



Generating answers with the top-k highest scored beams

Inference

(1) Beam Search



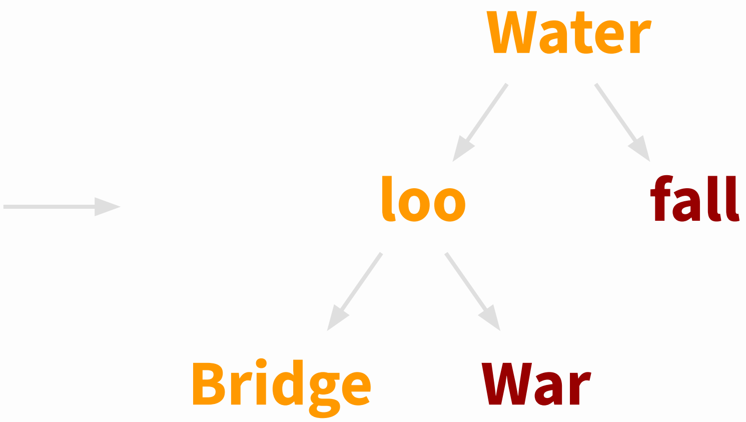
It may generate invalid items

How to ground the LLM outputs to real items?

Inference

(2) Constrained Beam Search

Valid items:
Waterloo Bridge, Waterfall Story, and Waterloo War



Constrained search tree

Inference

(2) Constrained Beam Search

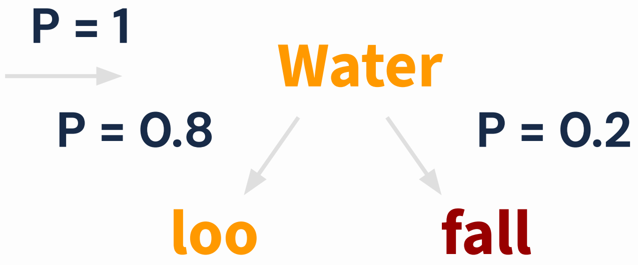
I have watched Titanic, Roman Holiday, …
Gone with the wind. Predict the next movie
I will watch:

$\xrightarrow{P = 1}$ Water

Inference

(2) Constrained Beam Search

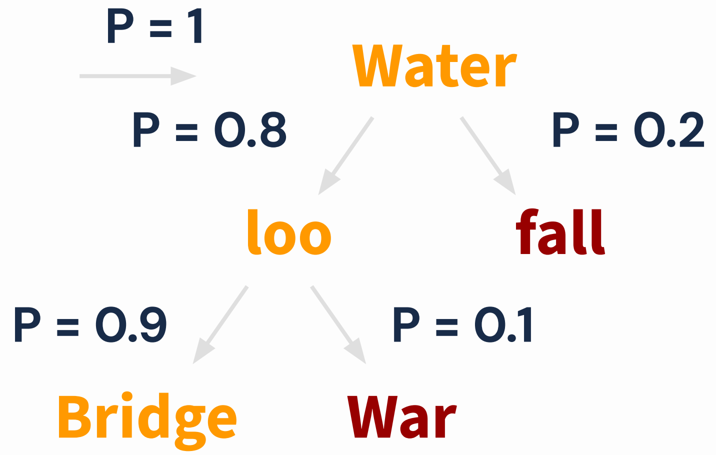
I have watched Titanic, Roman Holiday, … Gone with the wind. Predict the next movie I will watch:



Inference

(2) Constrained Beam Search

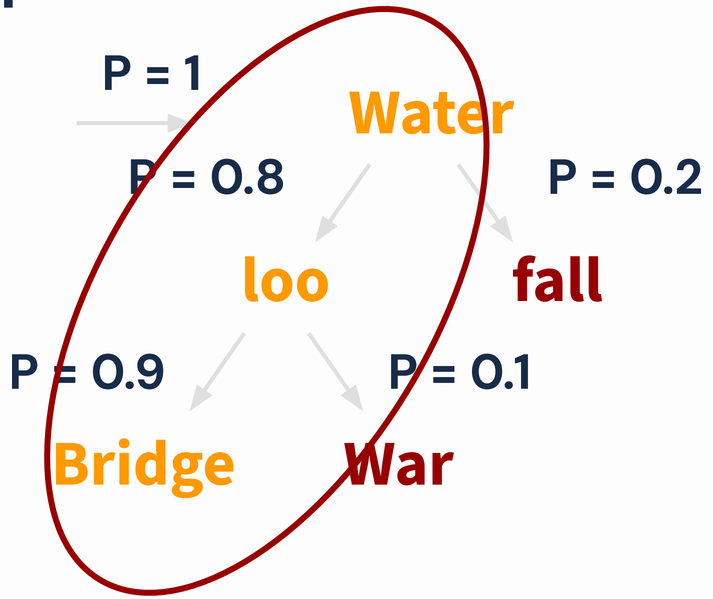
I have watched Titanic, Roman Holiday, … Gone with the wind. Predict the next movie I will watch:



Inference

(2) Constrained Beam Search

I have watched Titanic, Roman Holiday, … Gone with the wind. Predict the next movie I will watch:



Valid Item!

Inference

(3) Improved Constrained Beam Search (D3)

$$S(h_{\leq t}) = S(h_{\leq t-1}) + \log(p(h_t|x, h_{\leq t-1})),$$

$$S(h) = S(h)\boxed{/h_L^{\alpha},}$$

Length penalty in beam search;
Human does not like over long sentences.

Redundant for recommendation

Bao et al. Decoding Matters: Addressing Amplification Bias and Homogeneity Issue for LLM-based Recommendation. EMNLP 2024.

Inference

(3) Improved Constrained Beam Search (D3)

$$\mathcal{S}(h_{\leq t}) = \mathcal{S}(h_{\leq t-1}) + \log(p(h_t|x, h_{\leq t-1})),$$

$$\mathcal{S}(h) = \mathcal{S}(h)/\cancel{L},$$

Remove length penalty

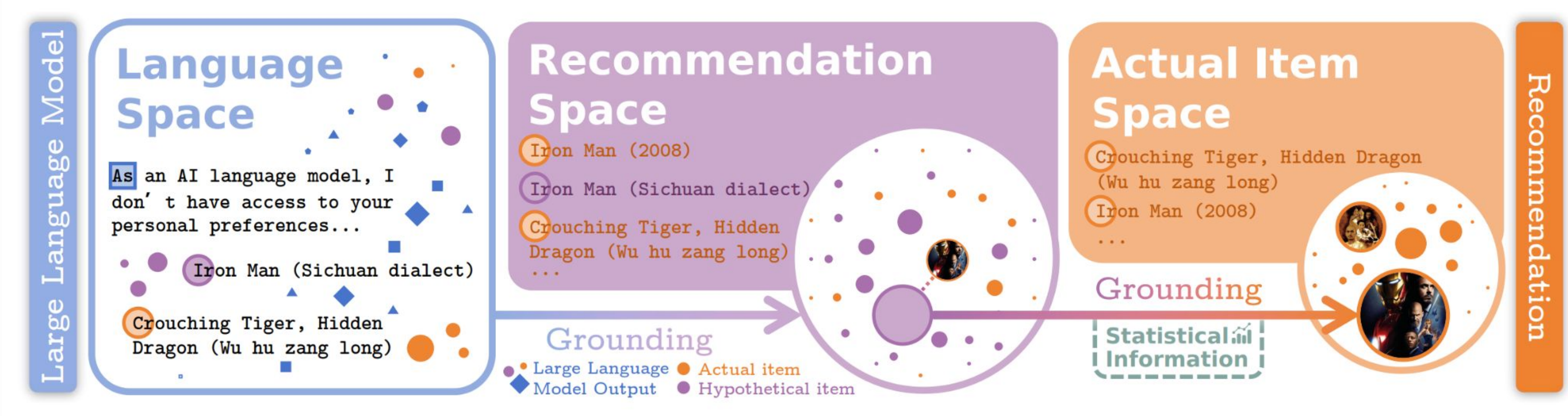
	Instruments	Books	CDs	Sports	Toys	Games
Baseline	0.1062	0.0308	0.0956	0.1171	0.0965	0.0610
D^3	**0.1111**	**0.0354**	**0.1190**	**0.1215**	**0.1025**	**0.0767**
- RLN	0.1093	0.0353	0.1000	0.1200	0.0975	0.0659
- TFA	0.1086	0.0309	0.1115	0.1192	0.1006	0.0732

Imp when removing

Bao et al. Decoding Matters: Addressing Amplification Bias and Homogeneity Issue for LLM-based Recommendation. EMNLP 2024.

Inference

(4) Dense Retrieval Grounding (BIGRec)



Retrieve real items by generated text

Bao et al. A Bi-Step Grounding Paradigm for Large Language Models in Recommendation Systems. TORS.

Part 1: LLM as Sequential Recommender

(1) Early efforts: using LLMs in a zero-shot setting

(2) Aligning LLMs for recommendation

(Multi-task) instruction tuning

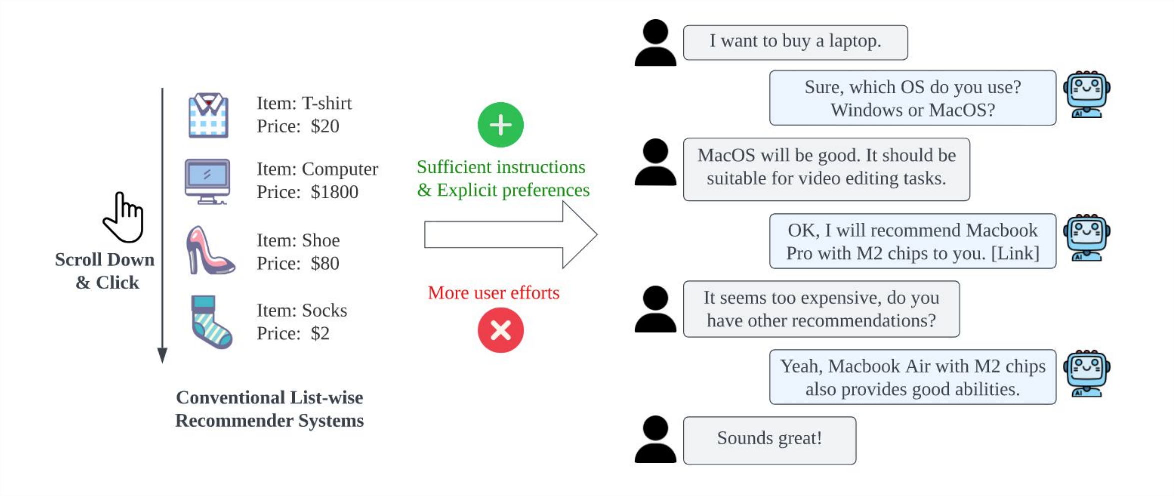
(3) Training objective: SFT, DPO, RL;

Inference: (constrained) beam search, retrieval;

Application 1:
LLM as Conversational Recommender

Conversational Recommender System (CRS)

- Recommendations with multiple turns conversation
- Interactive; engaging users in the loop



Chen et al. All Roads Lead to Rome: Unveiling the Trajectory of Recommender Systems Across the LLM Era. arXiv.2407.10081

Paradigms of CRS before the era of LLM

Features: Task-specific conversational recommenders, trained on limited conversation data.

Paradigms of CRS before the era of LLM

Features: Task-specific conversational recommenders, trained on limited conversation data.

- Lack of world knowledge.
- Requirement of complicated strategies.
- Lack of generalization capabilities.

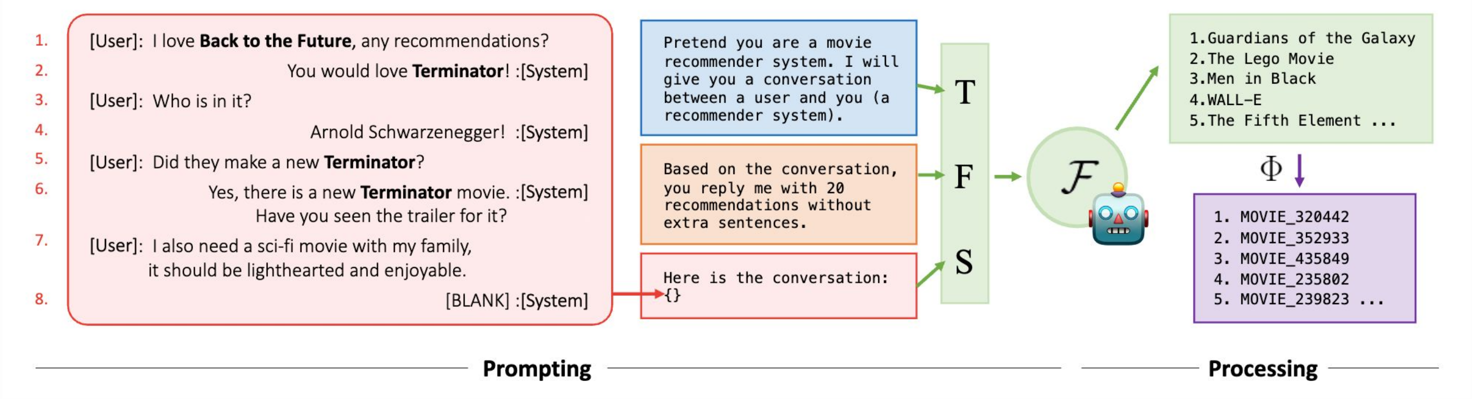
Example

LLM as conversational recommender

https://www.amazon.com/Rufus/

LLM as Conversational Recommender

LLMs as zero-shot CRS

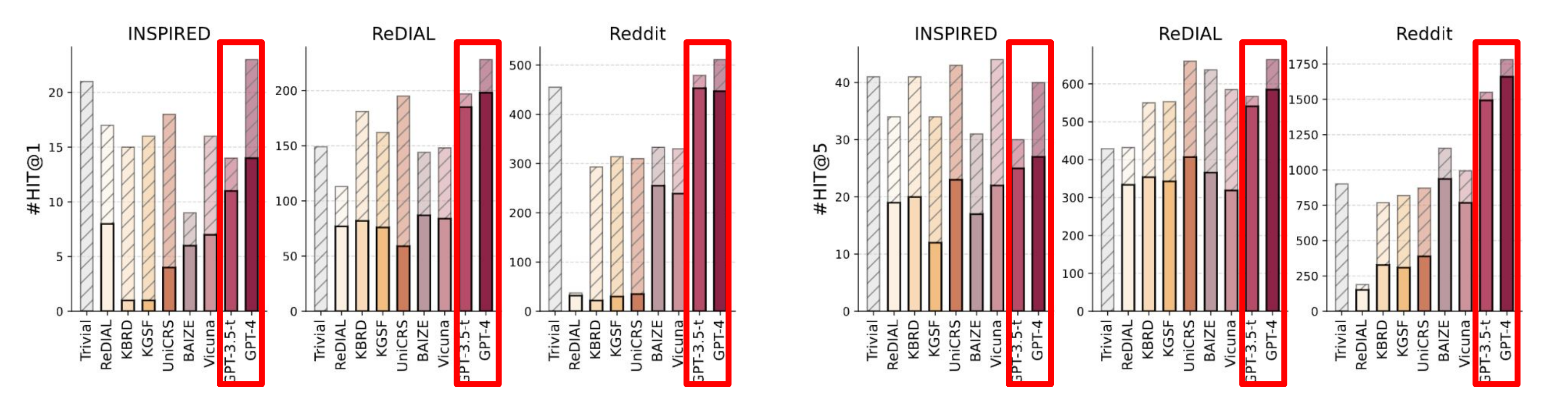


How powerful are LLMs for zero-shot CRS?

He et al. Large Language Models as Zero-Shot Conversational Recommenders. CIKM 2023.

LLM as Conversational Recommender

LLMs as zero-shot CRS

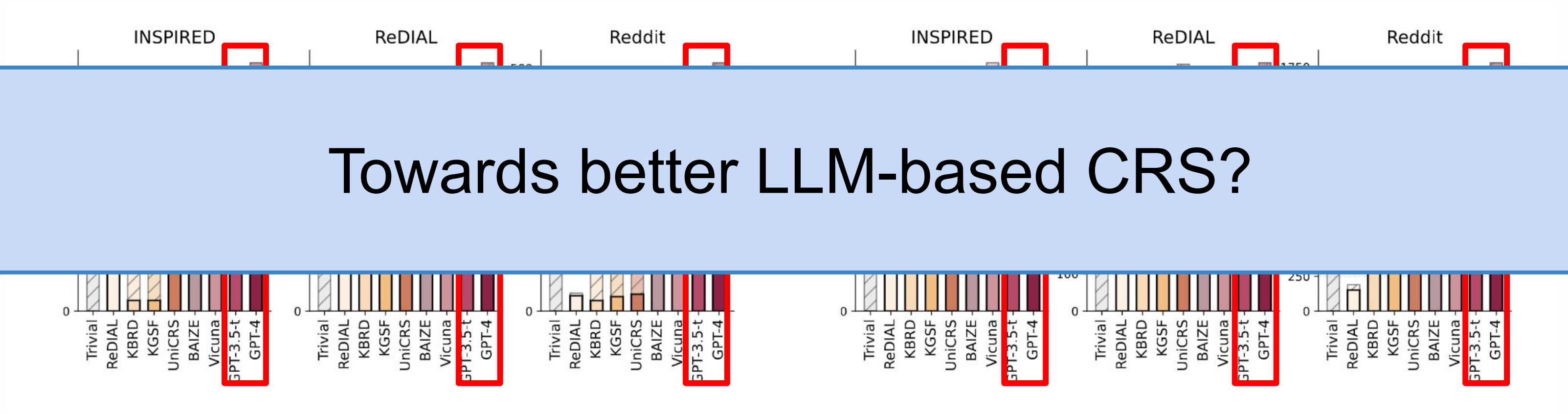


Can surpass traditional CRSs!

He et al. Large Language Models as Zero-Shot Conversational Recommenders. CIKM 2023.

LLM as Conversational Recommender

LLMs as zero-shot CRS



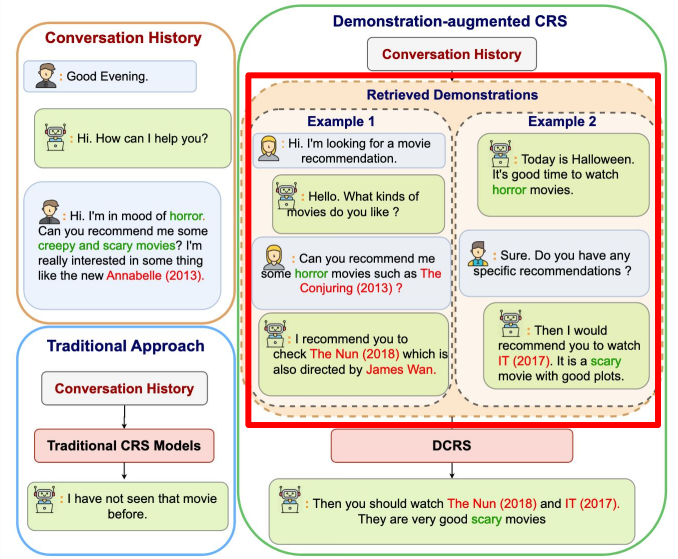
Towards better LLM-based CRS?

Can surpass traditional CRSs!

He et al. Large Language Models as Zero-Shot Conversational Recommenders. CIKM 2023.

LLM as Conversational Recommender

+ Demonstration

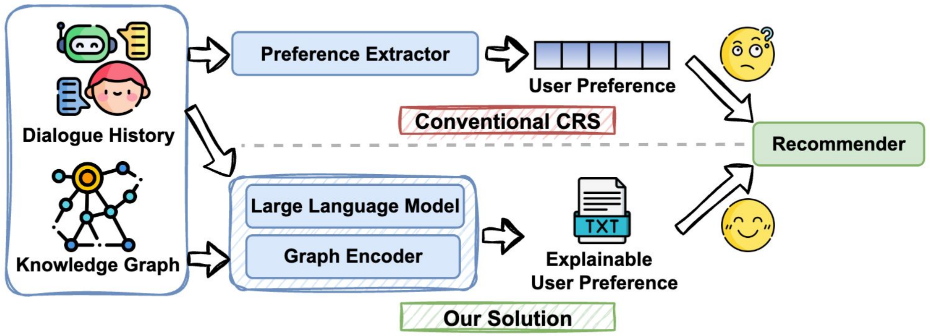


Prompting with previously successful conversation

Relevant conversation history helps!

LLM as Conversational Recommender

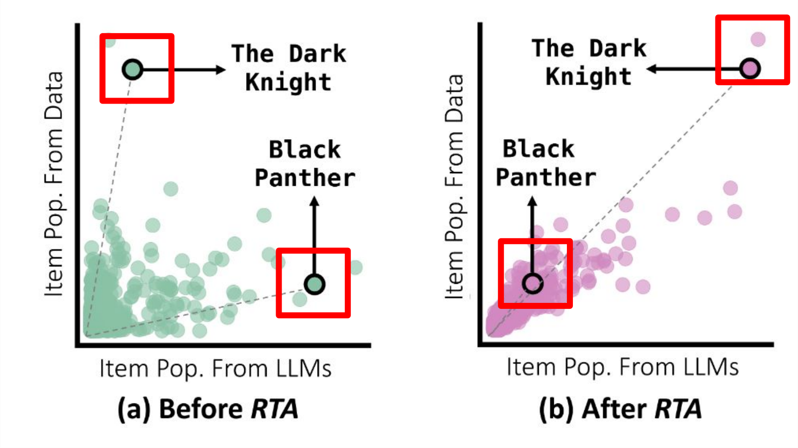
+ Knowledge graph



Recommendation-specific knowledge graph helps

LLM as Conversational Recommender

+ Collaborative information



Collaborative information (e.g., popularity) helps LLMs fit the real distribution in CRS

He et al. Reindex-Then-Adapt: Improving Large Language Models for Conversational Recommendation. WSDM 2025.

LLM as Conversational Recommender

Challenges – Datasets

Public datasets for CRS are limited, due to the scarcity of conversational products and real-world CRS datasets

LLM as Conversational Recommender

Challenges – Evaluation

Traditional metrics like NDCG and BLEU are often insufficient to assess user experience

LLM as Conversational Recommender

Challenges – Product

What is the form of LLM-based CRS products?

ChatBot? Search bar? Independent App?

Application 1:
LLM as Conversational Recommender

(1) LLMs are promising backbone models for CRS

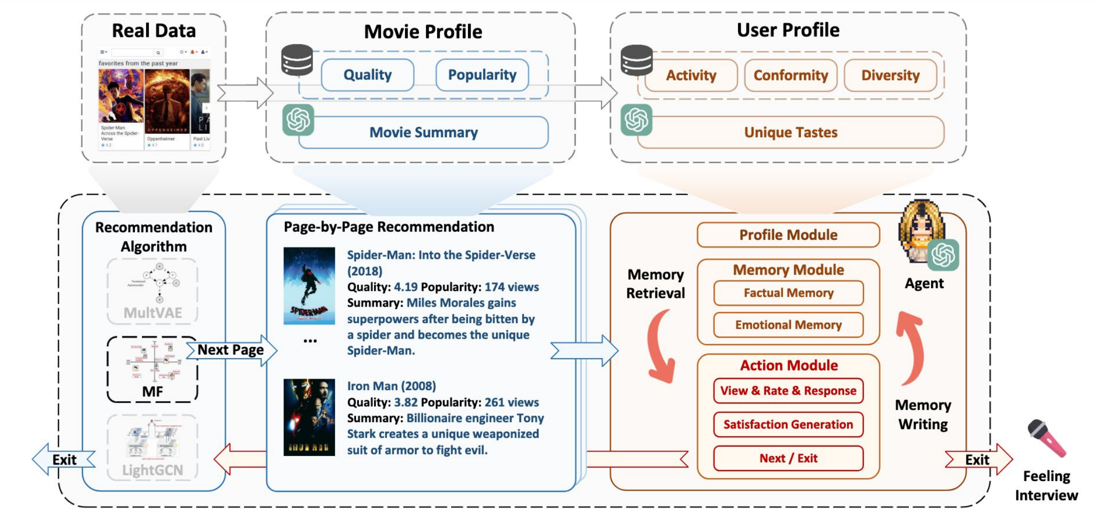
(2) Challenges in LLM-based CRSs:

dataset, evaluation, and product

Application 2:
LLM as User Simulator

LLM as User Simulator

Generative agents for recommendation



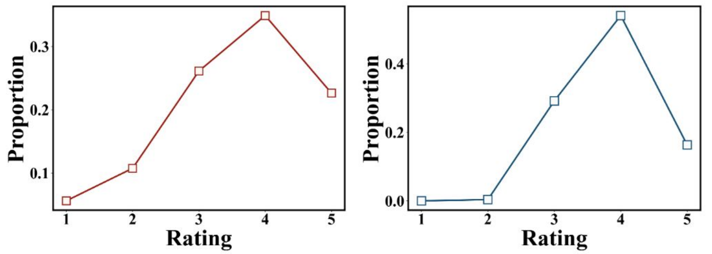
Realworld-like simulation paradigm

- 1000 users
- Page-by-page simulation

Zhang et al. On generative agents in recommendation. SIGIR 2024.

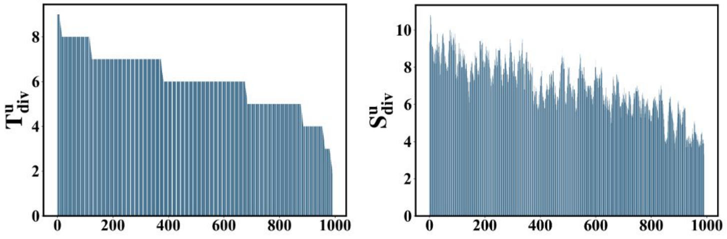
LLM as User Simulator

Generative agents for recommendation



(a) Distribution on MovieLens (b) Agent-simulated distribution

Aligned user preferences
& Recommender evaluation



(a) Ground-truth diversity (b) Simulated diversity

Table 2: Recommendation strategies evaluation.

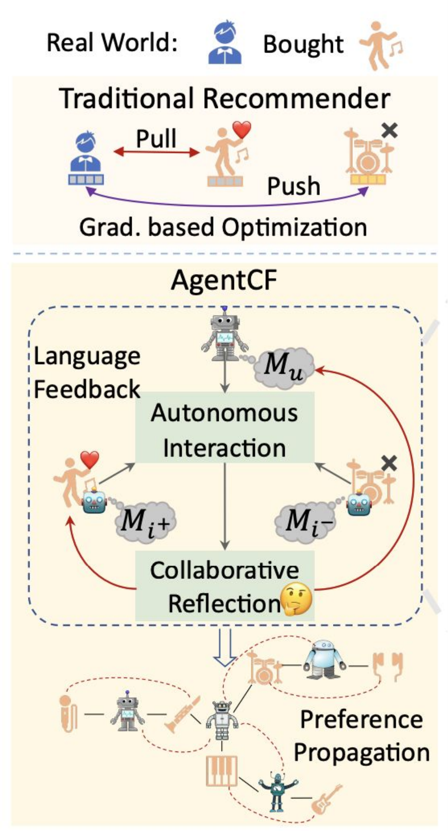
	$\overline{P}_{view}$	$\overline{N}_{like}$	$\overline{P}_{like}$	$\overline{N}_{exit}$	$\overline{S}_{sat}$
Random	0.312	3.3	0.269	2.99	2.93
Pop	0.398	4.45	0.360	3.01	3.42
MF	0.488	**6.07***	0.462	**3.17***	3.80
MultVAE	0.495	5.69	0.452	3.10	3.75
LightGCN	**0.502***	5.73	**0.465***	3.02	**3.85***

Zhang et al. On generative agents in recommendation. SIGIR 2024.

LLM as User Simulator

AgentCF

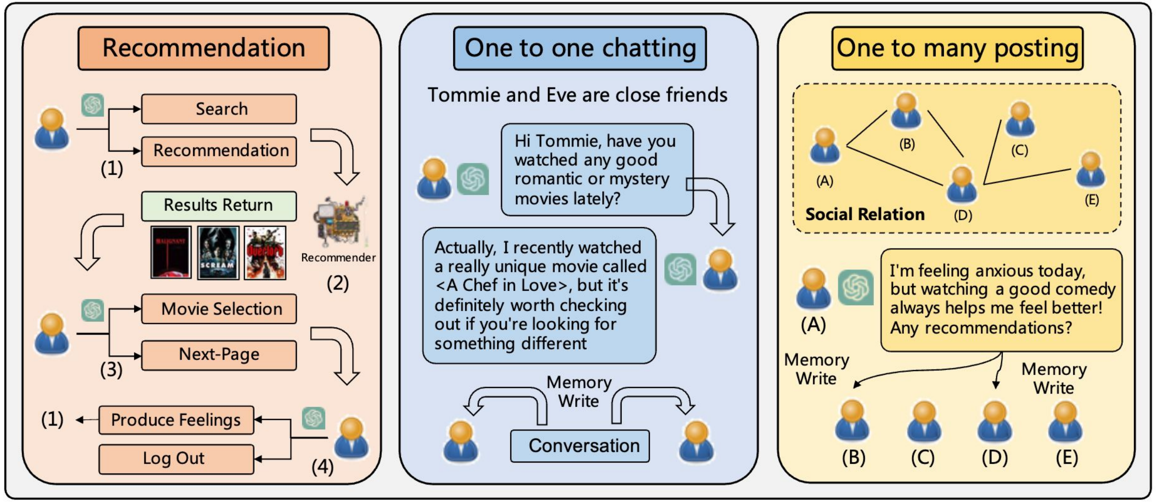
- Agents for both users and items
- Co-optimized by real collaborative filtering signals (user-item interactions)
- Memory updated by reflection



Zhang et al. AgentCF: Collaborative Learning with Autonomous Language Agents for Recommender Systems. WWW 2024.

LLM as User Simulator

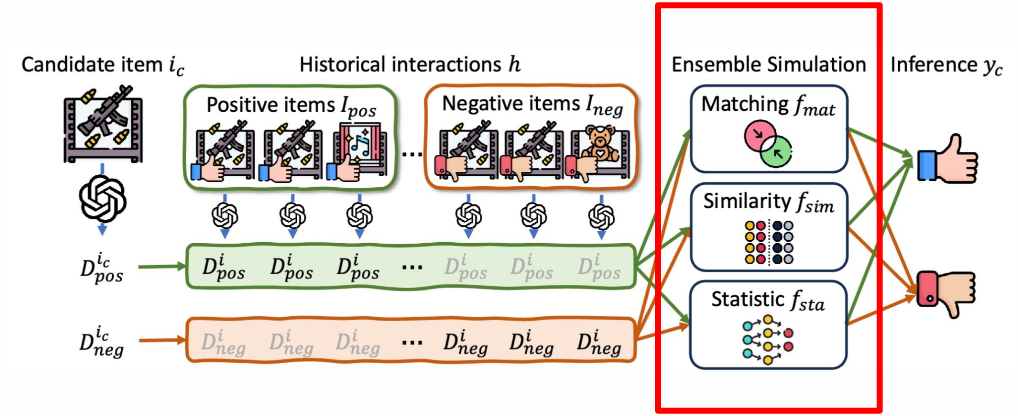
+ ## Social behaviors



Recommendation
Chat
Networking

LLM as User Simulator

+ Multi-facet simulation objective



Category matching
Fine-grained similarity
Statistic information

LLM-based Generative Rec

(1) Tokenize actions by text

Pros: distribution naturally aligned with LLMs

Cons: inefficient

(2) From zero-shot to instruction tuning

Training objectives: SFT, DPO, RL, …

Inference: constrained beam search, retrieval

(3) Applications

Conversational RS, User Simulator

03

Intro of Semantic IDs

Language Tokenization

Human-readable Data 👨 👩

Will be the 46th film overall in the

Tokenize

Machine-readable Data 🤖

8743, 307, 262, 6337, 400, 2646, 4045, 287, 262 9923, 21932, 1512, 11950

Detokenize

Marvel Cinematic Universe

Language Tokenization

Subword Tokenization

e.g., BPE (Byte-Pair Encoding) in gpt-5

Machine-readable Data 🤖

8743, 307, 262, 6337, 400, 2646, 4045, 287, 262, 9923, 21932, 1512, 11950

Will _be _the _46 th _film _overall _in _the _Marvel _Cinem atic _Universe

Language Tokenization

Subword Tokenization

e.g., BPE (Byte-Pair Encoding) in gpt-5

Machine-readable Data 🤖

8743, 307, 262, 6337, 400, 2646, 4045, 287, 262, 9923, 21932, 1512, 11950

Will _be _the _46 th _film _overall _in _the _Marvel _Cinem atic _Universe

Q1: Why not use **one token per word**?

Language Tokenization

Subword Tokenization

e.g., BPE (Byte-Pair Encoding) in gpt-5

Machine-readable Data 🤖

8743,	307,	262,	6337,	400,	2646,	4045,	287,	262	9923,	21932,	1512,	11950
Will	_be	_the	_46	th	_film	_overall	_in	_the	_Marvel	_Cinem	atic	_Universe

Q1: Why not use **one token per word**?

Large token vocabulary; Not robust enough;

Language Tokenization

Subword Tokenization

e.g., BPE (Byte-Pair Encoding) in gpt-5

Machine-readable Data 🤖

8743,	307,	262,	6337,	400,	2646,	4045,	287,	262		9923,	21932,	1512,	11950
Will	_be	_the	_46	th	_film	_overall	_in	_the		_Marvel	_Cinem	atic	_Universe

Q2: Why not use **one token per character**?

Language Tokenization

Subword Tokenization

e.g., BPE (Byte-Pair Encoding) in gpt-5

Machine-readable Data 🤖

8743, 307, 262, 6337, 400, 2646, 4045, 287, 262, 9923, 21932, 1512, 11950

Will _be _the _46 th _film _overall _in _the _Marvel _Cinem atic _Universe

Q2: Why not use **one token per character**?

Too many tokens per sentence (long context windows); Difficult to model;

Action Tokenization

Human-readable Data 👨 👩

Machine-readable Data 🤖

↓ **Tokenize**

i3487, i124, i19240, i772

One token per action? 🤔

Action Tokenization

Human-readable Data 👨 👩

Tokenize

Machine-readable Data 🤖

Premium Men's Short Sleeve Athletic Training T-Shirt Made of Lightweight Breathable Fabric, Ideal for Running, Gym Workouts, and Casual Sportswear in All Seasons; High-Performance Breathable Cotton Crew Socks for Men with Arch Support, Cushioned Heel and Toe, and Moisture Control, Perfect for Sports, Walking, and Everyday Comfort; Men's Loose-Fit Basketball Shorts with Elastic Drawstring Waistband, Quick-Dry Mesh Fabric, and Printed Number 11 for Professional and Recreational Play; Official Size 7 Composite Leather Basketball Designed for Indoor and Outdoor Use, Deep Channel Design for Enhanced Grip and Ball Control, Ideal for Training and Competitive Matches;

Text description of each action? 🤔

Action Tokenization

Can we find a sweet spot between

One token per action

i3487, i124, i19240, i772

Text description

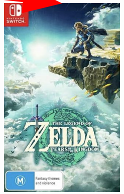
Premium Men's Short Sleeve Athletic Training T-Shirt Made of Lightweight Breathable Fabric, Ideal for Running, Gym Workouts, and Casual Sportswear in All Seasons; High-Performance Breathable Cotton Crew Socks for Men with Arch Support, Cushioned Heel and Toe, and Moisture Control, Perfect for Sports, Walking, and Everyday Comfort; Men's Loose-Fit Basketball Shorts with Elastic Drawstring Waistband, Quick-Dry Mesh Fabric, and Printed Number 11 for Professional and Recreational Play; Official Size 7 Composite Leather Basketball Designed for Indoor and Outdoor Use, Deep Channel Design for Enhanced Grip and Ball Control, Ideal for Training and Competitive Matches;

Semantic IDs

(also called: SemID or SID)

A few tokens that jointly index one item.

$\overbrace{\texttt{t3, t321, t643, t1011}}$



Semantic IDs

(also called: SemID or SID)

A few tokens that jointly index one item.

t3, $\boxed{\textbf{t321,}}$ t643, t1011

{t257, t258, ..., t320, t321, t322, ..., t511, t512}

Each token from a vocabulary shared by all items

Semantic IDs

(also called: SemID or SID)

A few tokens that jointly index one item.

`t3, t321, t643, t1011`

Can index maximally $256^4 \approx 4.3 \times 10^9$ items with 1024 tokens

(4 tokens per item, each from a vocabulary of 256)

Action Tokenization

Can we find a sweet spot between

One token per action

i3487, i124, i19240, i772

Text description of each action

Premium Men's Short Sleeve Athletic Training T-Shirt Made of Lightweight Breathable Fabric, Ideal for Running, Gym Workouts, and Casual Sportswear in All Seasons; High-Performance Breathable Cotton Crew Socks for Men with Arch Support, Cushioned Heel and Toe, and Moisture Control, Perfect for Sports, Walking, and Everyday Comfort; Men's Loose-Fit Basketball Shorts with Elastic Drawstring Waistband, Quick-Dry Mesh Fabric, and Printed Number 11 for Professional and Recreational Play; Official Size 7 Composite Leather …

Action Tokenization

Can we find a sweet spot between

One token per action

i3487, i124, i19240, i772

Semantic IDs

t34, t392, t600, t891, t21, t502, t592, t1002, t123, t403, t611, t821, t21, t502, t711, t1022

Text description of each action

Premium Men's Short Sleeve Athletic Training T-Shirt Made of Lightweight Breathable Fabric, Ideal for Running, Gym Workouts, and Casual Sportswear in All Seasons; High-Performance Breathable Cotton Crew Socks for Men with Arch Support, Cushioned Heel and Toe, and Moisture Control, Perfect for Sports, Walking, and Everyday Comfort; Men's Loose-Fit Basketball Shorts with Elastic Drawstring Waistband, Quick-Dry Mesh Fabric, and Printed Number 11 for Professional and Recreational Play; Official Size 7 Composite Leather …

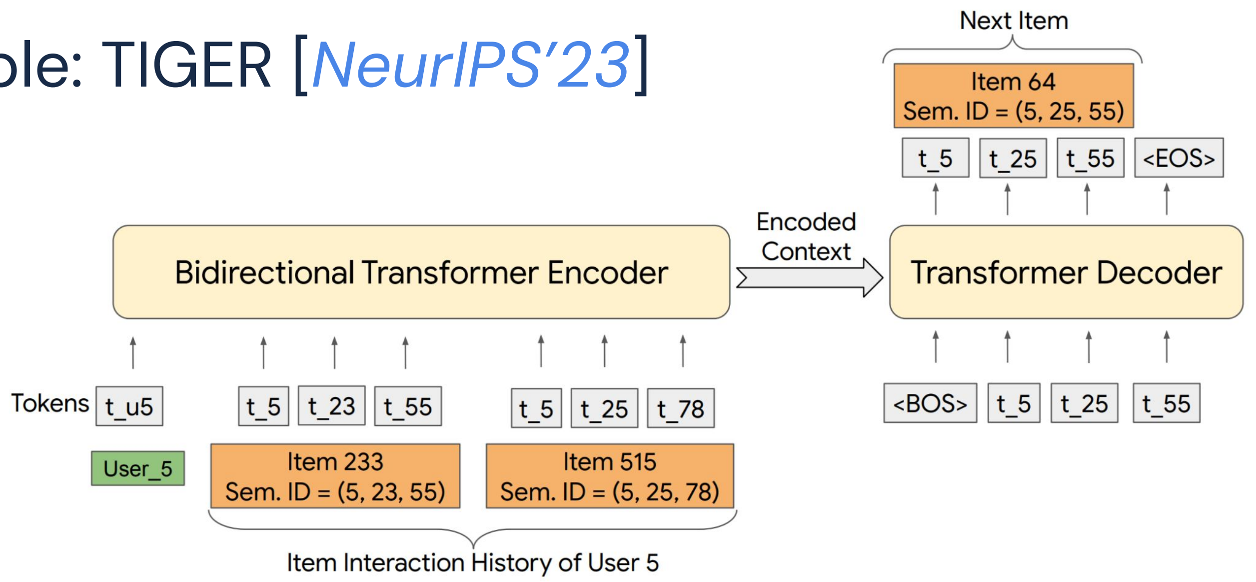
Action Tokenization

Can we find a sweet spot between

	One token per action	Semantic IDs	Text description of each action
Length	4	16	148
#Vocab Action Tokens	~100k	1024	0

Generative Models based on Semantic IDs

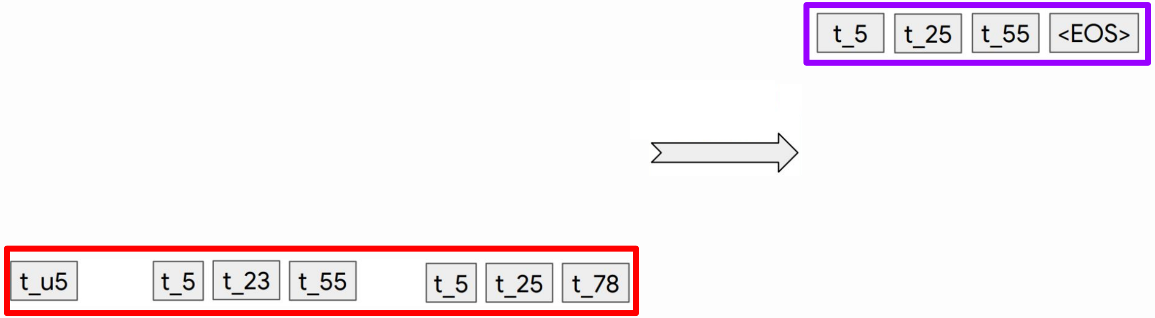
Example: TIGER [*NeurIPS'23*]



Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

Generative Models based on Semantic IDs

Example: TIGER [*NeurIPS'23*]



Recommendation as a **seq**-to-**seq** generation problem

Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

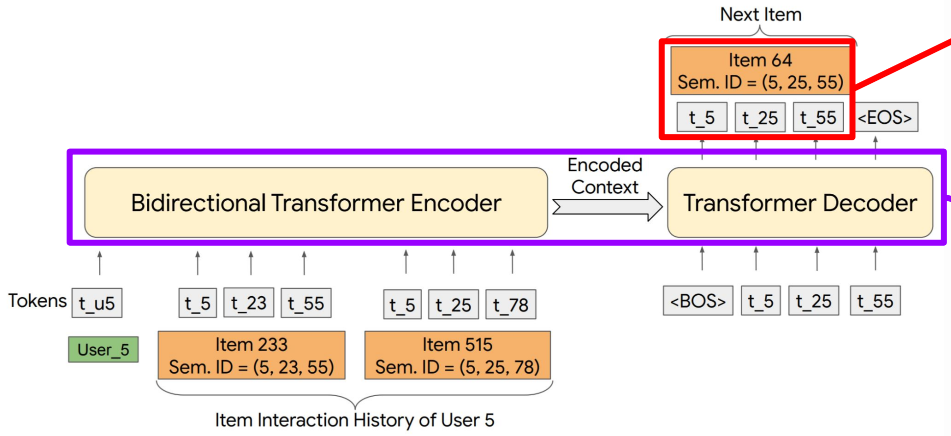
Generative Models based on Semantic IDs

Recommendation as a **seq**-to-**seq** generation problem

Input: user interacted items $\{c_{11}, c_{12}, c_{13}, c_{14}, c_{21}, c_{22}, \ldots\}$

Output: next item $\{c_{t1}, c_{t2}, c_{t3}, c_{t4}\}$

SemID-based Generative Recommendation



Part 1:

How to construct SIDs

Part 2:

How to build SID-based Generative Rec Models

Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

04

Semantic ID

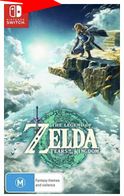
-based Generative Recommendation

Semantic IDs

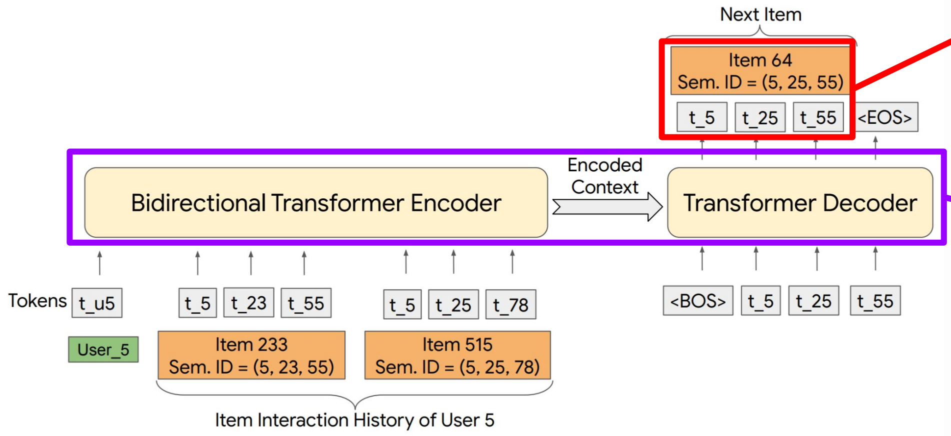
(also called: SemID or SID)

A few tokens that jointly index one item.

t3, t321, t643, t1011



SemID-based Generative Recommendation



Part 1:

How to construct SIDs

Part 2:

How to build SID-based Generative Rec Models

Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

Part 1: Semantic ID Construction

Semantic ID Construction

Input: all data associated with the item (description, title, interactions, features, …)

Output: mapping between items ⇔ Semantic IDs

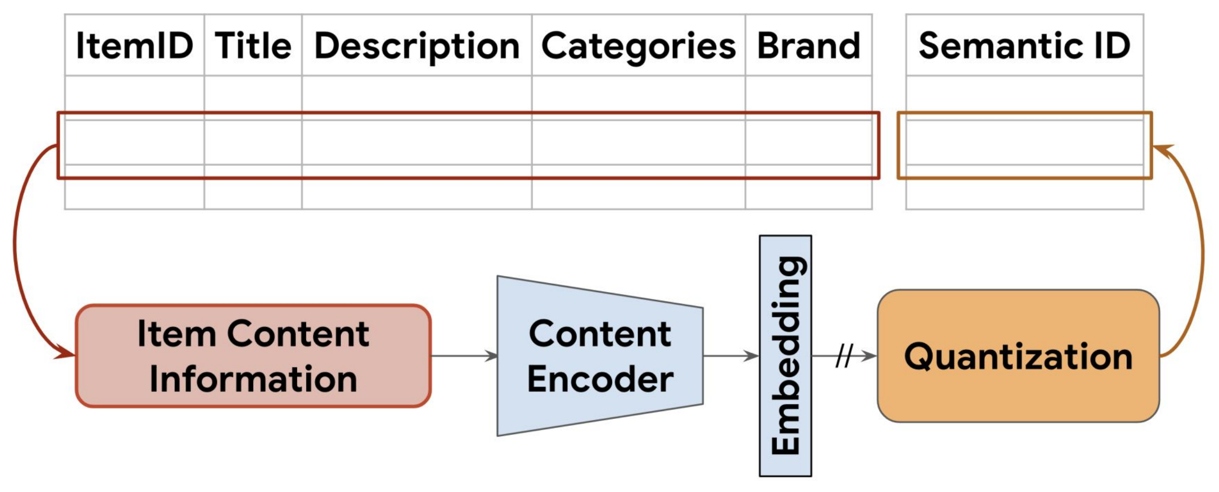
B097B2YWFX ⇔ {t3, t321, t643, t1011}

Part 1: Semantic ID Construction

(i) **First example**: TIGER and RQ-VAE-based SemIDs;

SemID Construction – First Example: TIGER

Input: concatenated text features

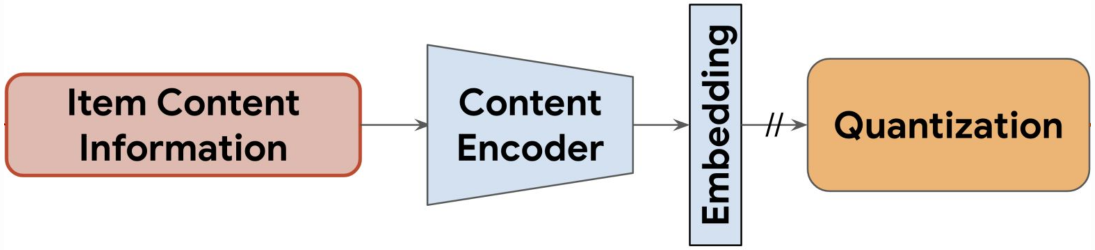


Output: mapping between items ⇔ Semantic IDs

Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

SemID Construction – First Example: TIGER

Text ➤ Vector ➤ IDs



SemID Construction – First Example: TIGER

1. Item Content Information (Text)

ItemID Title Description

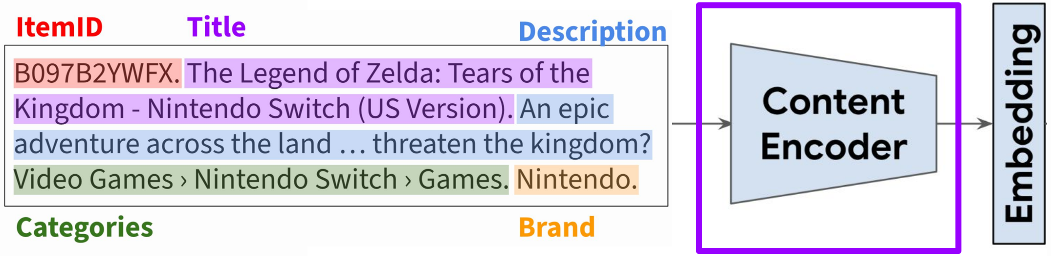
B097B2YWFX. The Legend of Zelda: Tears of the Kingdom - Nintendo Switch (US Version). An epic adventure across the land … threaten the kingdom? Video Games › Nintendo Switch › Games. Nintendo.

Categories Brand

SemID Construction – First Example: TIGER

2. Content Encoder + Embedding (Text ➤ Vector)

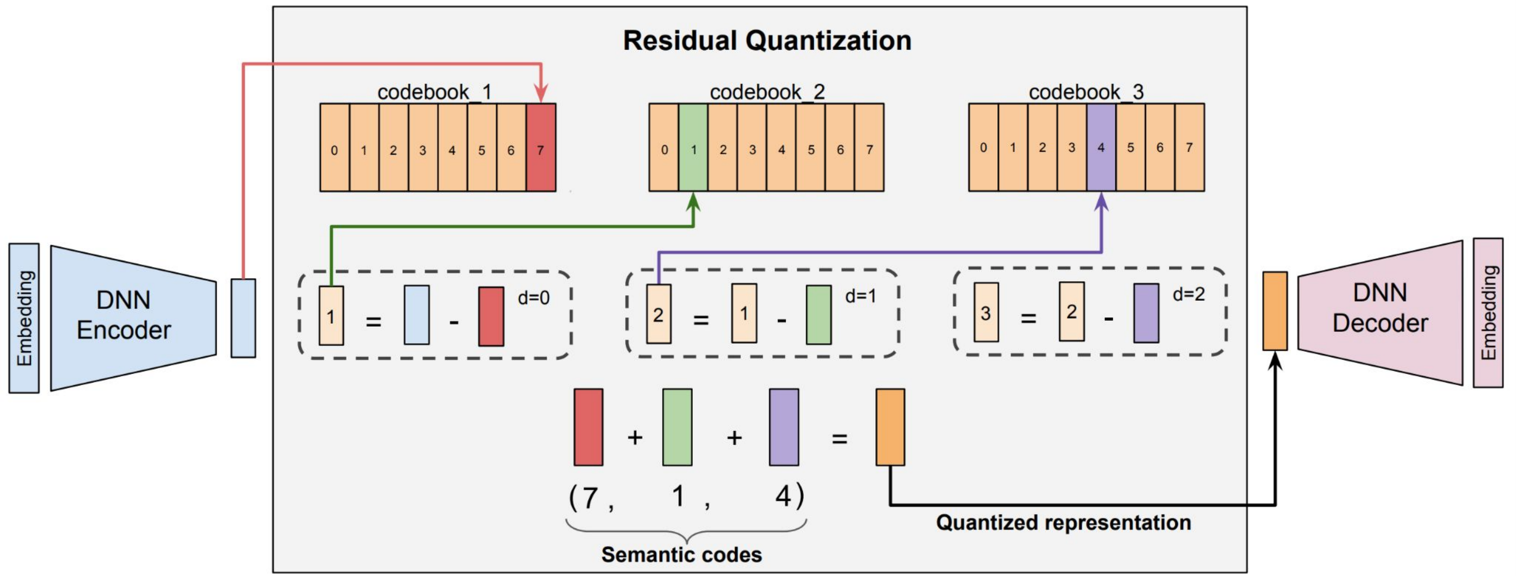
Pre-trained (fixed) sentence embedding model
(**SentenceT5**)



Ni et al. Sentence-T5: Scalable Sentence Encoders from Pre-trained Text-to-Text Models. Findings of ACL 2022.
Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

SemID Construction – First Example: TIGER

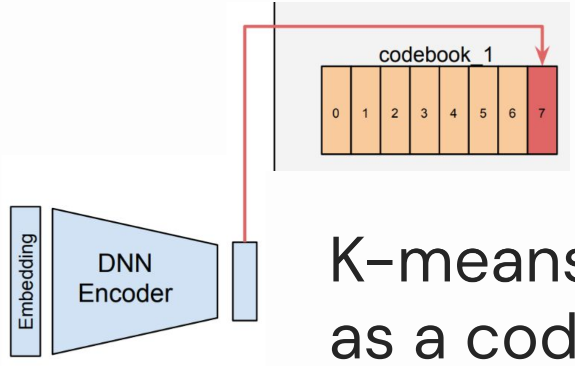
3. RQ-VAE Quantization (Vector ➤ IDs)



Zeghidour et al. SoundStream: An End-to-End Neural Audio Codec. TASLP 2022.
Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

SemID Construction – First Example: TIGER

3. RQ-VAE Quantization (Vector ➤ IDs)

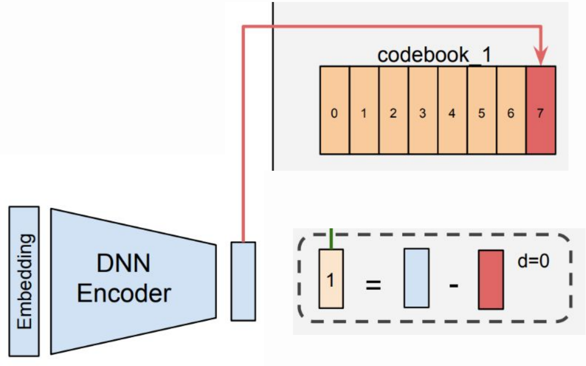


K-means: cluster center ID as a code in the codebook

Zeghidour et al. SoundStream: An End-to-End Neural Audio Codec. TASLP 2022.
Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

SemID Construction – First Example: TIGER

3. RQ-VAE Quantization (Vector ➤ IDs)

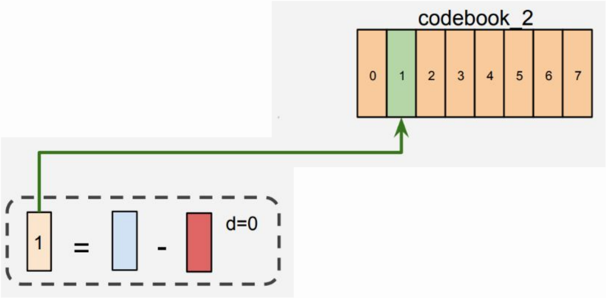


Residual of “input vector” and “clustering center vector”

Zeghidour et al. SoundStream: An End-to-End Neural Audio Codec. TASLP 2022.
Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

SemID Construction – First Example: TIGER

3. RQ-VAE Quantization (Vector ➤ IDs)

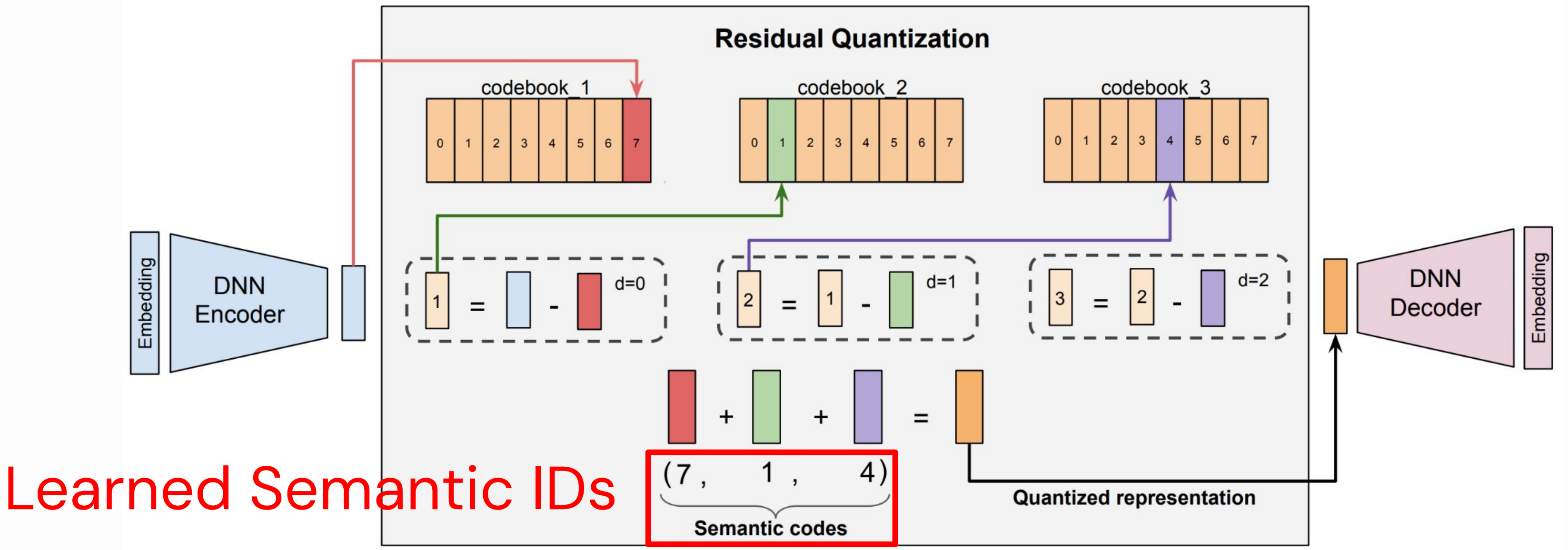


Residual as next level's input

Zeghidour et al. SoundStream: An End-to-End Neural Audio Codec. TASLP 2022.
Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

SemID Construction – First Example: TIGER

3. RQ-VAE Quantization (Vector ➤ IDs)

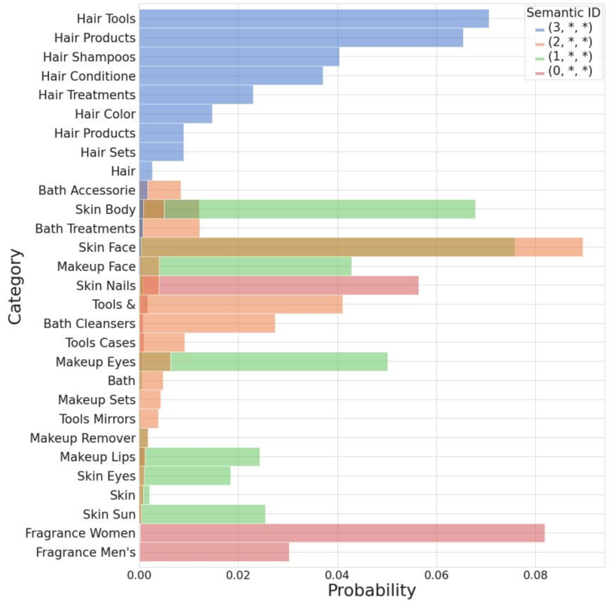


Zeghidour et al. SoundStream: An End-to-End Neural Audio Codec. TASLP 2022.
Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

SemID Construction – First Example: TIGER

Properties of **RQ-VAE**-based SemIDs

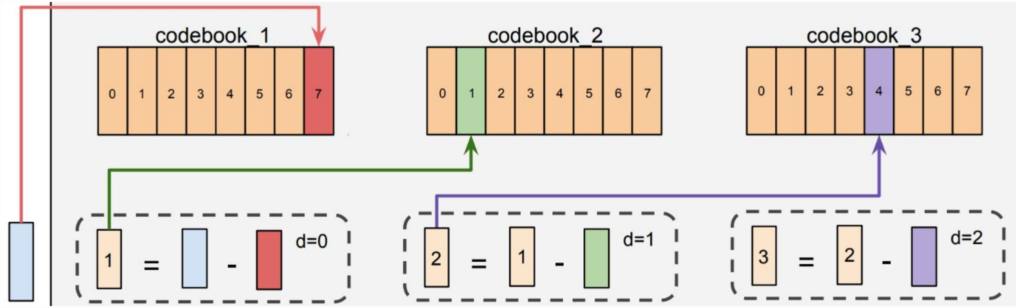
1. Semantic;



SemID Construction – First Example: TIGER

Properties of **RQ-VAE**-based SemIDs

1. Semantic;
2. Ordered / sequential dependent;



Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

SemID Construction – First Example: TIGER

Collisions

(12, 24, 52)

(12, 24, 52)

SemID Construction – First Example: TIGER

Collisions

(12, 24, 52, 0)

(12, 24, 52, 1)

One extra token
to avoid conflicts

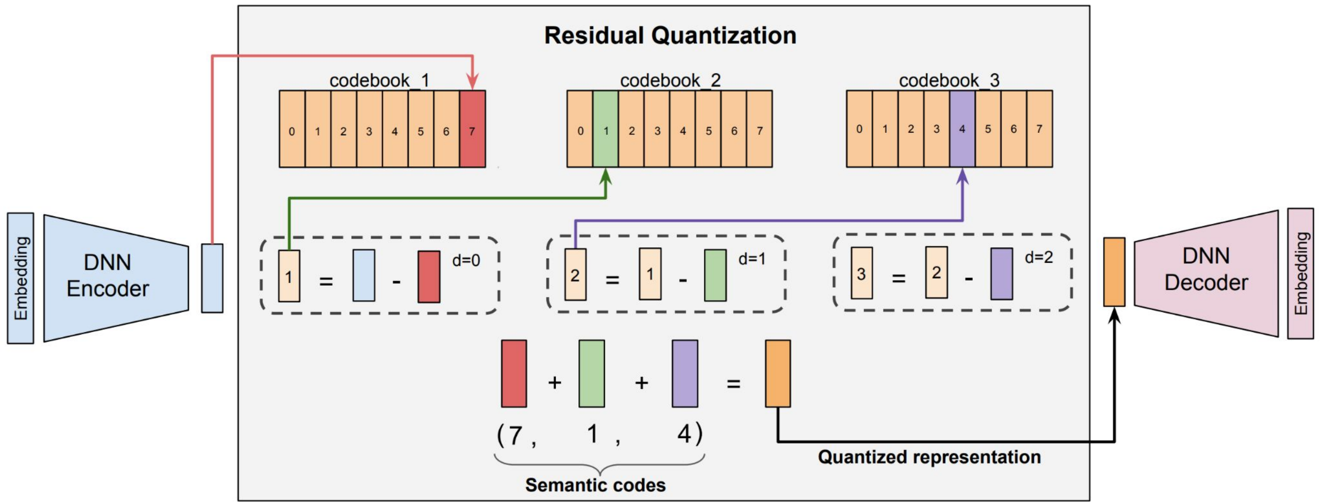
Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

Part 1: Semantic ID Construction

(i) First example: TIGER and RQ-VAE-based SemIDs;
(ii) **Techniques** to construct SemIDs;

Techniques to Construct SemIDs

Residual Quantization: RQ-VAE



Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

Techniques to Construct SemIDs

Residual Quantization: RQ-VAE

Issues:

- Enc-Dec Training Unstable
- Unbalanced IDs

Techniques to Construct SemIDs

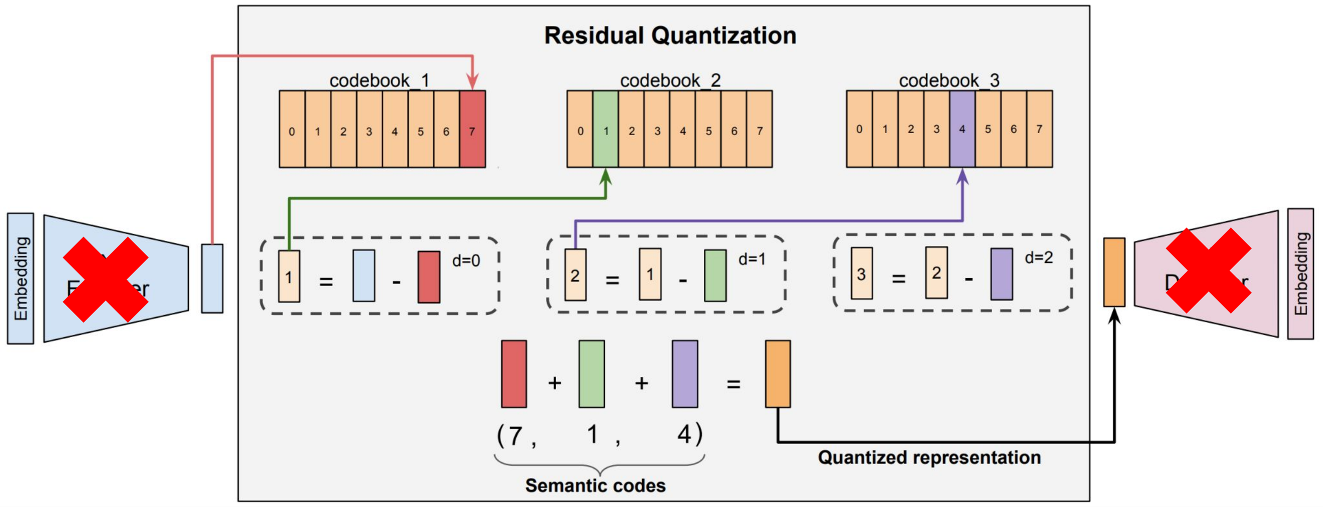
Residual Quantization: RQ-VAE

Issues:

- Enc-Dec Training Unstable
- Unbalanced IDs

Techniques to Construct SemIDs

Variants of Residual Quantization: RQ



Techniques to Construct SemIDs

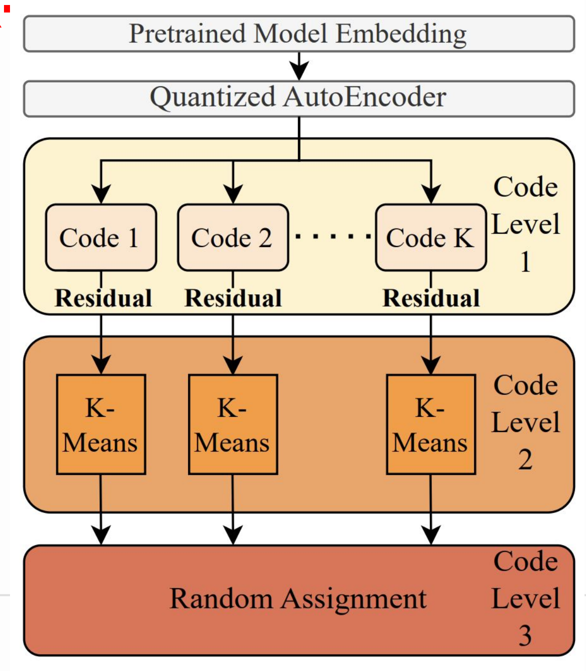
Residual Quantization: RQ-VAE

Issues:

- Enc-Dec Training Unstable
- Unbalanced IDs

Techniques to Construct SemIDs

Variants of Residual Quantization: VQ-VAE + K-Means



MBGen [CIKM'24]:

- Layer 1: VQ-VAE
- Layer 2: K-Means on residuals
- Layer 3: Random hash

Liu et al. Multi-Behavior Generative Recommendation. CIKM 2024.

Techniques to Construct SemIDs

Variants of Residual Quantization: RQ + K-Means

Algorithm 1: Balanced K-means Clustering

Input: Item set $\mathcal{V}$, number of clusters K

Compute $w \leftarrow |\mathcal{V}|/K$

Initialize centroids $C_l = \{\mathbf{c}_1^l, \ldots, \mathbf{c}_K^l\}$ with random selection;

repeat

  Initialize unassigned set $\mathcal{U} \leftarrow \mathcal{V}$

  for *each cluster* $k \in \{1, \ldots, K\}$ **do**

    Sort $\mathcal{U}$ by ascending distance from centroid $\mathbf{c}_k^l$;

    $\boxed{\text{Assign } \mathcal{V}_k \leftarrow \mathcal{U}[0 : w-1];}$

    Update centroid $\mathbf{c}_k^l \leftarrow \frac{1}{w} \sum_{\mathbf{r}^l \in \mathcal{V}_k} \mathbf{r}^l$;

    Remove assigned items $\mathcal{U} \leftarrow \mathcal{U} \setminus \mathcal{V}_k$;

  end

until *Assignment convergence*;

Output: Optimized codebook $C_l = \{\mathbf{c}_1^l, \ldots, \mathbf{c}_K^l\}$

OneRec:

Limit the max #items per cluster

Deng et al. OneRec: Unifying Retrieve and Rank with Generative Recommender and Iterative Preference Alignment. arXiv:2502.18965.

Techniques to Construct SemIDs

Variants of Residual Quantization

Table 1: Performance of GR models with by SIDs generated by different tokenization algorithms.

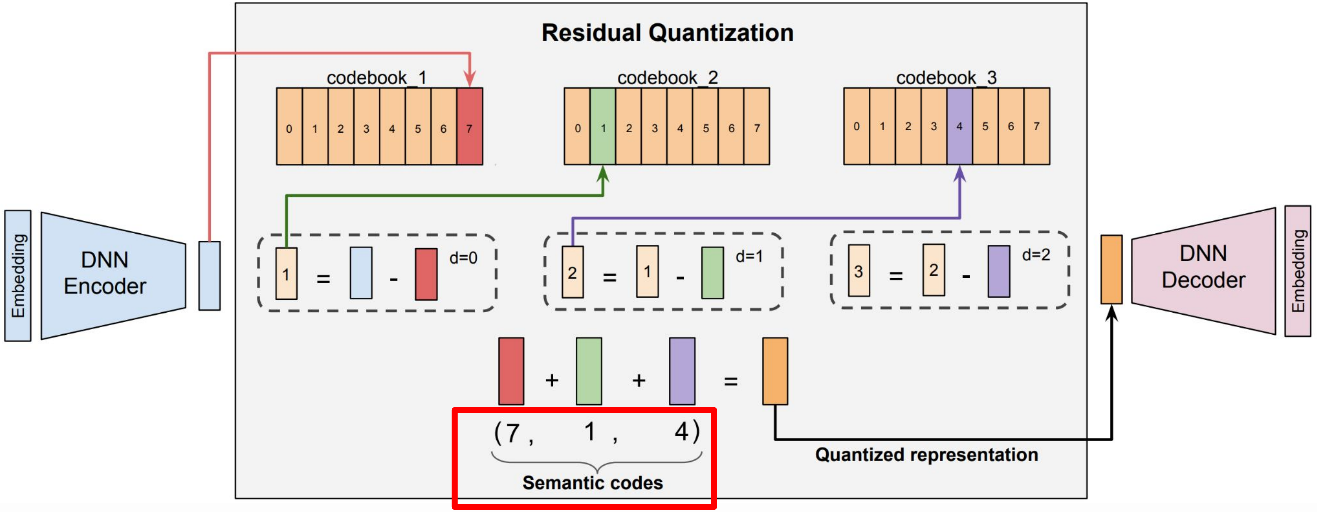
Methods	Recommendation Performance (Recall@5/Recall@10/NDCG@5/NDCG@10)											
	Beauty				Toys				Sports			
RK-Means	**0.0422**	**0.0639**	0.0277	0.0347	**0.0376**	**0.0577**	**0.0243**	**0.0308**	**0.0236**	**0.0353**	**0.0153**	**0.0191**
R-VQ	**0.0422**	0.0638	**0.0282**	**0.0351**	0.0327	0.0493	0.0209	0.0262	0.0234	0.0352	0.0151	0.0189
RQ-VAE	0.0404	0.0593	0.0268	0.0329	0.0342	0.0514	0.0224	0.0280	0.0205	0.0312	0.0132	0.0166

https://github.com/snap-research/GRID

Ju et al. Generative Recommendation with Semantic IDs: A Practitioner's Handbook. CIKM 2025.

Techniques to Construct SemIDs

Are the orders among tokens necessary?



Techniques to Construct SemIDs

Are the orders among tokens necessary?

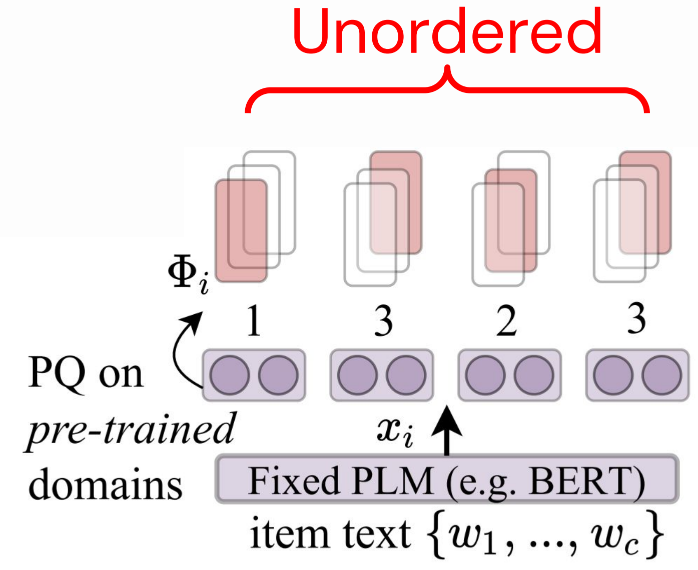
Orders

<-> autoregressive generation

<-> SemIDs have to be short

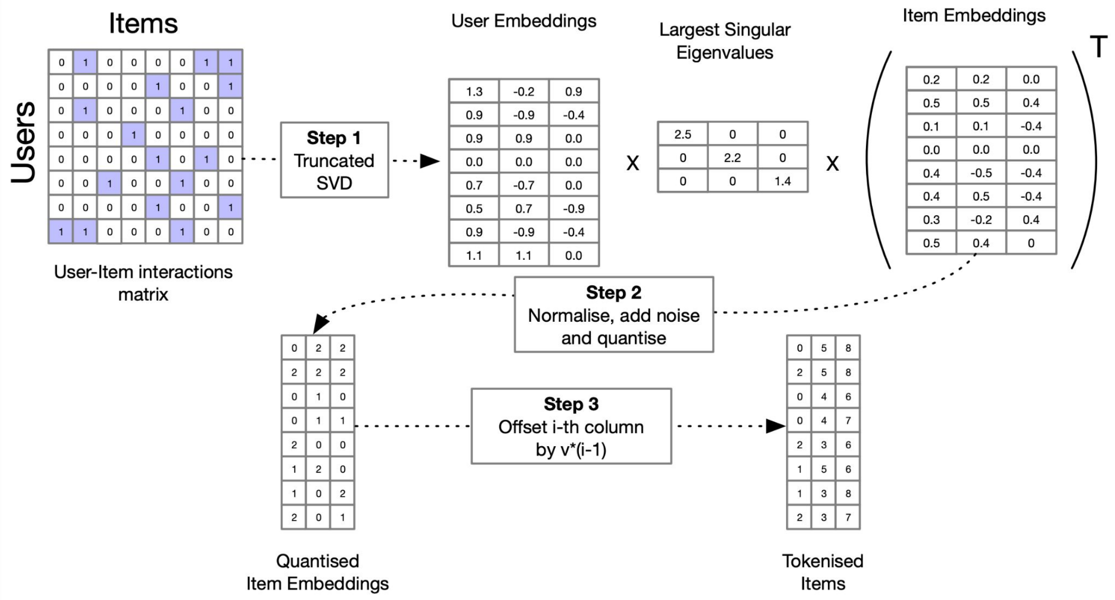
Techniques to Construct SemIDs

Product Quantization



Techniques to Construct SemIDs

Product Quantization



Early exploration on PQ-based SIDs, but performance is not ideal

Autoregressive generation needs orders

Petrov et al. Generative Sequential Recommendation with GPTRec. Gen-IR@SIGIR 2023.

Techniques to Construct SemIDs

Product Quantization: RPG [KDD'25]

Long semantic IDs + parallel generation

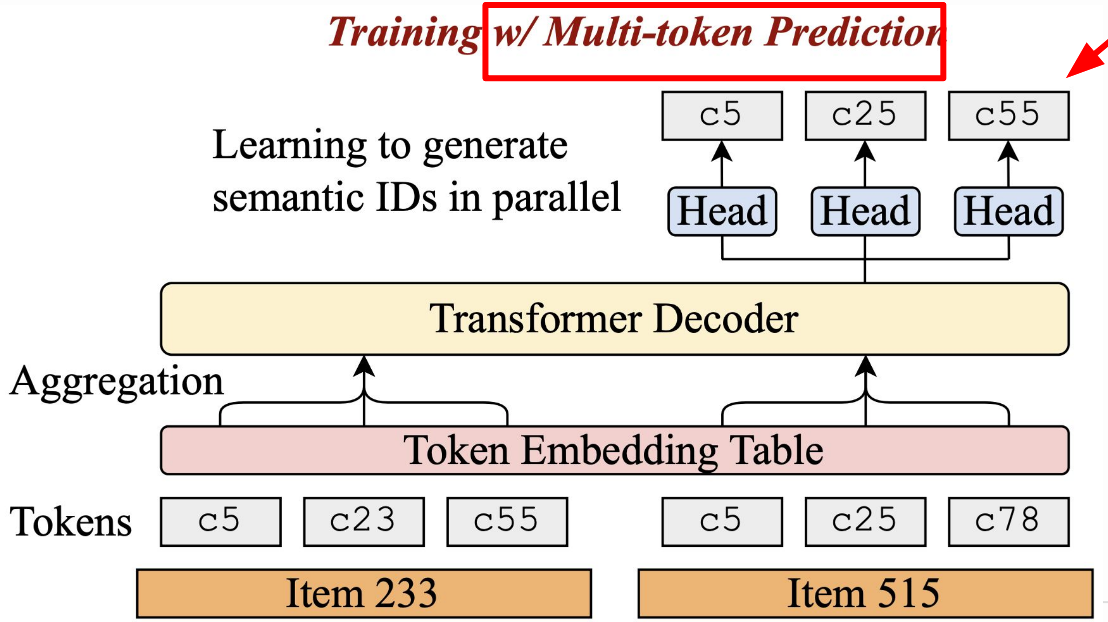
Original (#digit=4): t34, t392, t600, t891

RPG (#digit=64): t34, t392, t600, t891, t1221, t1495, t1556, t1924, t2296, t2511, t2715, t3060, t3255, t3439, t3655, t3984, t4167, t4400, t4736, t4939, t5278, t5426, t5669, t6057, t6385, t6451, t6837, t7134, t7329, t7528, t7924, t8162, t8325, t8479, t8711, t9007, t9420, t9472, t9980, t10154, t10364, t10662, t10784, t11105, t11377, t11642, t11848, t12261, t12334, t12585, t12963, t13306, t13367, t13722, t13973, t14143, t14506, t14696, t14995, t15331, t15406, t15813, t16034, t16251

Hou et al. Generating Long Semantic IDs in Parallel for Recommendation. KDD 2025.

Techniques to Construct SemIDs

Product Quantization: RPG [KDD'25]



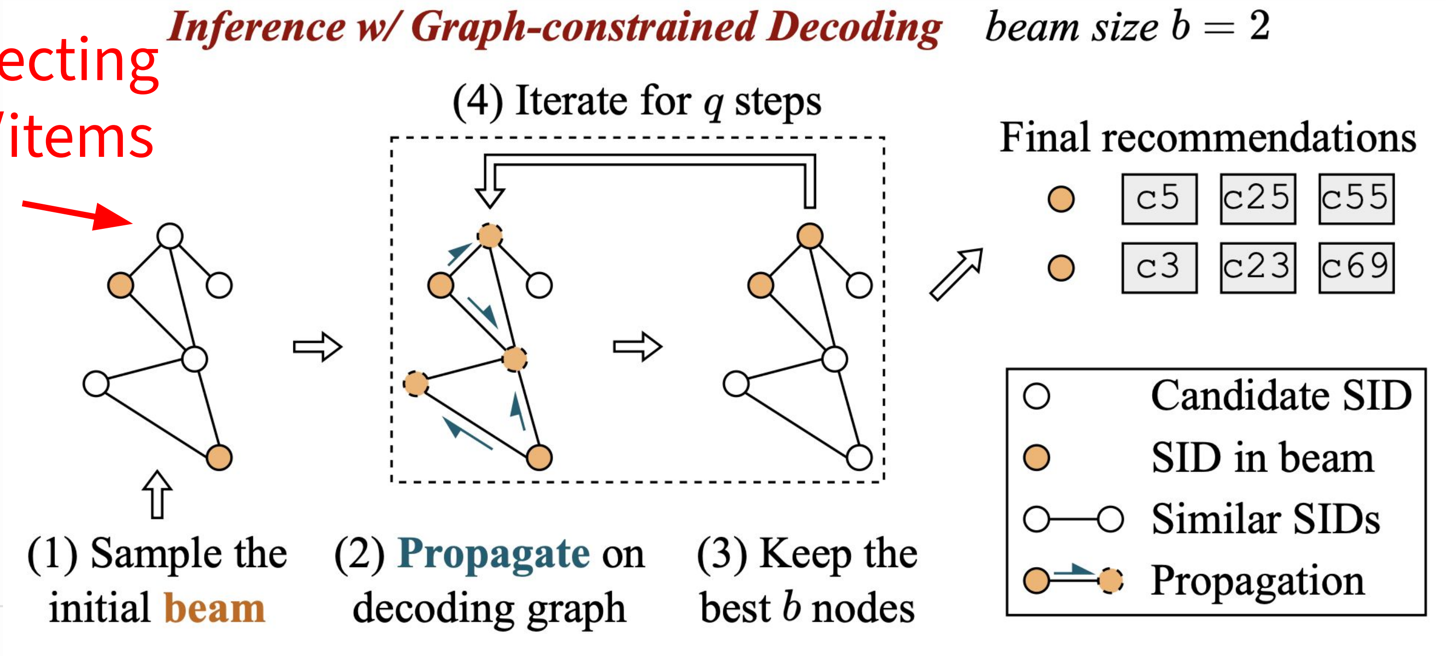
Using *OPQ* to produce semIDs w/o sequential dependencies

Hou et al. Generating Long Semantic IDs in Parallel for Recommendation. KDD 2025.

Techniques to Construct SemIDs

Product Quantization: RPG [KDD'25]

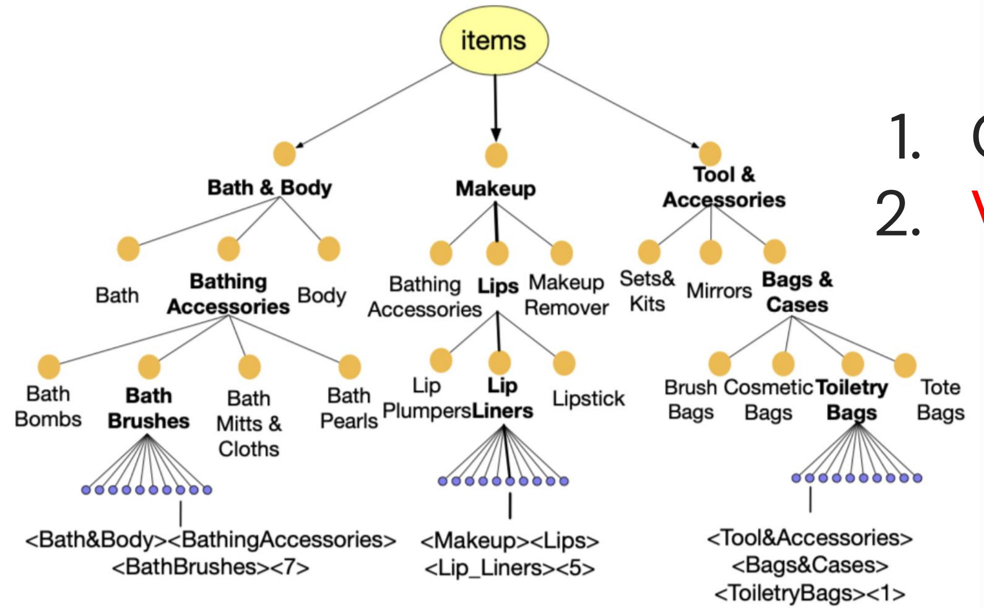
Graph: connecting similar SIDs/items (offline)



Hou et al. Generating Long Semantic IDs in Parallel for Recommendation. KDD 2025.

Techniques to Construct SemIDs

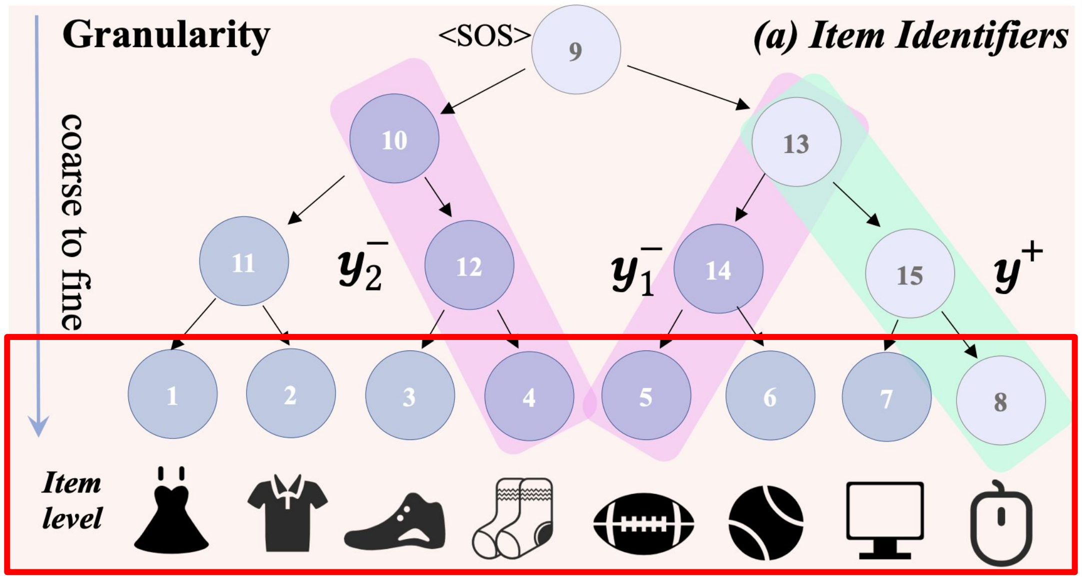
Hierarchical Clustering (Heuristics-based)



1. Ordered;
2. Variable-length SemIDs;

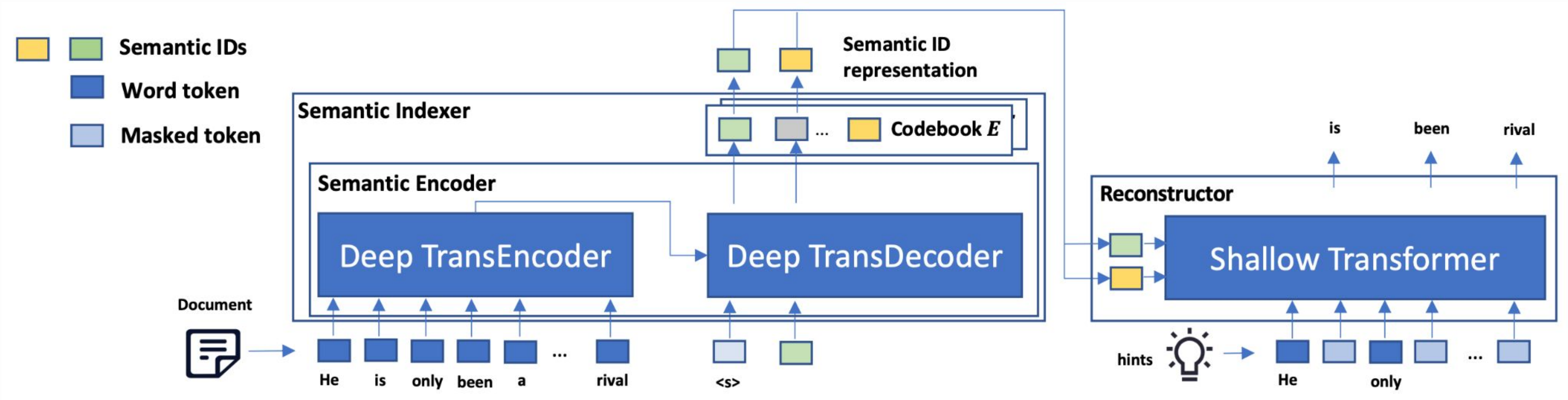
Techniques to Construct SemIDs

Hierarchical Clustering (Latent-based)



Techniques to Construct SemIDs

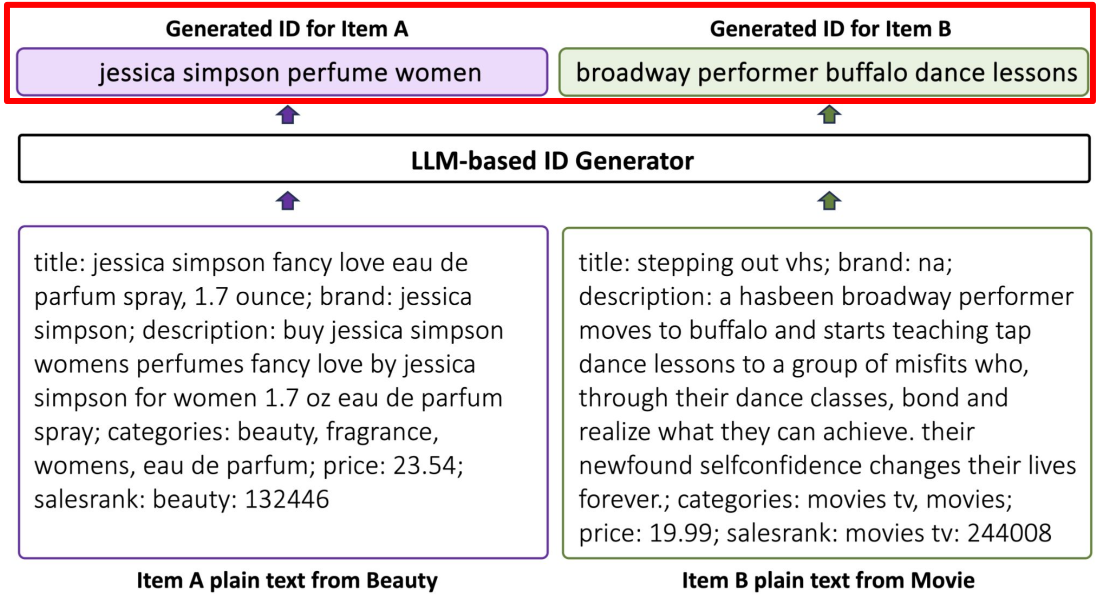
Language Model-based ID Generator



Natural language as inputs;
SemIDs as outputs

Techniques to Construct SemIDs

Language Model-based ID Generator



Words as SemIDs (like tagging)

Tan et al. IDGenRec: LLM-RecSys Alignment with Textual ID Learning. SIGIR 2024.

Summary of Techniques to Construct SemIDs

- Residual Quantization (ordered)
- Product Quantization (unordered)
- Hierarchical Clustering
- LM-based ID Generator

Part 1: Semantic ID Construction

(i) First example: TIGER and RQ-VAE-based SemIDs;
(ii) Techniques to construct SemIDs;
(iii) **Inputs** for SemID construction;

Inputs for SemID Construction

Input: all data associated with the item

Inputs for SemID Construction

Input: all data associated with the item

What exactly does “all data” mean? 🤷

Inputs for SemID Construction

Text or Multimodal Features

Text/Visual/Acoustic ➤ **Vector** ➤ **IDs**

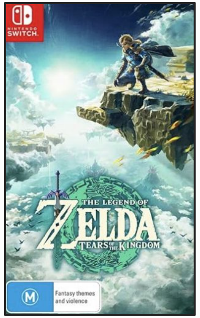
Pretrained Encoder Quantization

ItemID **Title** **Description**

B097B2YWFX. The Legend of Zelda: Tears of the Kingdom - Nintendo Switch (US Version). An epic adventure across the land … threaten the kingdom? Video Games › Nintendo Switch › Games. Nintendo.

Categories **Brand**

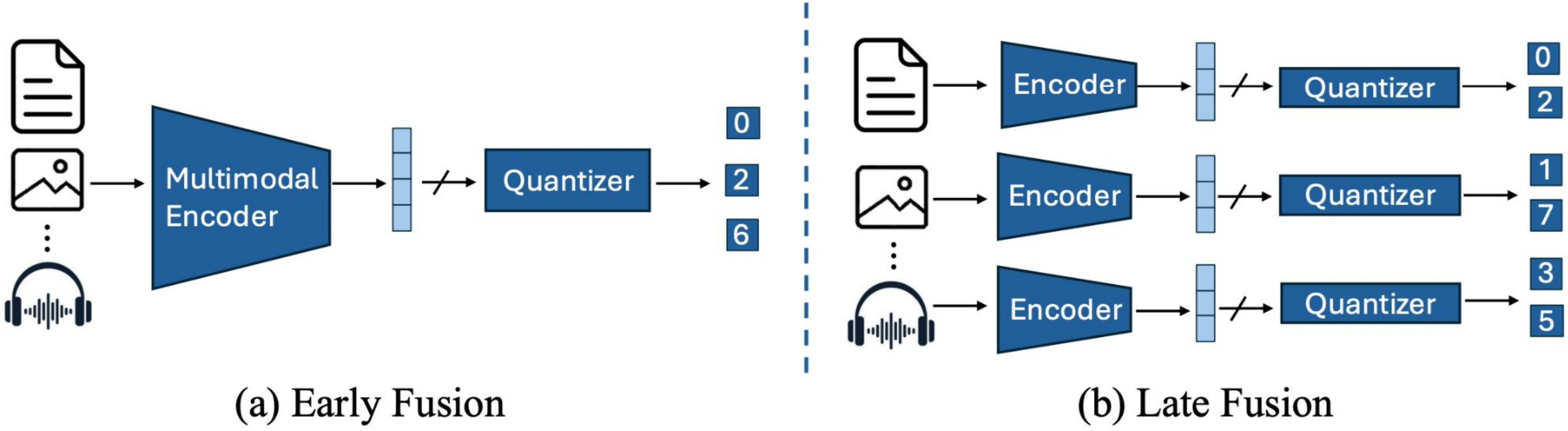
Text



Multimodal

Inputs for SemID Construction

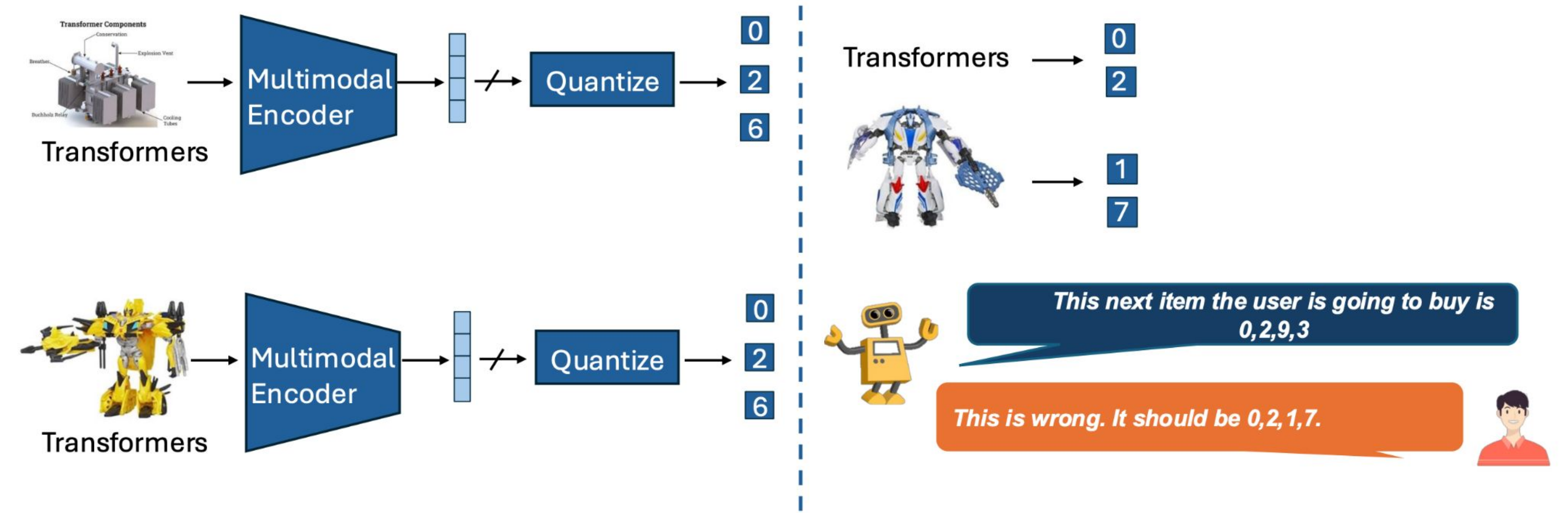
Text or Multimodal Features



Zhu et al. Beyond Unimodal Boundaries: Generative Recommendation with Multimodal Semantics. arXiv:2503.23333.

Inputs for SemID Construction

Text or Multimodal Features



(a) Early Fusion: Modality Sensitivity (b) Late Fusion: Modality Correspondence

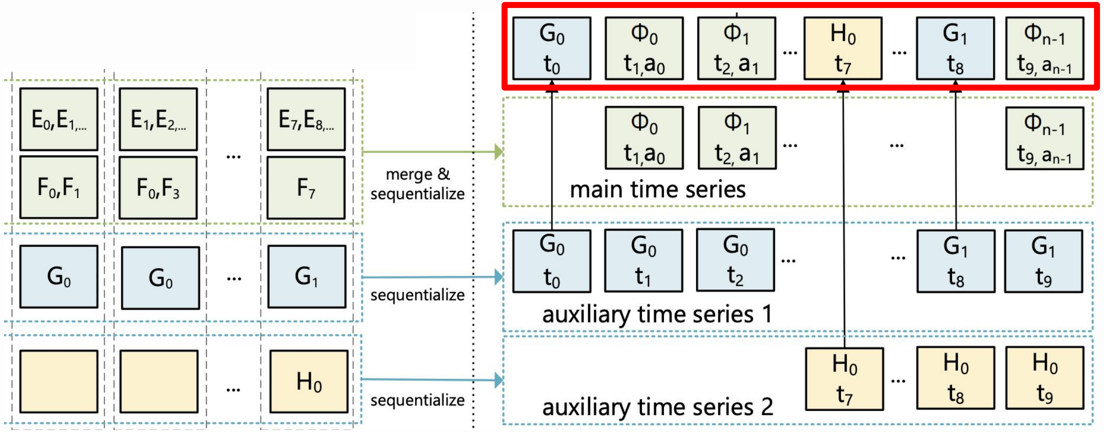
Zhu et al. Beyond Unimodal Boundaries: Generative Recommendation with Multimodal Semantics. arXiv:2503.23333.

Inputs for SemID Construction

Categorical Features

Categorical Features ➤ IDs

Merge & Sequentialize



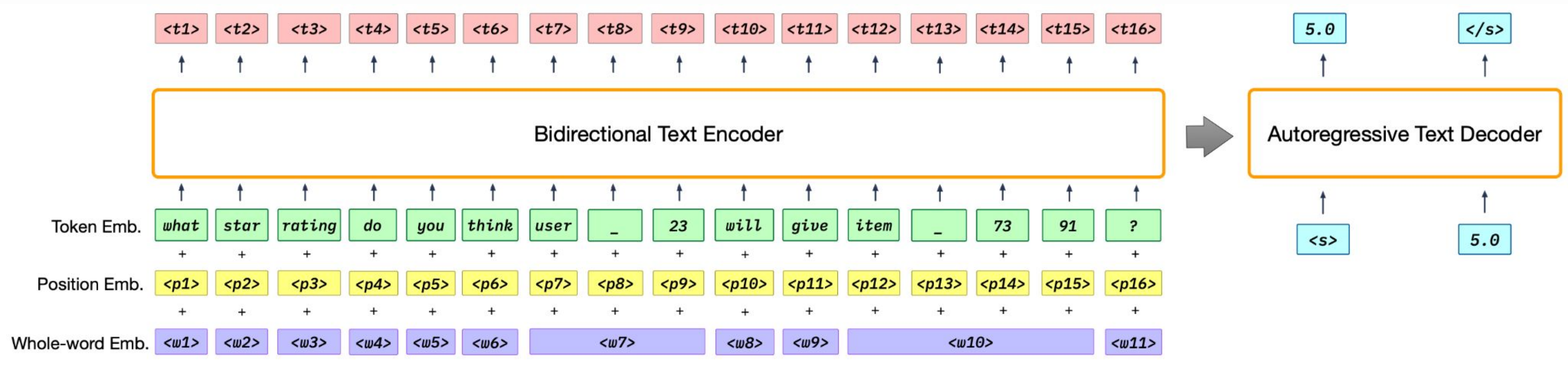
Zhai et al. Actions Speak Louder than Words: Trillion-Parameter Sequential Transducers for Generative Recommendations. ICML 2024.

Inputs for SemID Construction

No Features

Item ID ➤ IDs

Text Tokenizer



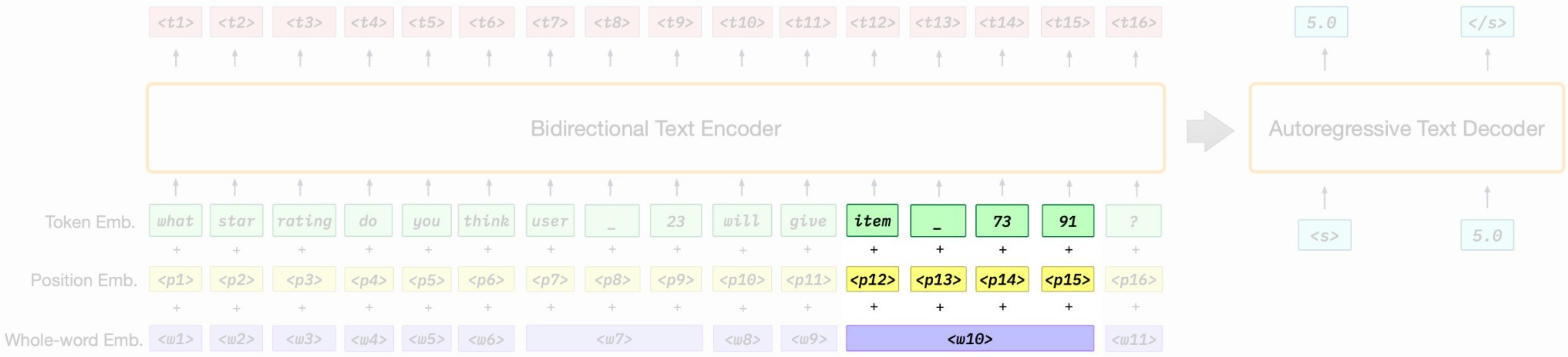
Geng et al. Recommendation as Language Processing (RLP): A Unified Pretrain, Personalized Prompt & Predict Paradigm (P5). RecSys 2022.

Inputs for SemID Construction

No Features

Item ID ➤ IDs

Text Tokenizer

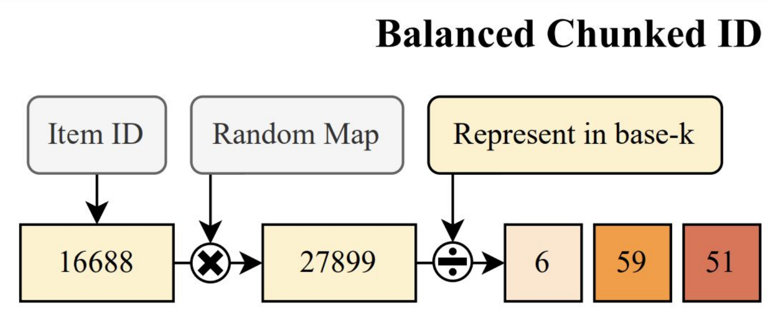


Geng et al. Recommendation as Language Processing (RLP): A Unified Pretrain, Personalized Prompt & Predict Paradigm (P5). RecSys 2022.

Inputs for SemID Construction

No Features

Random IDs



Inputs for SemID Construction

Input: all data associated with the item

(1) Item Metadata

Text / Multimodal / Categorical / No Features

Inputs for SemID Construction

Input: all data associated with the item

(1) Item Metadata

Text / Multimodal / Categorical / No Features

(2) Item Metadata + Behaviors

Inputs for SemID Construction

Input: all data associated with the item

(1) Item Metadata

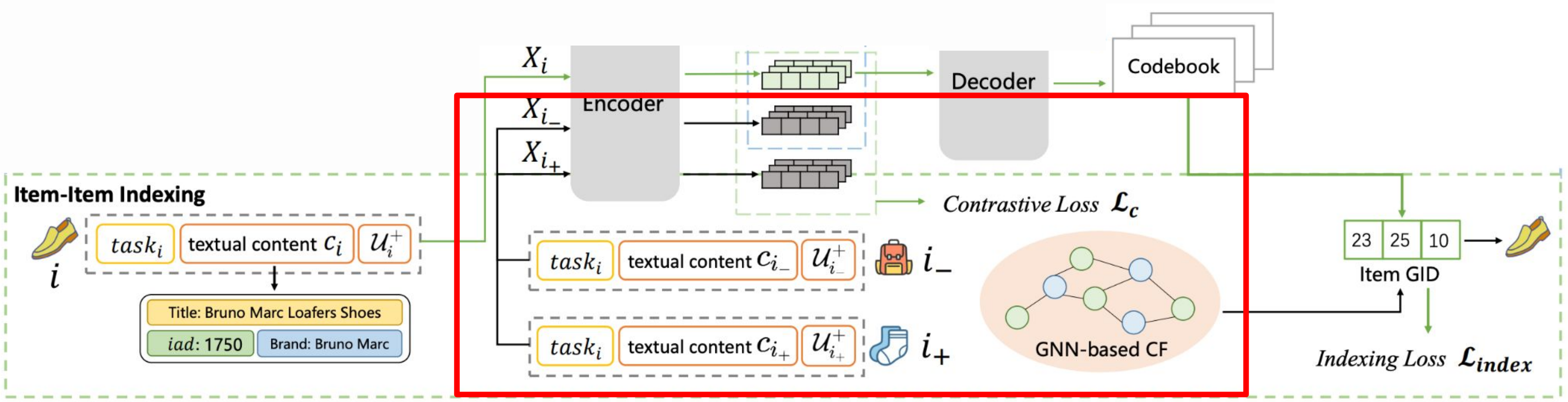
Text / Multimodal / Categorical / No Features

(2) Item Metadata + Behaviors

But how?

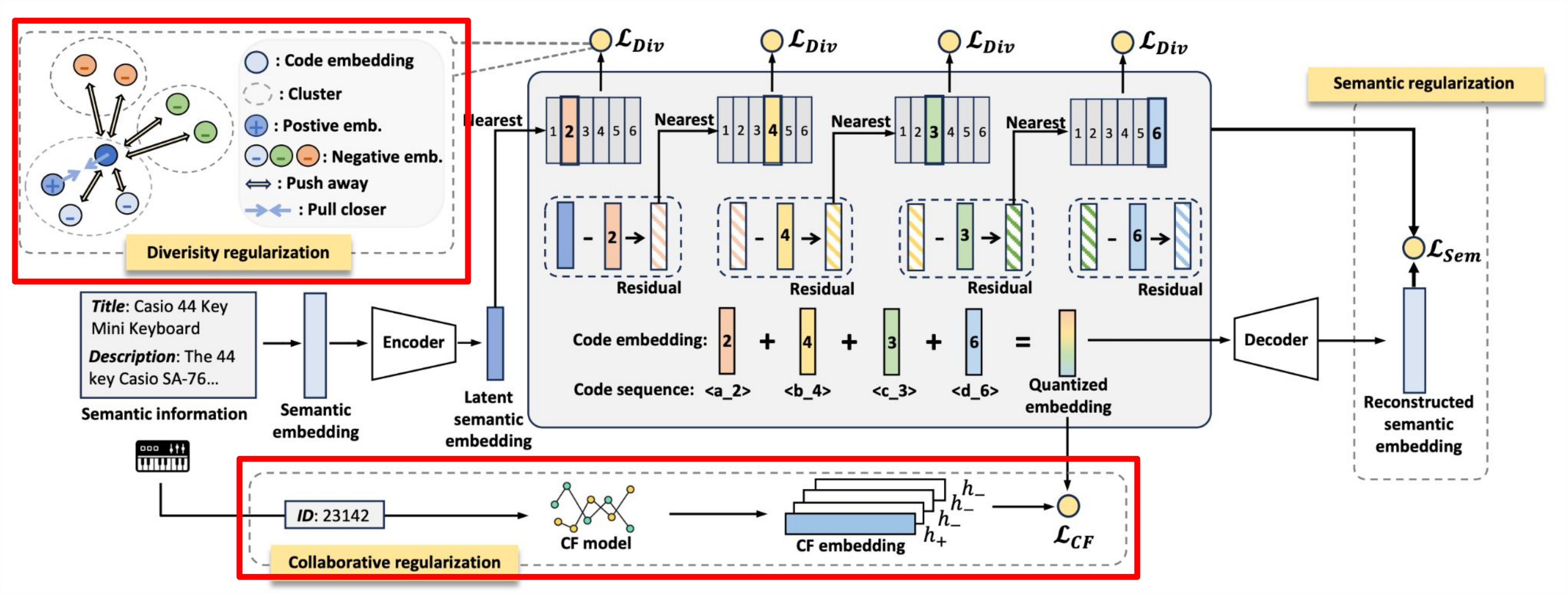
Inputs for SemID Construction

Residual Quantization + Item-level Regularization



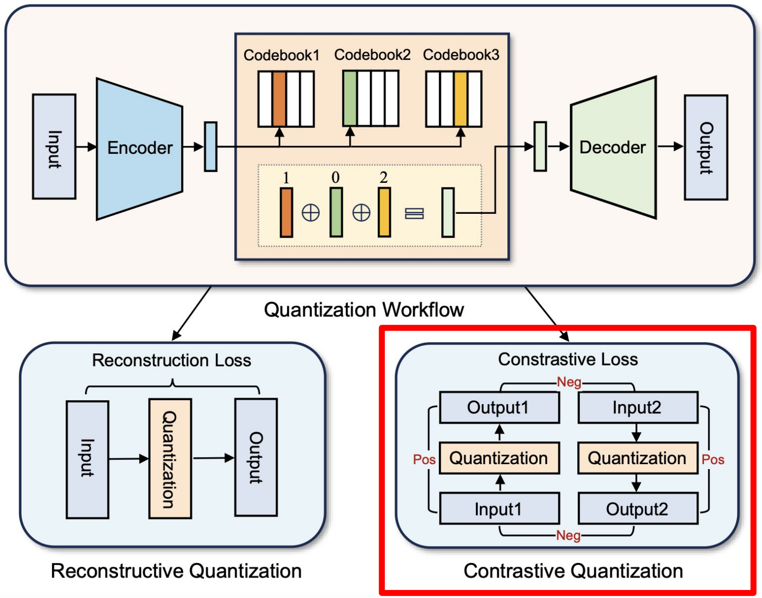
Inputs for SemID Construction

Residual Quantization + Item-level Regularization



Inputs for SemID Construction

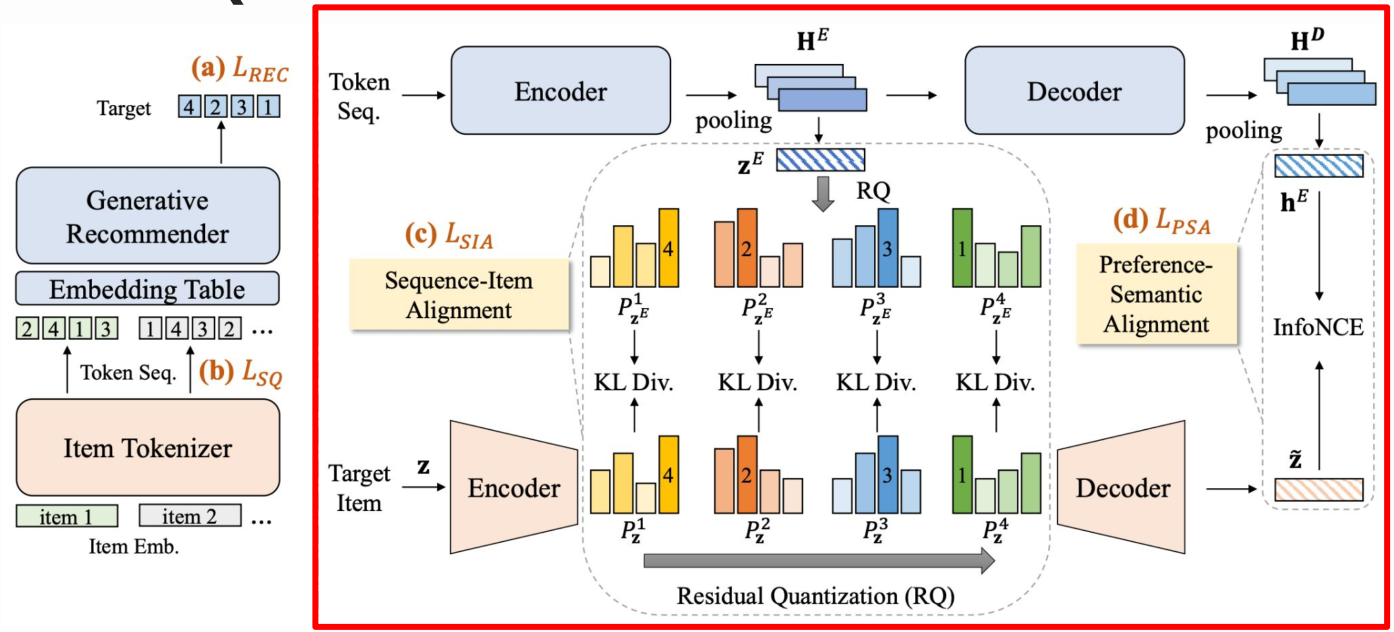
Residual Quantization + Item-level Regularization



Zhu et al. CoST: Contrastive Quantization based Semantic Tokenization for Generative Recommendation. RecSys 2024.

Inputs for SemID Construction

Residual Quantization + Recommendation Loss

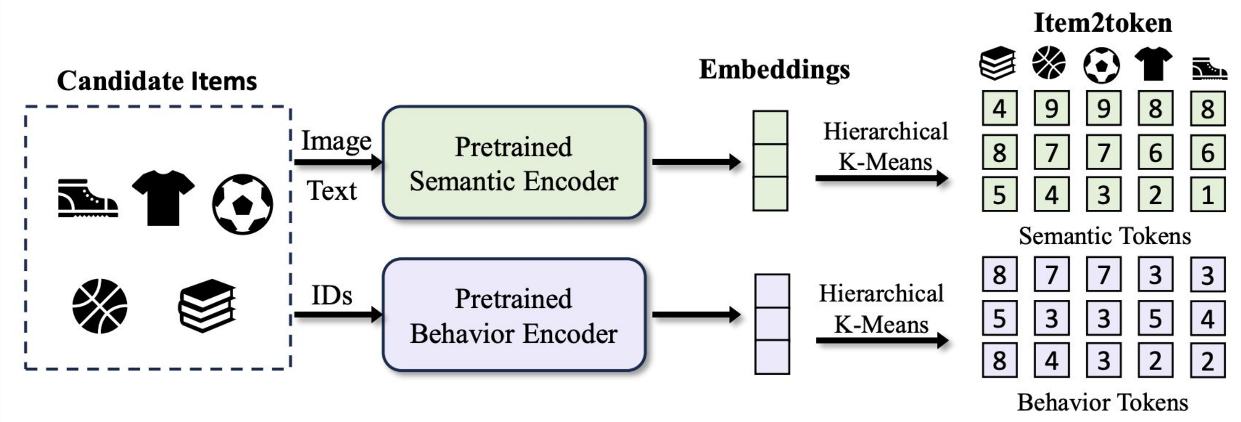


Liu et al. End-to-End Learnable Item Tokenization for Generative Recommendation. arXiv:2409.05546.

Inputs for SemID Construction

Item Metadata + Behaviors

Fused Semantic IDs

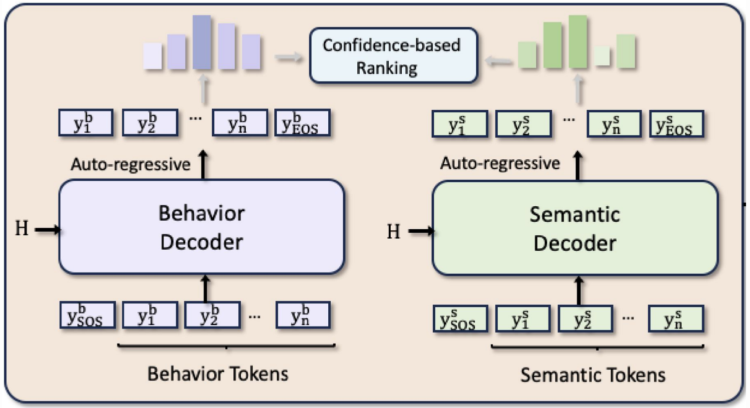


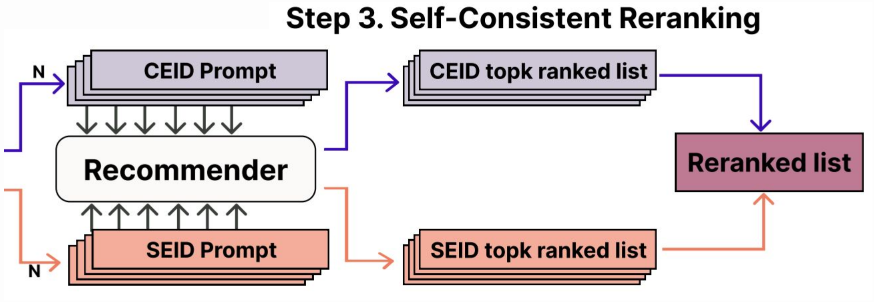
Wang et al. EAGER: Two-Stream Generative Recommender with Behavior-Semantic Collaboration. KDD 2024.

Inputs for SemID Construction

Item Metadata + Behaviors

Fused Semantic IDs + Two-stream Generation





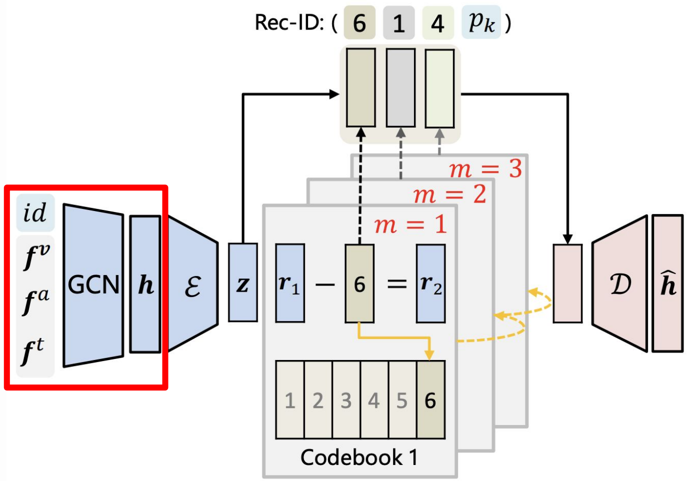
Wang et al. EAGER: Two-Stream Generative Recommender with Behavior-Semantic Collaboration. KDD 2024.
Kim et al. SC-Rec: Enhancing Generative Retrieval with Self-Consistent Reranking for Sequential Recommendation. arXiv:2408:08686.

Inputs for SemID Construction

Item Metadata + Behaviors

Fused Representations

User–Item Graph +
Semantic Features

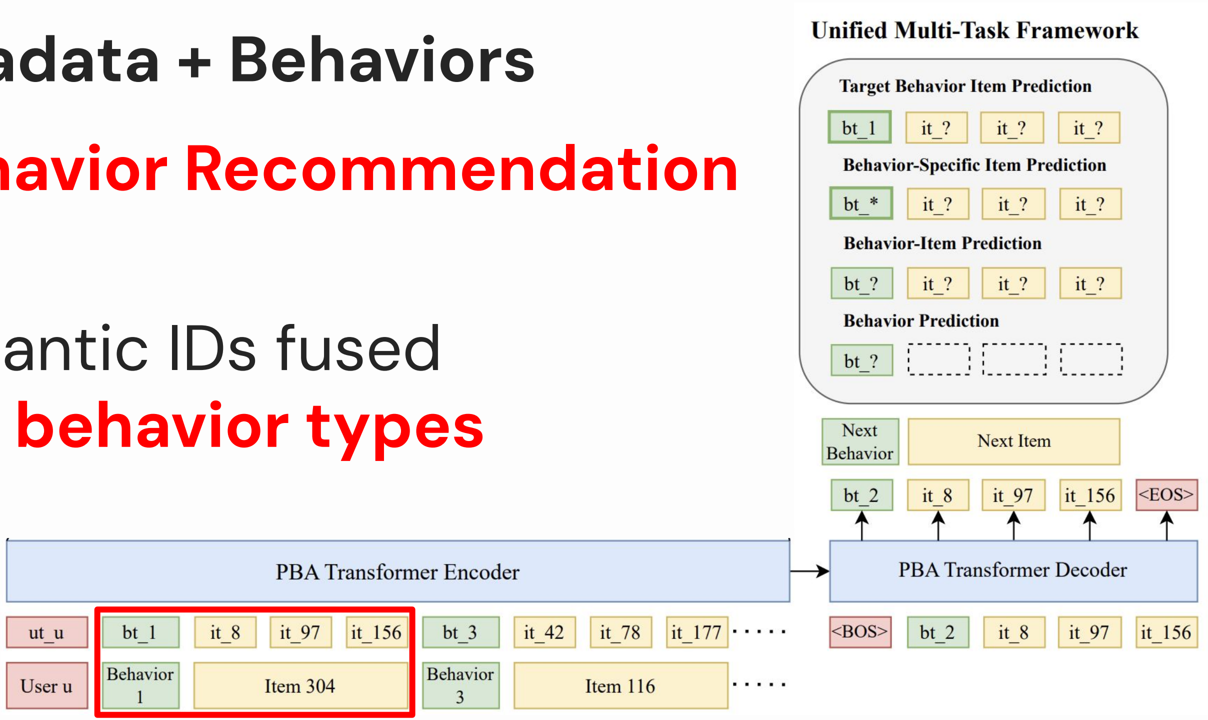


Inputs for SemID Construction

Item Metadata + Behaviors

Multi-Behavior Recommendation

Semantic IDs fused
with **behavior types**



Liu et al. Multi-Behavior Generative Recommendation. CIKM 2024.

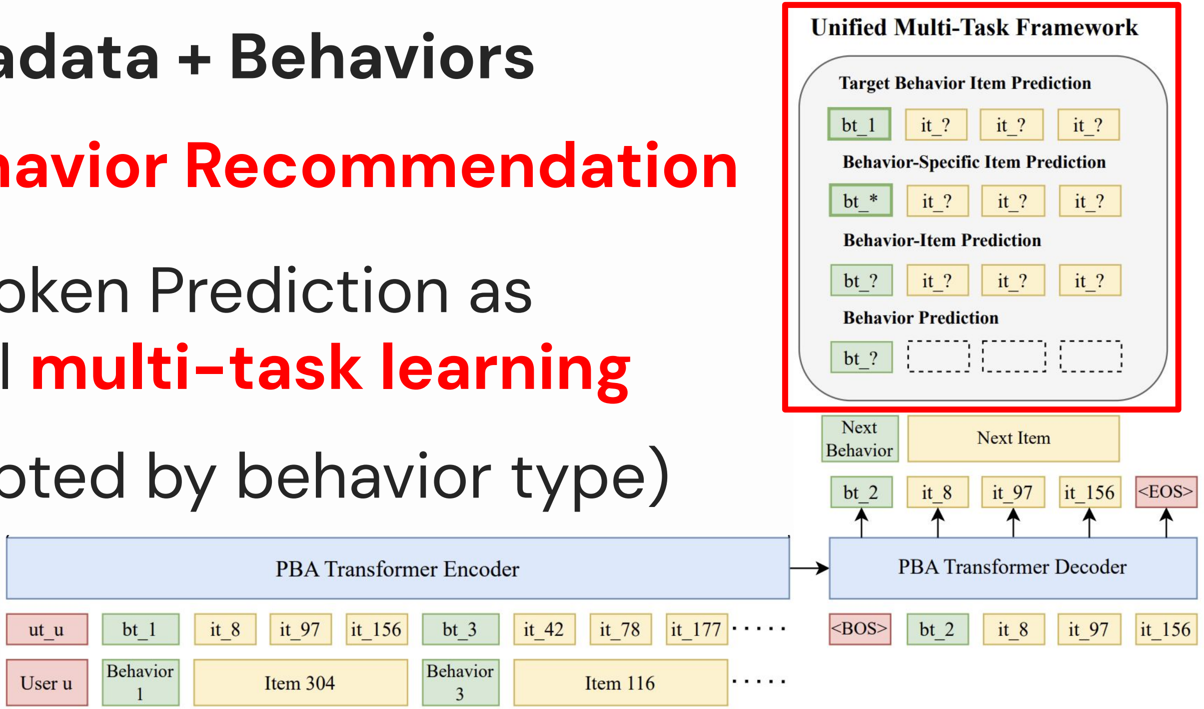
Inputs for SemID Construction

Item Metadata + Behaviors

Multi-Behavior Recommendation

Next Token Prediction as natural **multi-task learning**

(prompted by behavior type)



Liu et al. Multi-Behavior Generative Recommendation. CIKM 2024.

Inputs for SemID Construction

Context-independent

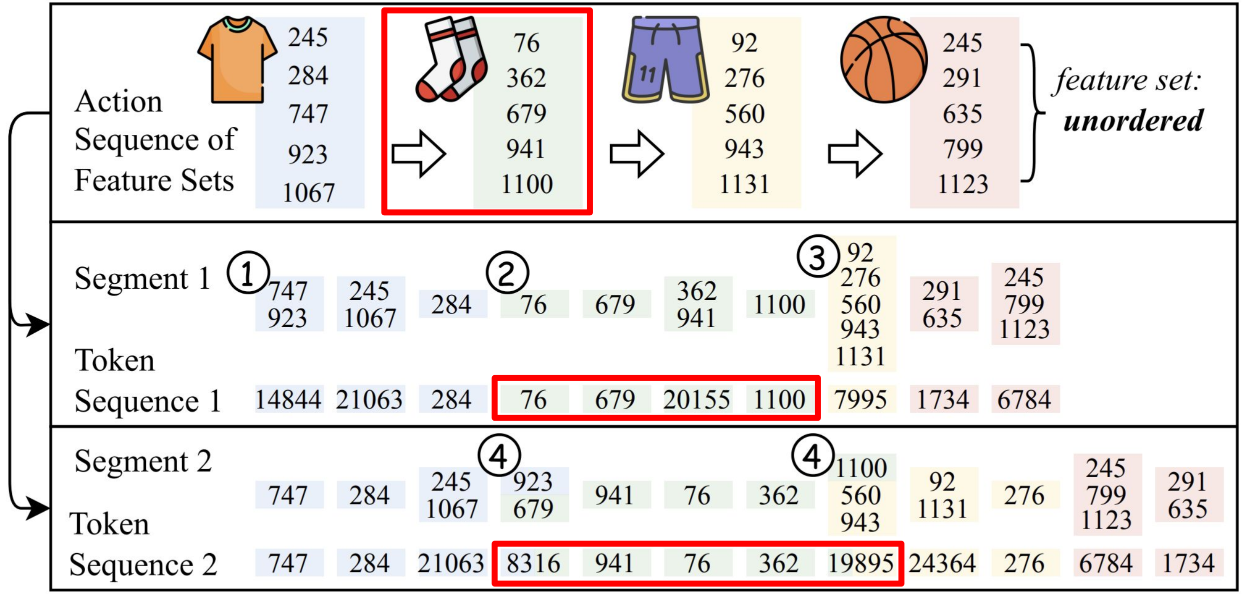
Action Tokenization	Example	Contextual	Unordered
Product Quantization	VQ-Rec (Hou et al., 2023)	✗	✓
Hierarchical Clustering	P5-CID (Hua et al., 2023)	✗	✗
Residual Quantization	TIGER (Rajput et al., 2023)	✗	✗
Text Tokenization	LMIndexer (Jin et al., 2024)	✗	✗
Raw Features	HSTU (Zhai et al., 2024)	✗	✗
SentencePiece	SPM-SID (Singh et al., 2024)	✗	✗

Same item ⇒ fixed semIDs in all sequences

Hou et al. ActionPiece: Contextually Tokenizing Action Sequences for Generative Recommendation. ICML 2025.

Inputs for SemID Construction

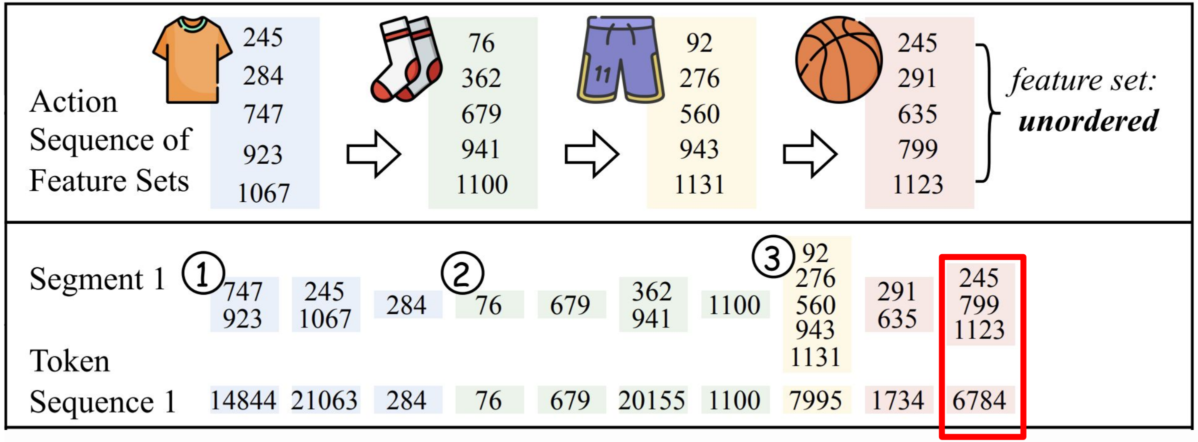
Context-independent ⇒ Context-aware



Same item ⇒ different semIDs based on context

Inputs for SemID Construction

Context-independent ⇒ Context-aware



Core Idea:

Merge frequently co-occurring features as new tokens

(**ActionPiece**: "WordPiece" tokenization for generative rec)

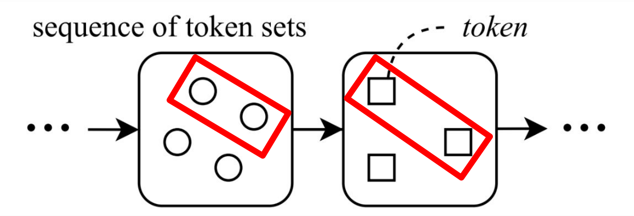
Inputs for SemID Construction

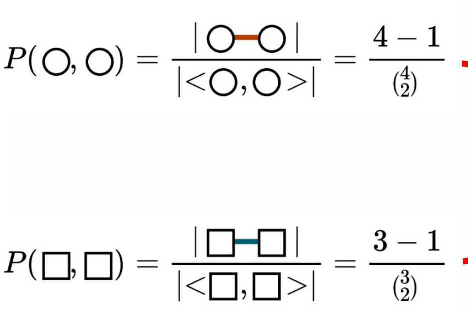
Context-independent ⇒ Context-aware

Algorithm 1 ActionPiece Vocabulary Construction

input Sequence corpus $\mathcal{S}'$, initial tokens $\mathcal{V}_0$, target size Q
output Merge rules $\mathcal{R}$, constructed vocabulary $\mathcal{V}$

Initialize vocabulary $\mathcal{V} \leftarrow \mathcal{V}_0$ # each initial token corresponds to one unique item feature
$\mathcal{R} \leftarrow \emptyset$
while $|\mathcal{V}| < Q$ **do**
  # ***Count:*** accumulate weighted token co-occurrences
  $\text{count}(\cdot,\cdot) \leftarrow \text{Count}(\mathcal{S}', \mathcal{V})$ # Algorithm 2
  # ***Update:*** Merge a frequent token pair into a new token
  Select $(c_u, c_v) \leftarrow \arg\max_{(c_i, c_j)} \text{count}(c_i, c_j)$
  $\mathcal{S}' \leftarrow \text{Update}(\mathcal{S}', \{(c_u, c_v) \to c_{\text{new}}\})$ # Algorithm 3
  $\mathcal{R} \leftarrow \mathcal{R} \cup \{(c_u, c_v) \to c_{\text{new}}\}$ # new merge rule
  $\mathcal{V} \leftarrow \mathcal{V} \cup \{c_{\text{new}}\}$ # add new token to the vocabulary
end while
return $\mathcal{R}, \mathcal{V}$





Features co-occurring within items

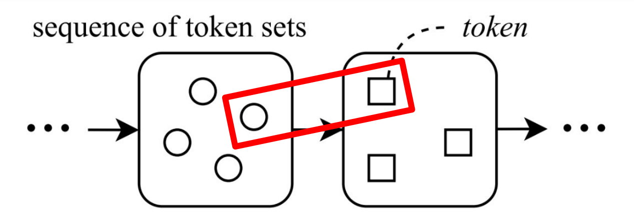
Inputs for SemID Construction

Context-independent ⇒ Context-aware

Algorithm 1 ActionPiece Vocabulary Construction

input Sequence corpus $\mathcal{S}'$, initial tokens $\mathcal{V}_0$, target size Q
output Merge rules $\mathcal{R}$, constructed vocabulary $\mathcal{V}$

Initialize vocabulary $\mathcal{V} \leftarrow \mathcal{V}_0$ # each initial token corresponds to one unique item feature
$\mathcal{R} \leftarrow \emptyset$
while $|\mathcal{V}| < Q$ **do**
  # ***Count:*** accumulate weighted token co-occurrences
  $\text{count}(\cdot,\cdot) \leftarrow \text{Count}(\mathcal{S}', \mathcal{V})$ # Algorithm 2
  # ***Update:*** Merge a frequent token pair into a new token
  Select $(c_u, c_v) \leftarrow \arg\max_{(c_i, c_j)} \text{count}(c_i, c_j)$
  $\mathcal{S}' \leftarrow \text{Update}(\mathcal{S}', \{(c_u, c_v) \rightarrow c_{\text{new}}\})$ # Algorithm 3
  $\mathcal{R} \leftarrow \mathcal{R} \cup \{(c_u, c_v) \rightarrow c_{\text{new}}\}$ # new merge rule
  $\mathcal{V} \leftarrow \mathcal{V} \cup \{c_{\text{new}}\}$ # add new token to the vocabulary
end while
return $\mathcal{R}, \mathcal{V}$



$$P(\bigcirc, \bigcirc) = \frac{|\bigcirc\!-\!\bigcirc|}{|<\bigcirc, \bigcirc>|} = \frac{4-1}{\binom{4}{2}}$$

$$P(\bigcirc, \square) = \frac{|\bigcirc\!-\!\square|}{|\bigcirc| \times |\square|} = \frac{1}{4 \times 3}$$

$$P(\square, \square) = \frac{|\square\!-\!\square|}{|<\square, \square>|} = \frac{3-1}{\binom{3}{2}}$$

Features co-occurring within or across items

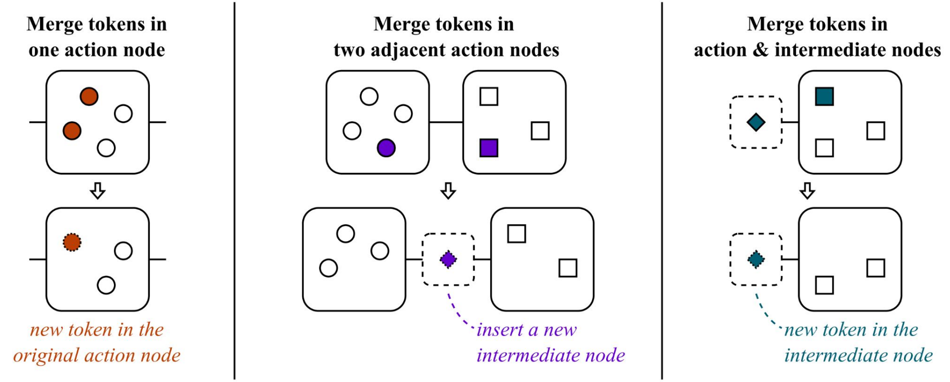
Inputs for SemID Construction

Context-independent ⇒ Context-aware

Algorithm 1 ActionPiece Vocabulary Construction

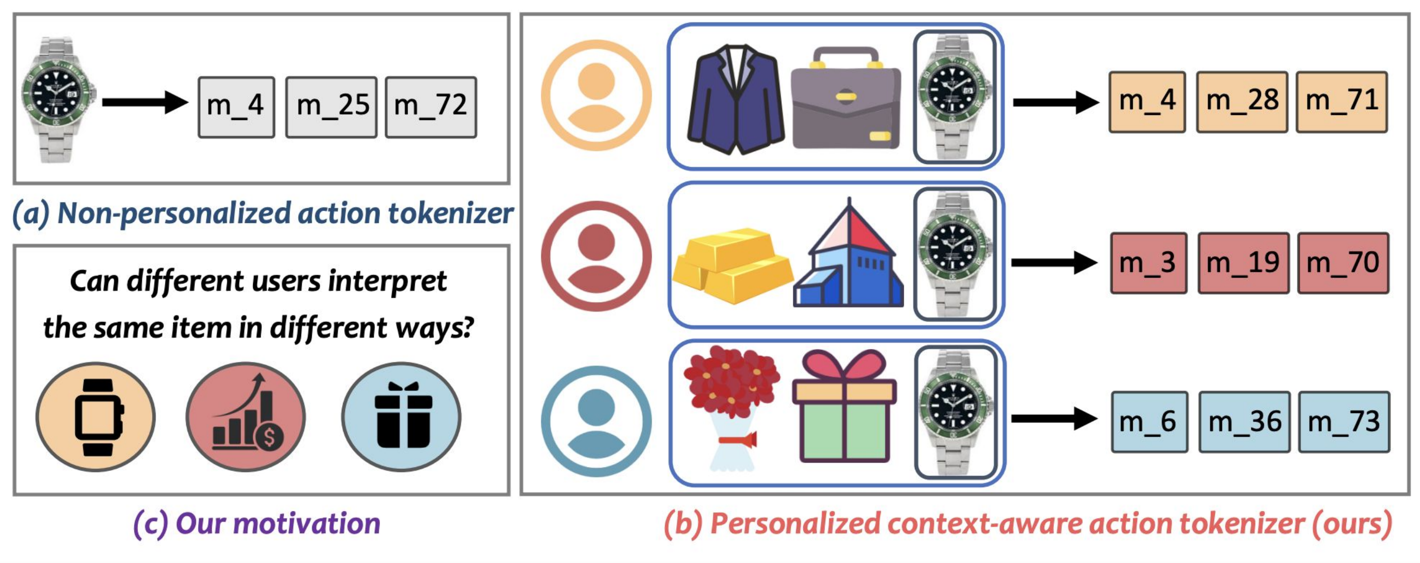
input Sequence corpus $\mathcal{S}'$, initial tokens $\mathcal{V}_0$, target size Q
output Merge rules $\mathcal{R}$, constructed vocabulary $\mathcal{V}$

Initialize vocabulary $\mathcal{V} \leftarrow \mathcal{V}_0$ # each initial token corresponds to one unique item feature
$\mathcal{R} \leftarrow \emptyset$
while $|\mathcal{V}| < Q$ **do**
  # ***Count:*** accumulate weighted token co-occurrences
  $\text{count}(\cdot,\cdot) \leftarrow \text{Count}(\mathcal{S}', \mathcal{V})$ # Algorithm 2
  # ***Update:*** Merge a frequent token pair into a new token
  Select $(c_u, c_v) \leftarrow \arg\max_{(c_i, c_j)} \text{count}(c_i, c_j)$
  $\mathcal{S}' \leftarrow \text{Update}(\mathcal{S}', \{(c_u, c_v) \rightarrow c_{\text{new}}\})$ # Algorithm 3
  $\mathcal{R} \leftarrow \mathcal{R} \cup \{(c_u, c_v) \rightarrow c_{\text{new}}\}$ # new merge rule
  $\mathcal{V} \leftarrow \mathcal{V} \cup \{c_{\text{new}}\}$ # add new token to the vocabulary
end while
return $\mathcal{R}, \mathcal{V}$



Inputs for SemID Construction

Context-independent ⇒ Context-aware ⇒ Personalized



Zhong et al. Pctx: Tokenizing Personalized Context for Generative Recommendation. arXiv:2510.21276.

Inputs for SemID Construction

Input: all data associated with the item

(1) Item Metadata

Text / Multimodal / Categorical / No Features

(2) Item Metadata + Behaviors

Regularization / Fusion

Context-independent ⇒ Context-aware

Part 1 Summary – SemID Construction

(1) First Example: TIGER

(2) Construction Techniques

(3) Inputs

Part 1 Summary - SemID Construction

(1) First Example: TIGER

(2) Construction Techniques

RQ, PQ, Clustering, LM-based generator

(3) Inputs

Part 1 Summary – SemID Construction

(1) First Example: TIGER

(2) Construction Techniques

RQ, PQ, Clustering, LM-based generator

(3) Inputs

Item Metadata (Text, Multimodal)

+ Behaviors (Regularization, Fusion, Context)

Part 2: SemID-based Generative Recommendation Model Architecture

SemID-based Recommender Architecture

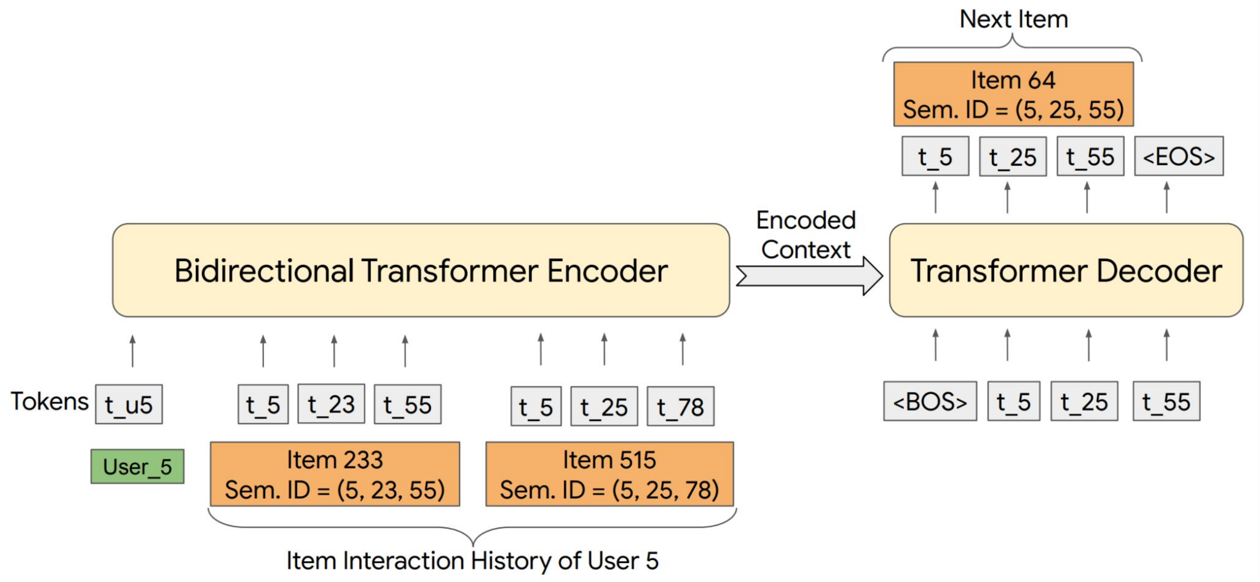
Recommendation as a **seq**-to-**seq** generation problem

Input: user interacted items $\{c_{11}, c_{12}, c_{13}, c_{14}, c_{21}, c_{22}, \ldots\}$

Output: next item $\{c_{t1}, c_{t2}, c_{t3}, c_{t4}\}$

SemID-based Recommender Architecture

Architecture: Encoder-Decoder

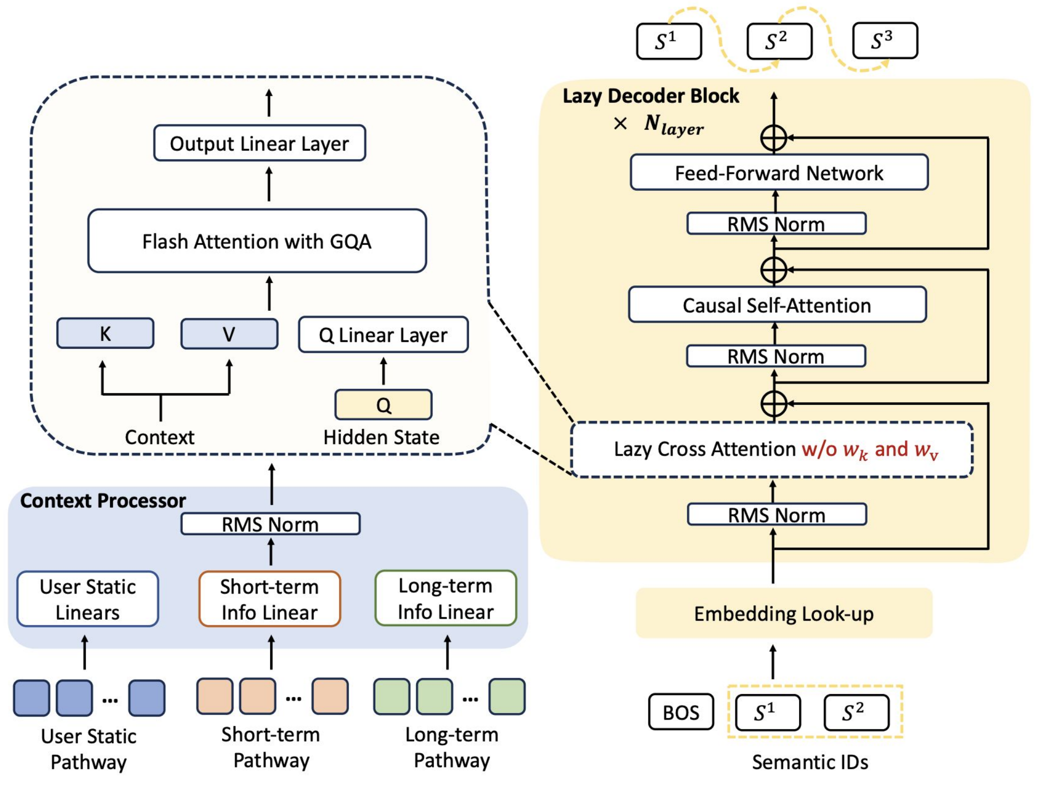


Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

SemID-based Recommender Architecture

Architecture:

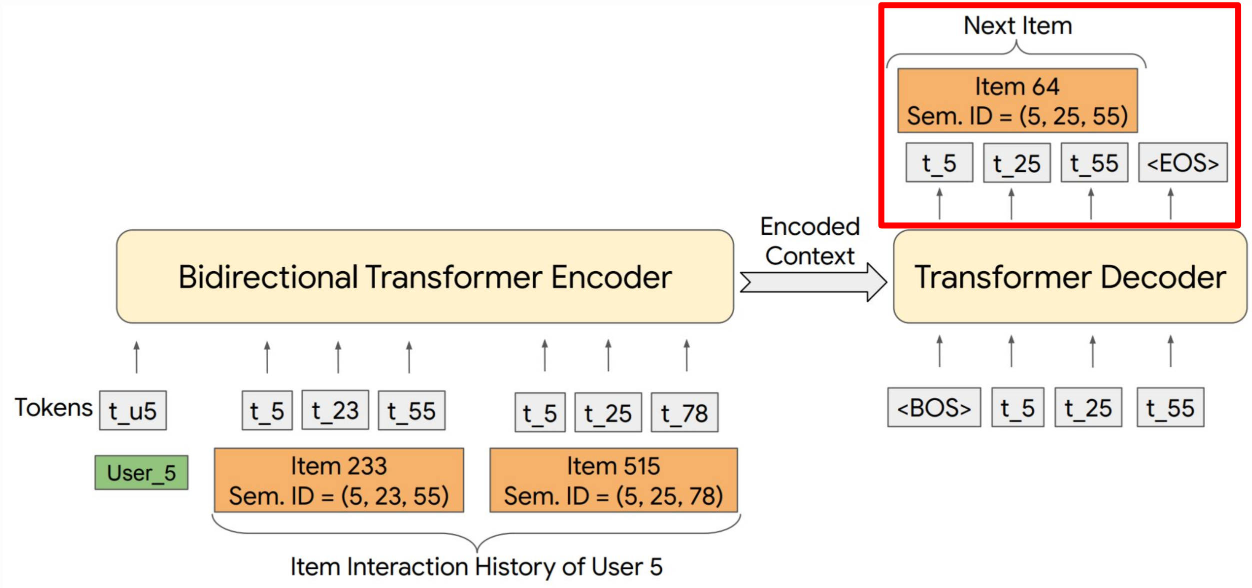
Decoder-Only?



Zhou et al. OneRec-V2 Technical Report. arXiv:2508.20900.

SemID-based Recommender Architecture

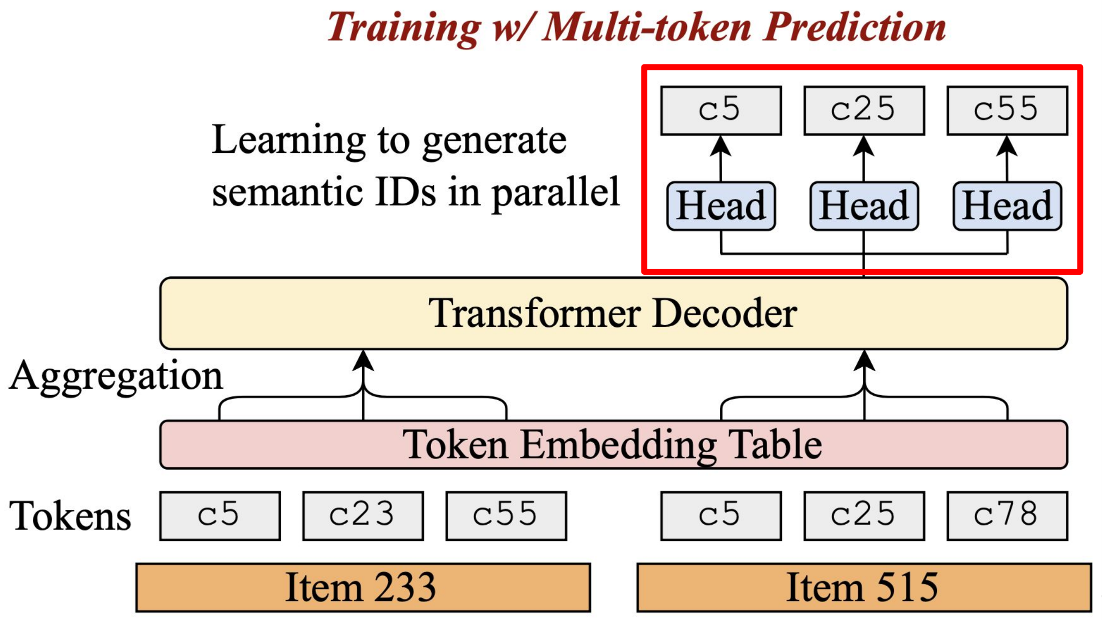
Objective: Next-Token Prediction (w/ RQ)



Rajputet al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

SemID-based Recommender Architecture

Objective: Multi-Token Prediction (w/ PQ)



Hou et al. Generating Long Semantic IDs in Parallel for Recommendation. KDD 2025.

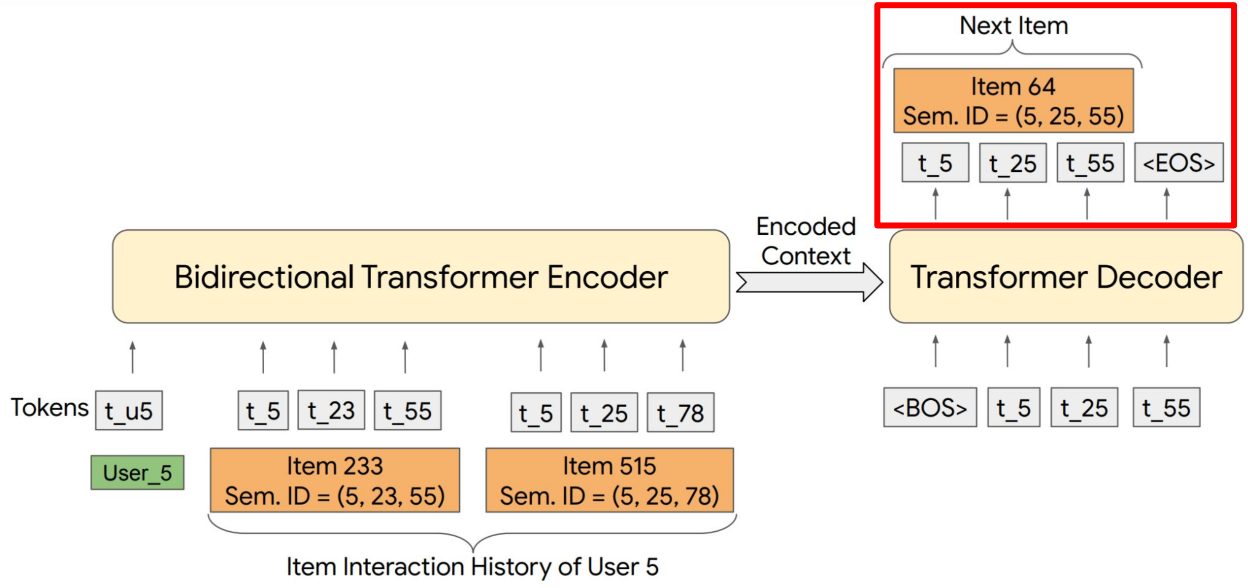
SemID-based Recommender Architecture

Objective: RL

Will introduce later at "**Align w/ LLMs**"

SemID-based Recommender Architecture

Inference: How to get a ranking list?

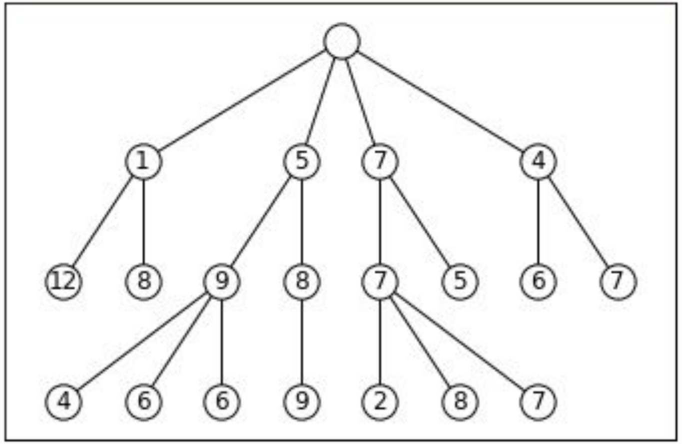


Rajput et al. Recommender Systems with Generative Retrieval. NeurIPS 2023.

SemID-based Recommender Architecture

Inference: How to get a ranking list?

(Constrained) Beam Search

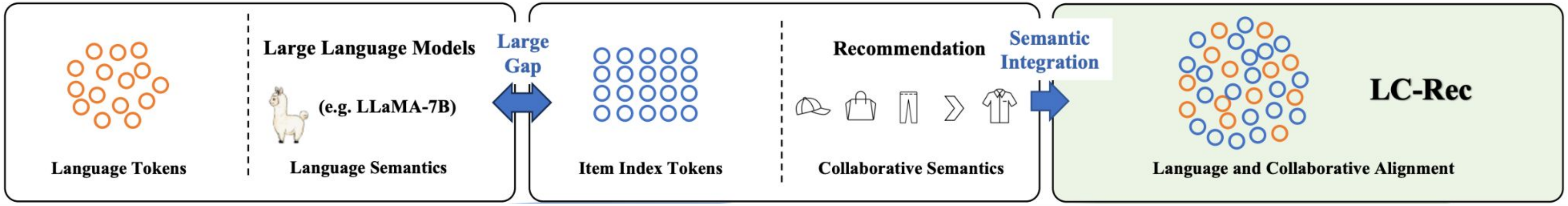


SemID-based Recommender Architecture

Inference: Can we do dense retrieval-like grounding?

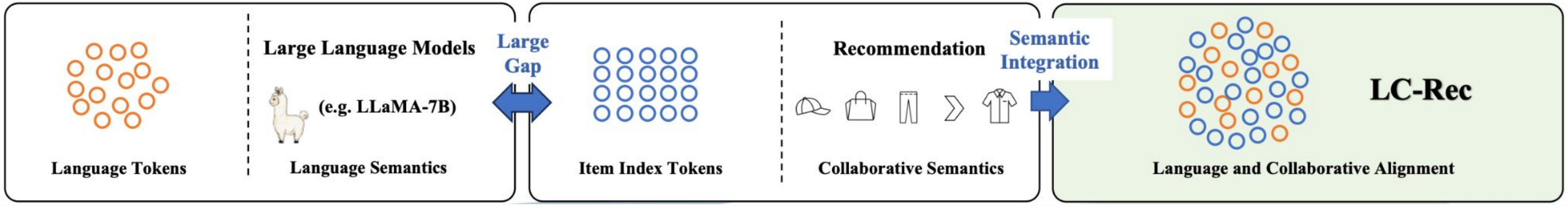
SemID-based Recommender Architecture

Align with LLMS - LC-Rec



SemID-based Recommender Architecture

Align with LLMS - LC-Rec



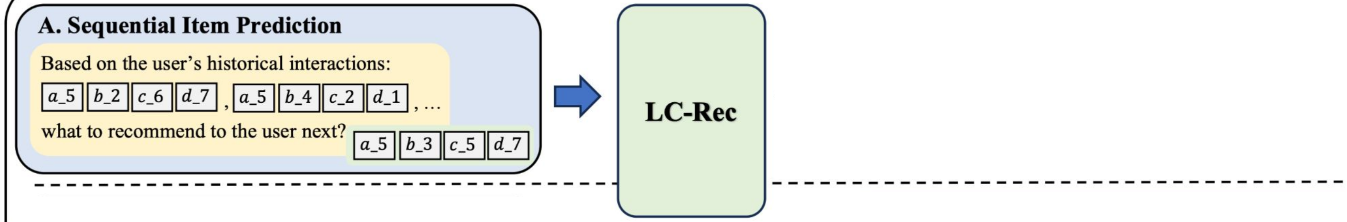
Core Idea:

Construct instructions containing both Semantic IDs and language tokens

Zheng et al. Adapting Large Language Models by Integrating Collaborative Semantics for Recommendation. ICDE 2024.

SemID-based Recommender Architecture

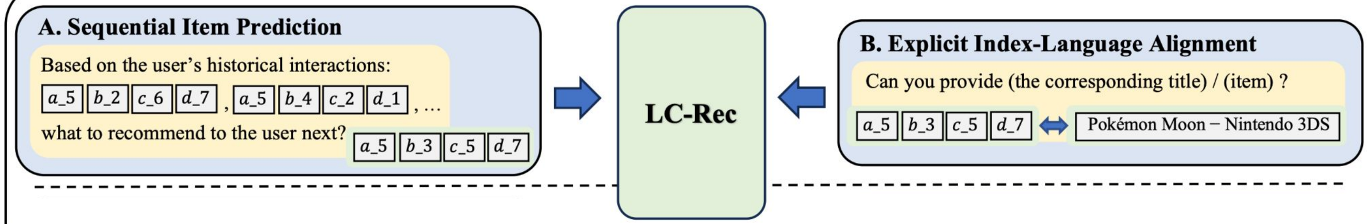
Align with LLMS - LC-Rec



SemID-based seq2seq task

SemID-based Recommender Architecture

Align with LLMS - LC-Rec

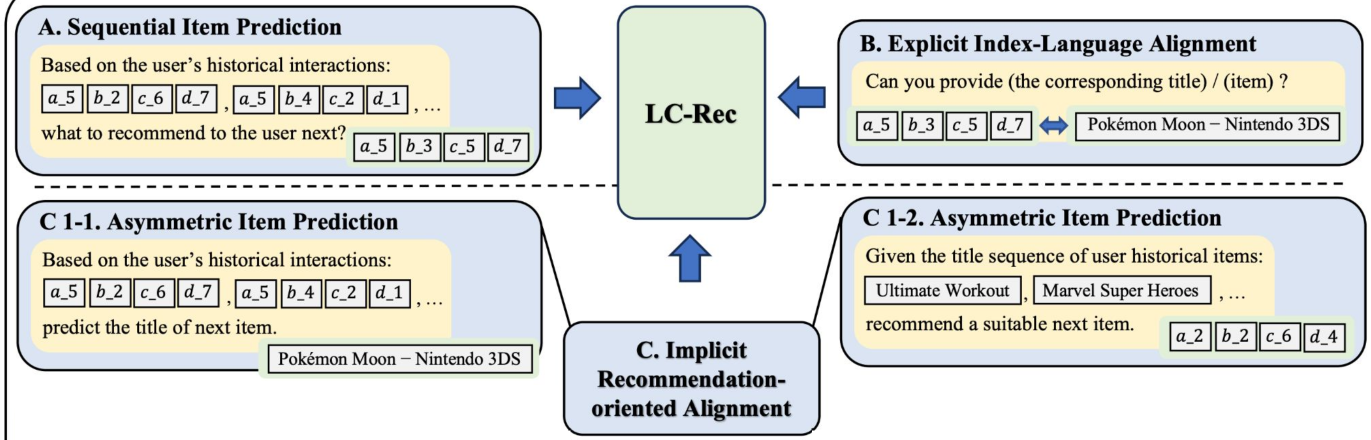


Translation between SemIDs and titles

Zheng et al. Adapting Large Language Models by Integrating Collaborative Semantics for Recommendation. ICDE 2024.

SemID-based Recommender Architecture

Align with LLMS - LC-Rec

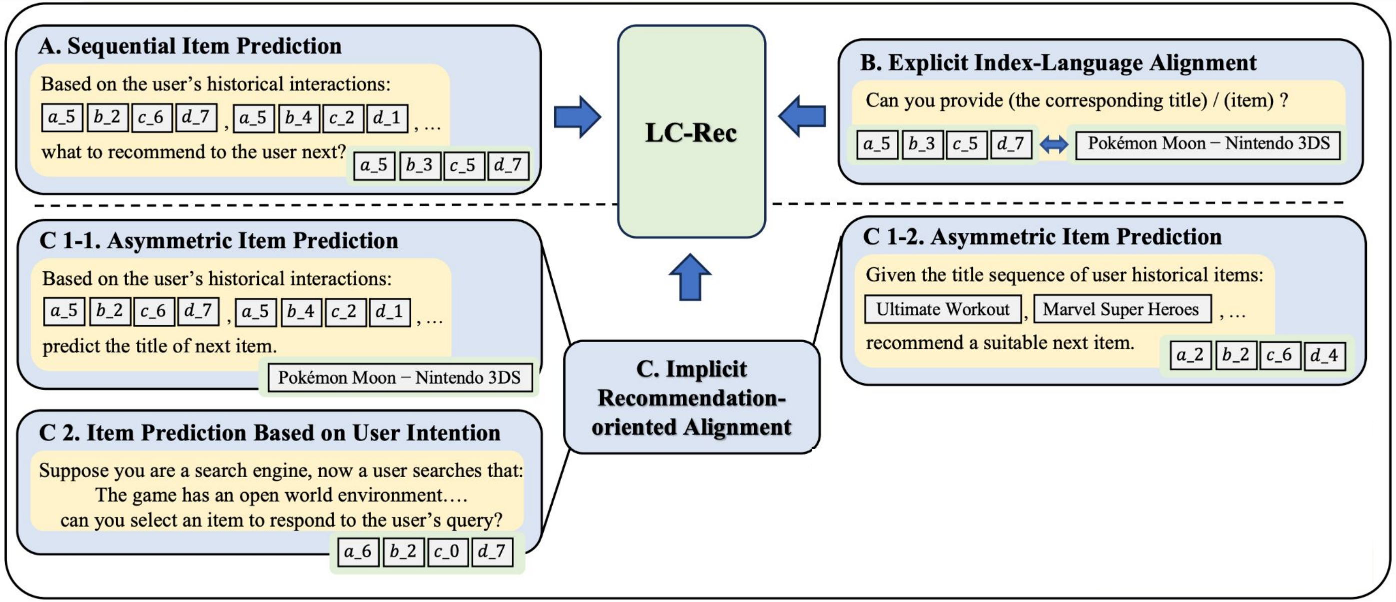


Implicit Translation between SemIDs and titles

Zheng et al. Adapting Large Language Models by Integrating Collaborative Semantics for Recommendation. ICDE 2024.

SemID-based Recommender Architecture

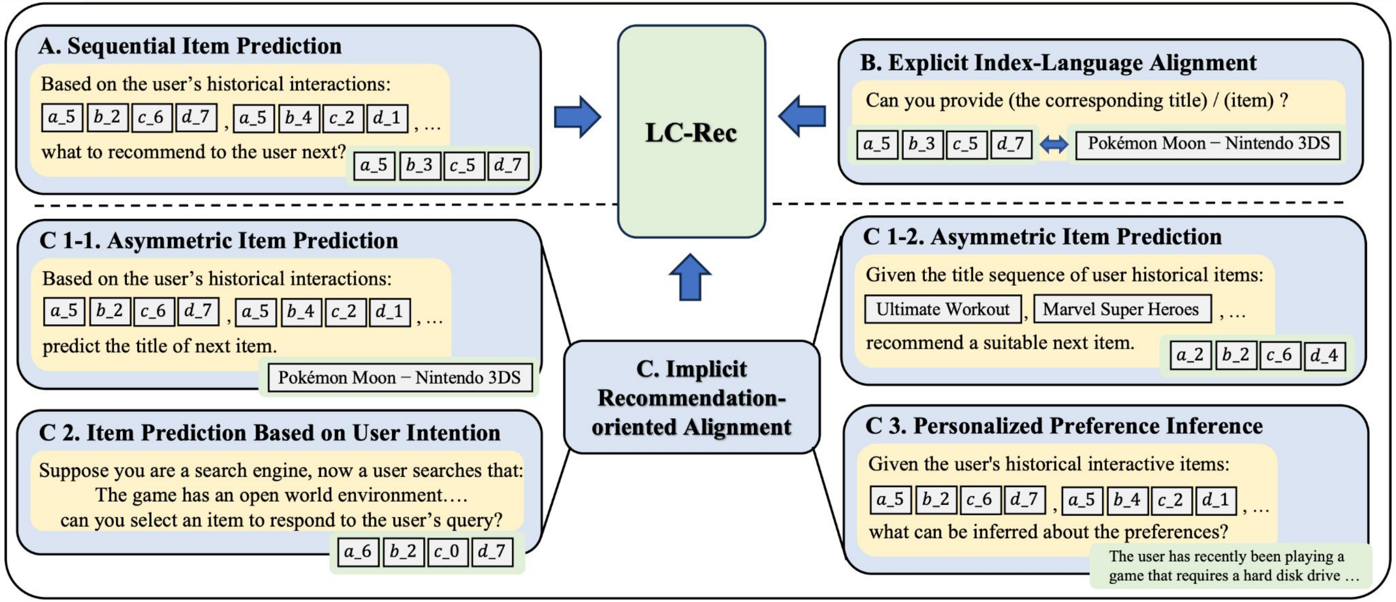
Align with LLMS - LC-Rec



Zheng et al. Adapting Large Language Models by Integrating Collaborative Semantics for Recommendation. ICDE 2024.

SemID-based Recommender Architecture

Align with LLMS - LC-Rec

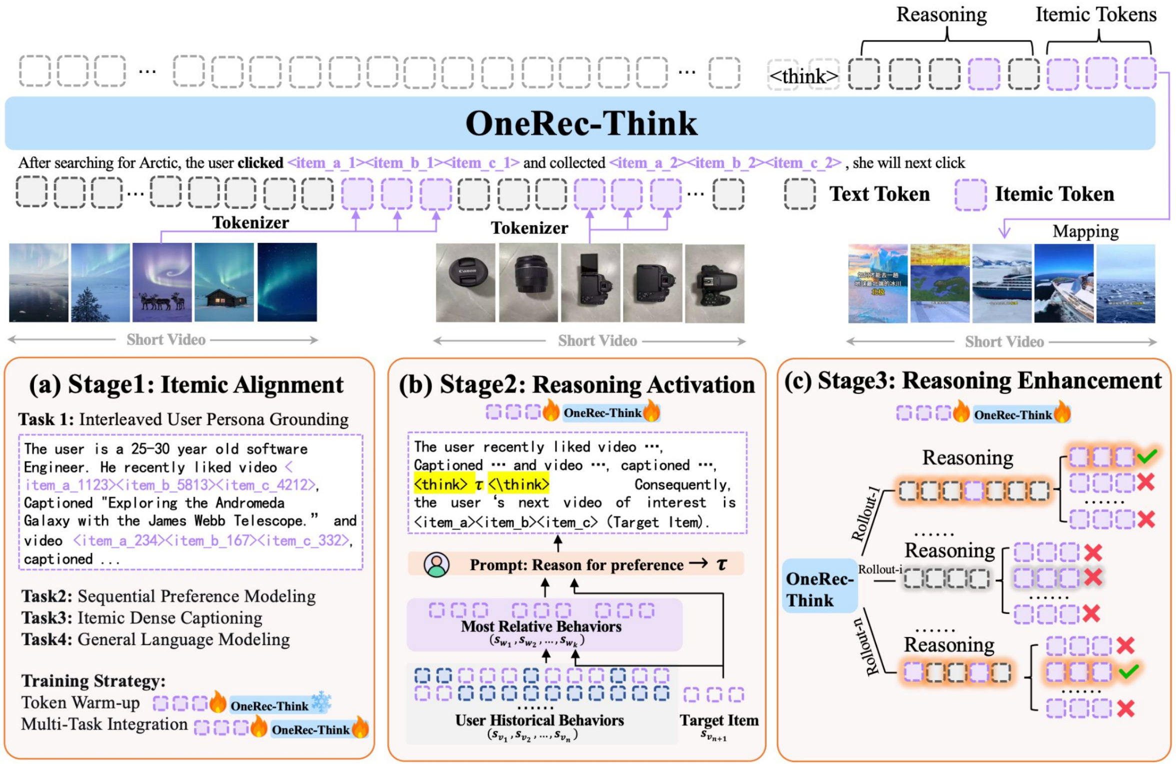


Zheng et al. Adapting Large Language Models by Integrating Collaborative Semantics for Recommendation. ICDE 2024.

SemID-based Recommender Architecture

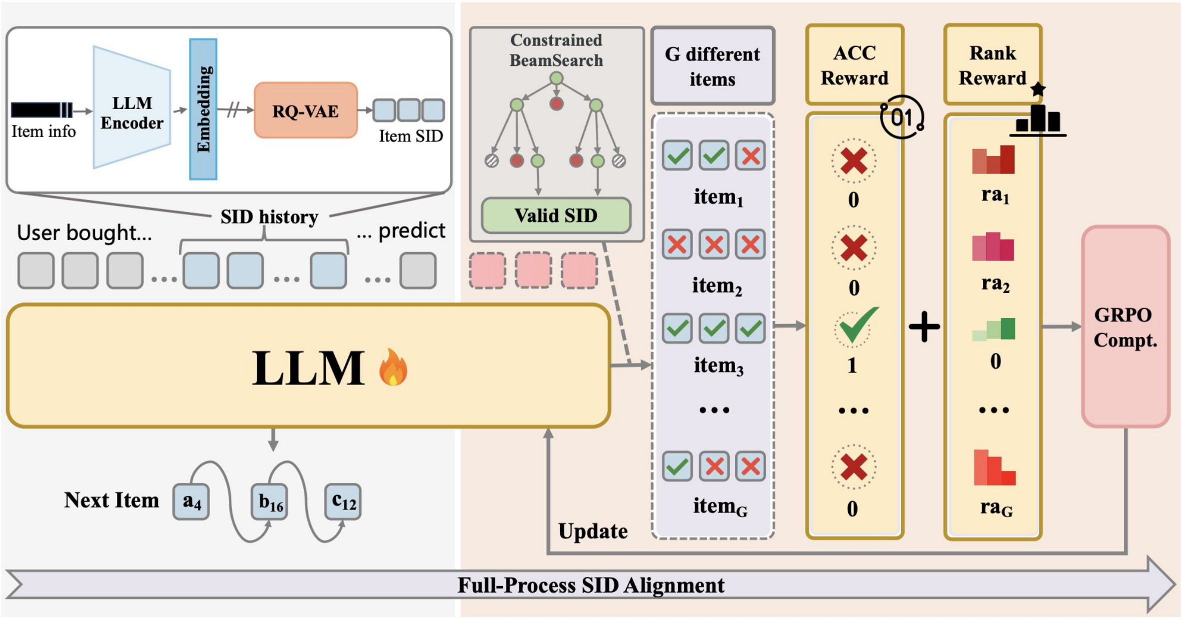
Align with LLMS

OneRec-Think



SemID-based Recommender Architecture

Align with LLMS: MiniOneRec



Kong et al. MiniOneRec: An Open-Source Framework for Scaling Generative Recommendation. arXiv:2510.24431.

Part 2 Summary – Architecture

(1) Train from Scratch

(2) Align with LLMs

Part 2 Summary – Architecture

(1) Train from Scratch

Objective (NTP, MTP, RL)

Inference (Beam Search)

(2) Align with LLMs

Part 2 Summary – Architecture

(1) Train from Scratch

Objective (NTP, MTP, RL)

Inference (Beam Search)

(2) Align with LLMs

LC-Rec / OneRec-Think / MiniOneRec

05

Summary

Open Challenges and Beyond

Action Tokenization

Human-readable Data 👨 👩

Machine-readable Data 🤖

Tokenize

?

Action Tokenization

Human-readable Data 👨 👩

Machine-readable Data 🤖

Tokenize

i3487, i124, i19240, i772

One token per action? 🤔

Action Tokenization

Human-readable Data 👨 👩

⇩ **Tokenize**

Machine-readable Data 🤖

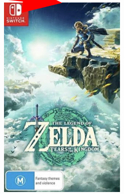
Premium Men's Short Sleeve Athletic Training T-Shirt Made of Lightweight Breathable Fabric, Ideal for Running, Gym Workouts, and Casual Sportswear in All Seasons; High-Performance Breathable Cotton Crew Socks for Men with Arch Support, Cushioned Heel and Toe, and Moisture Control, Perfect for Sports, Walking, and Everyday Comfort; Men's Loose-Fit Basketball Shorts with Elastic Drawstring Waistband, Quick-Dry Mesh Fabric, and Printed Number 11 for Professional and Recreational Play; Official Size 7 Composite Leather Basketball Designed for Indoor and Outdoor Use, Deep Channel Design for Enhanced Grip and Ball Control, Ideal for Training and Competitive Matches;

Text description of each action? 🤔

Semantic IDs
(also called: SemID or SID)

A few tokens that jointly index one item.

$\overbrace{\texttt{t3, t321, t643, t1011}}$



Action Tokenization

Can we find a sweet spot between

One token per action	Semantic IDs	Text description of each action
i3487, i124, i19240, i772	t34, t392, t600, t891, t21, t502, t592, t1002, t123, t403, t611, t821, t21, t502, t711, t1022	Premium Men's Short Sleeve Athletic Training T-Shirt Made of Lightweight Breathable Fabric, Ideal for Running, Gym Workouts, and Casual Sportswear in All Seasons; High-Performance Breathable Cotton Crew Socks for Men with Arch Support, Cushioned Heel and Toe, and Moisture Control, Perfect for Sports, Walking, and Everyday Comfort; Men's Loose-Fit Basketball Shorts with Elastic Drawstring Waistband, Quick-Dry Mesh Fabric, and Printed Number 11 for Professional and Recreational Play; Official Size 7 Composite Leather …

Open Challenges

Part 1: What becomes harder?

Comparing to traditional RecSys, what challenges may large generative models face?

Open Challenges

Part 1: What becomes harder?

Comparing to traditional RecSys, what challenges may large generative models face?

Part 2: What becomes possible?

What new opportunities may large generative models unlock for recommender systems?

Part 1: What Becomes Harder?

Comparing to traditional RecSys, what challenges may large generative models face?

Cold-Start Recommendation

Do semantic ID-based models really good at cold-start recommendation?

Cold-Start Recommendation

Do semantic ID-based models really good at cold-start recommendation?

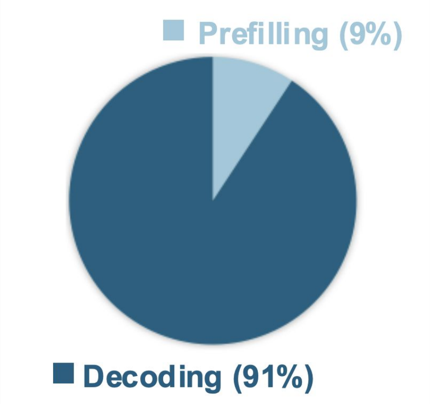
Model	#Params. (M)	Video Games						Cell Phones and Accessories					
		Overall		In-Sample (39.7%)		Unseen (60.3%)		Overall		In-Sample (31.8%)		Unseen (68.2%)	
		R@50	**N@50**	**R@50**	**N@50**	**R@50**	**N@50**	**R@50**	**N@50**	**R@50**	**N@50**	**R@50**	**N@50**
UniSRec	2.90	0.0621	0.0200	0.1386	0.0461	0.0118	0.0029	0.0233	0.0077	0.0604	0.0211	0.0060	0.0014
Recformer	233.73	**0.0740**	0.0218	0.1082	0.0333	**0.0514**	**0.0142**	0.0236	0.0070	0.0340	0.0103	**0.0188**	**0.0055**
TIGER	13.26	0.0584	0.0193	**0.1472**	**0.0486**	-	-	0.0232	0.0078	**0.0730**	<u>0.0245</u>	-	-
TIGER_C	13.26	0.0611	0.0198	<u>0.1447</u>	<u>0.0482</u>	0.0061	0.0011	0.0233	0.0078	0.0691	0.0238	0.0019	0.0003
$\textbf{SpecGR}_\textbf{Aux}$	16.16	<u>0.0726</u>	<u>0.0220</u>	0.1399	0.0436	0.0283	0.0078	<u>0.0269</u>	<u>0.0084</u>	0.0722	0.0230	0.0058	0.0015
SpecGR++	13.28	0.0717	**0.0225**	0.1323	0.0439	<u>0.0318</u>	<u>0.0084</u>	**0.0275**	**0.0090**	**0.0730**	**0.0246**	<u>0.0063</u>	<u>0.0017</u>

Ding et al. Inductive Generative Recommendation via Retrieval-based Speculation. AAAI 2026.

Inference Efficiency

Retrieval Models: **K Nearest Neighbor Search**

Generative Models (e.g., AR models): **Beam Search**

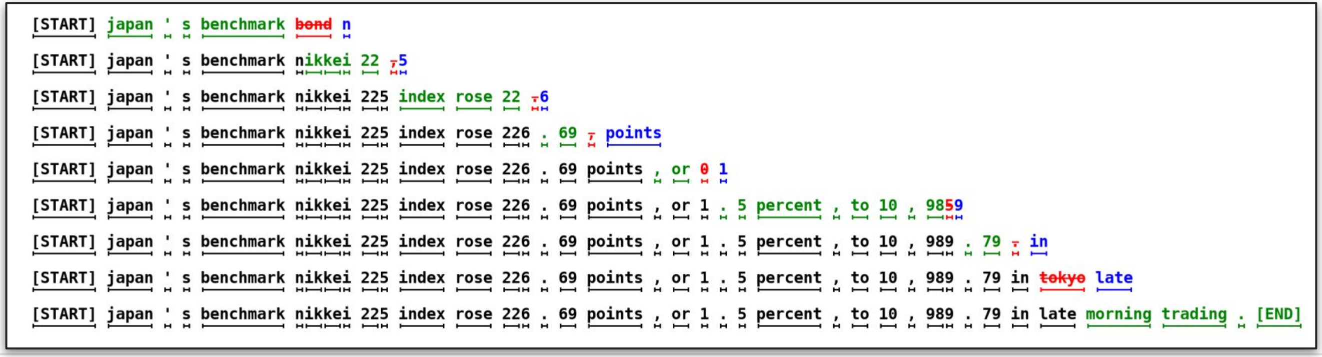


Lin et al. Efficient Inference for Large Language Model-based Generative Recommendation. ICLR 2025.

Inference Efficiency

How to accelerate LLMs? **Speculative Decoding**

- Use a “cheap” model to generate candidates
- “Expensive” model can accept or reject (and perform inference if necessary)

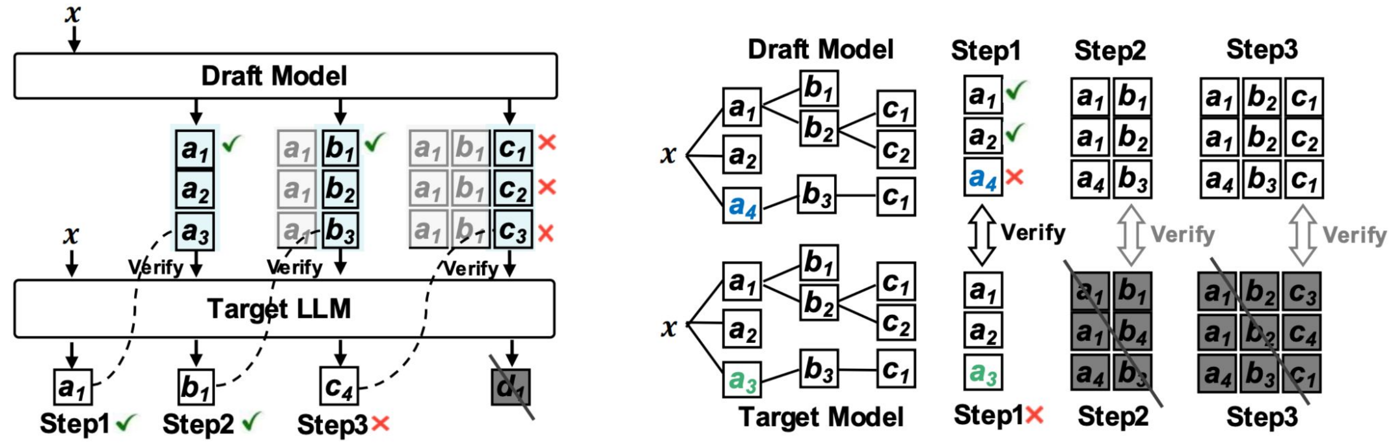


Leviathan et al. Fast Inference from Transformers via Speculative Decoding. ICML 2023.

Inference Efficiency

Speculative decoding for generative rec? ❌

N-to-K verification



(b) N-to-1 Verification of Traditional SD (N=3)
(c) N-to-K Verification of SD with Beam Search (N=K=3)

Lin et al. Efficient Inference for Large Language Model-based Generative Recommendation. ICLR 2025.

Inference Efficiency

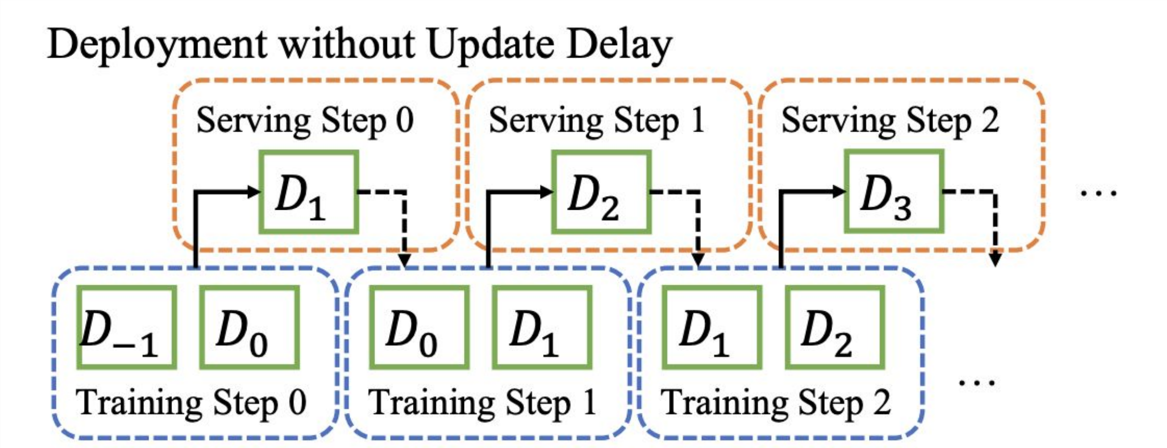
In addition to single-model acceleration methods, what about "serving throughout"?

Example: vLLM offers solutions for high-throughput and memory-efficient inference and serving

What's unique for generative rec?

Timely Model Update

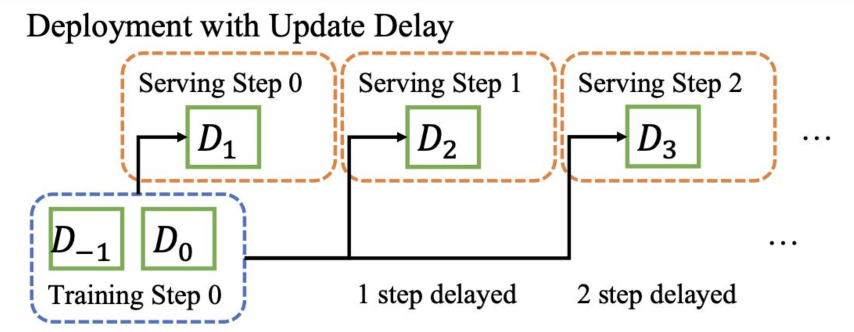
Recommendation models favor timely updates

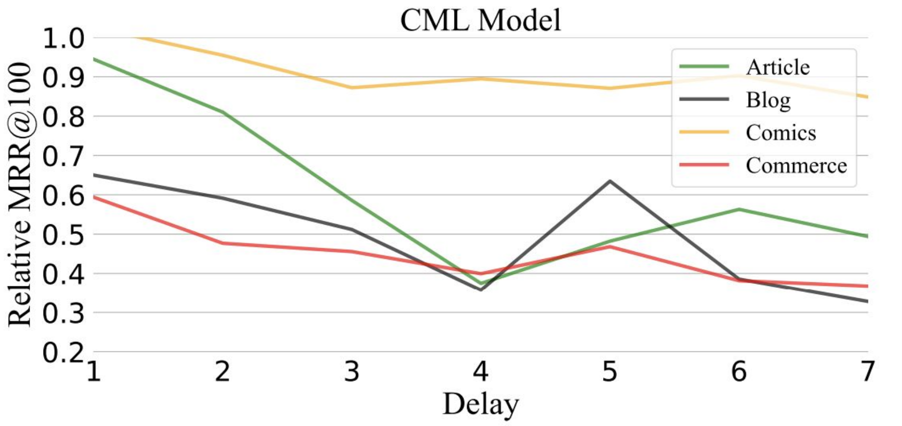


Lee et al. How Important is Periodic Model Update in Recommender Systems? SIGIR 2023.

Timely Model Update

Delayed updates lead to performance degradation





Lee et al. How Important is Periodic Model Update in Recommender Systems? SIGIR 2023.

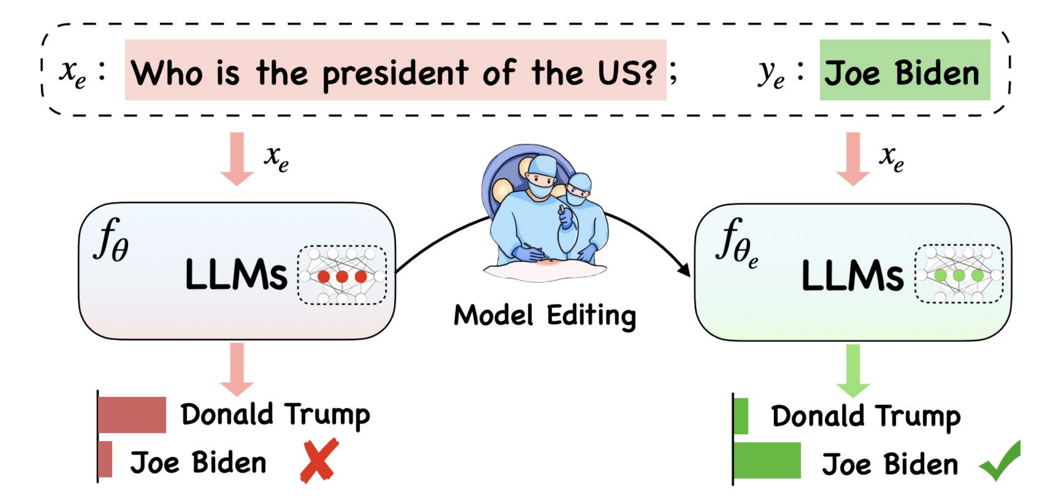
Timely Model Update

How to update large generative rec models timely?

(Frequently retraining large generative models may be resource consuming)

Timely Model Update

How to update large generative rec models timely?

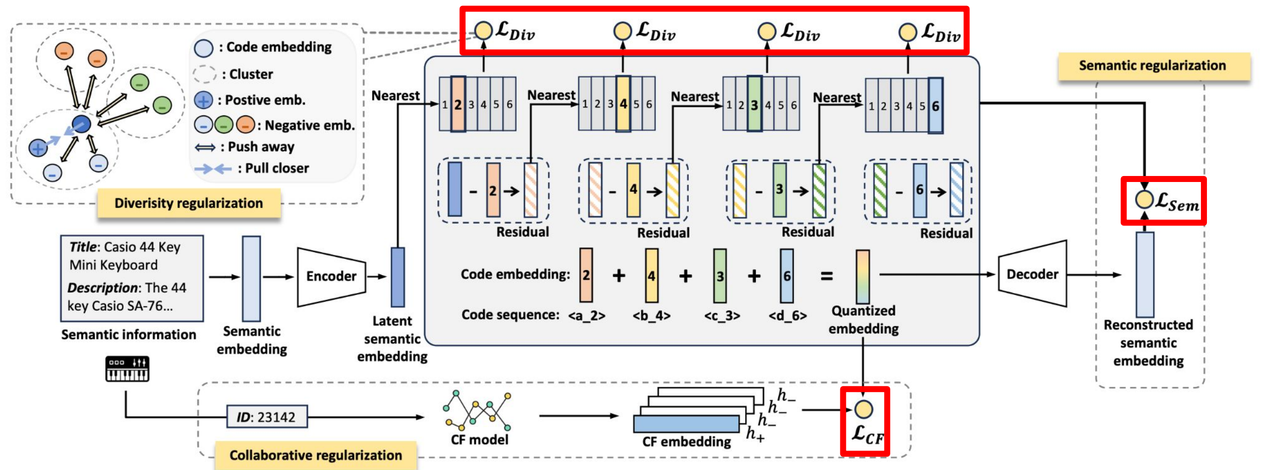


Knowledge editing?

Yao et al. Editing Large Language Models: Problems, Methods, and Opportunities. EMNLP 2023.

Item Tokenization

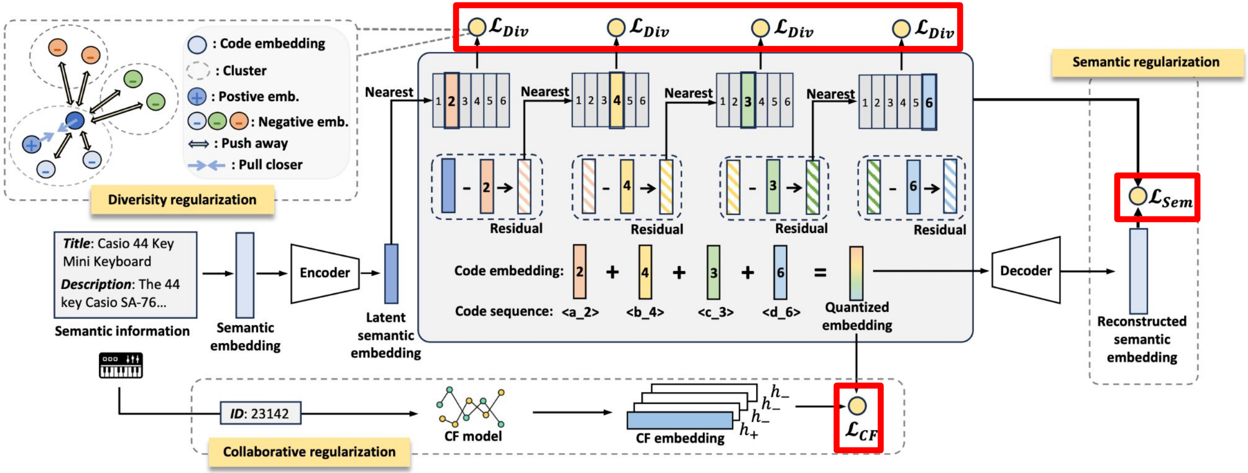
Multiple objectives for optimizing item tokenization …



Item Tokenization

Multiple objectives for optimizing item tokenization …

But **none** of them is **directly related to rec performance**



Item Tokenization

reconstruction loss ≠ downstream performance

How to connect tokenization objective with recommendation performance?

Zipf's distribution? Entropy? Linguistic metrics?

Item Tokenization

Language Tokenization

2014~2015:

Word / Char

Context-independent ⇒ Context-aware

Item Tokenization

Language Tokenization

2014~2015:
Word / Char

2016~present:
BPE / WordPiece

Context-independent ⇒ Context-aware

Item Tokenization

Language Tokenization

2014~2015:
Word / Char

2016~present:
BPE / WordPiece

Context-independent ⇒ Context-aware

SemID Construction

2023~2024:
RQ / PQ / Clustering /
LM-based Generator

Item Tokenization

Language Tokenization

2014~2015:
Word / Char

2016~present:
BPE / WordPiece

Context-independent ⇒ Context-aware

SemID Construction

2023~2024:
RQ / PQ / Clustering / LM-based Generator

2025:
ActionPiece / Pctx / **?**

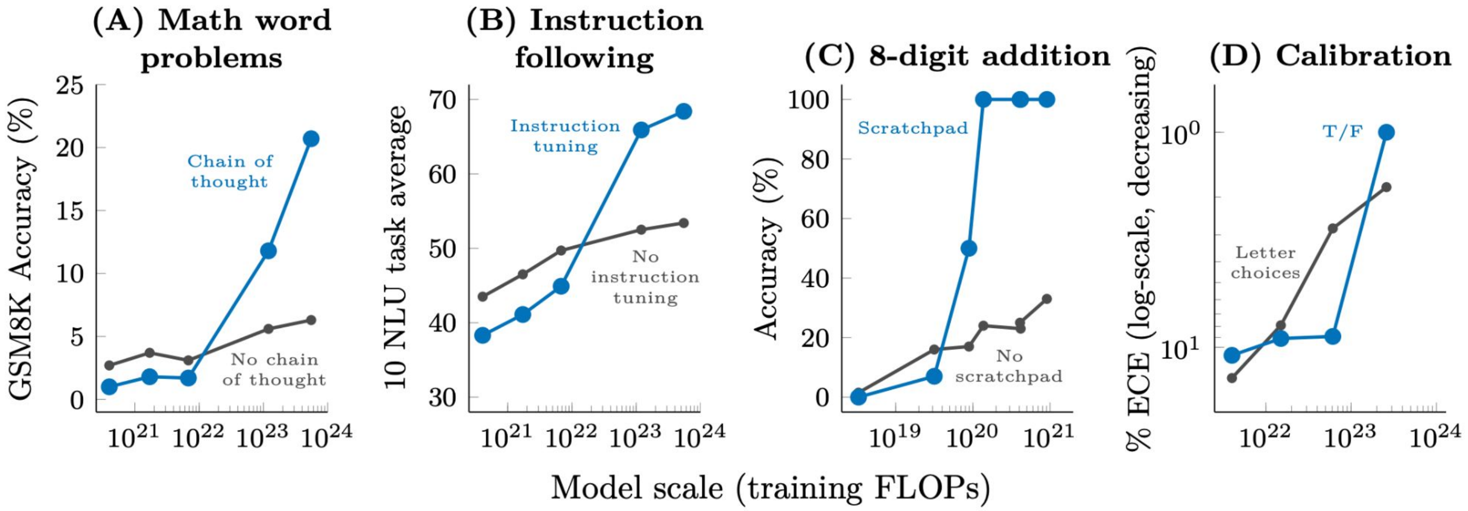
Hou et al. ActionPiece: Contextually Tokenizing Action Sequences for Generative Recommendation. arXiv:2502.13581.

Part 2: What Becomes Possible?

What new opportunities may large generative models unlock for recommender systems?

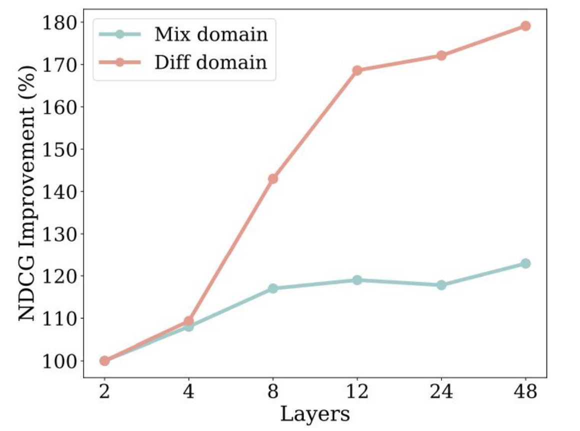
Emergent Ability

Abilities not present in smaller models but is present in larger models

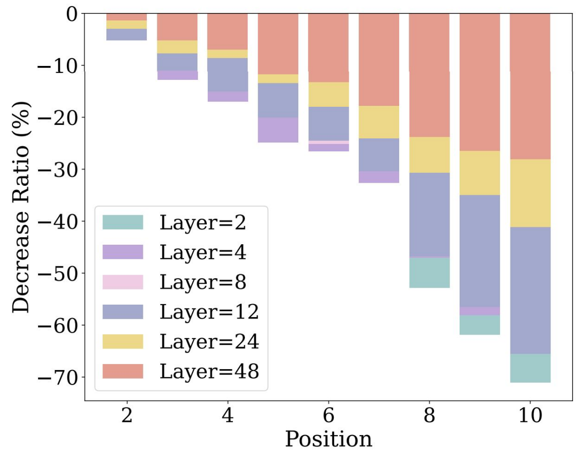


Emergent Ability

Do we have emergent abilities in large generative recommendation models?



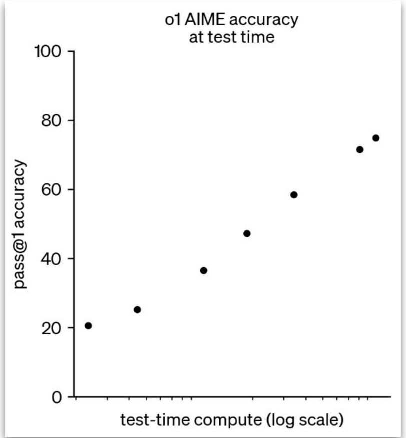
Cross-domain



Trajectory Prediction

Test-time Scaling & Reasoning

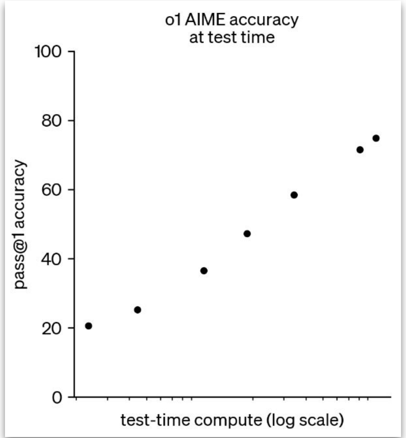
There have been explorations on model / data scaling of recommendation models



Test-time scaling is still under exploration

Test-time Scaling & Reasoning

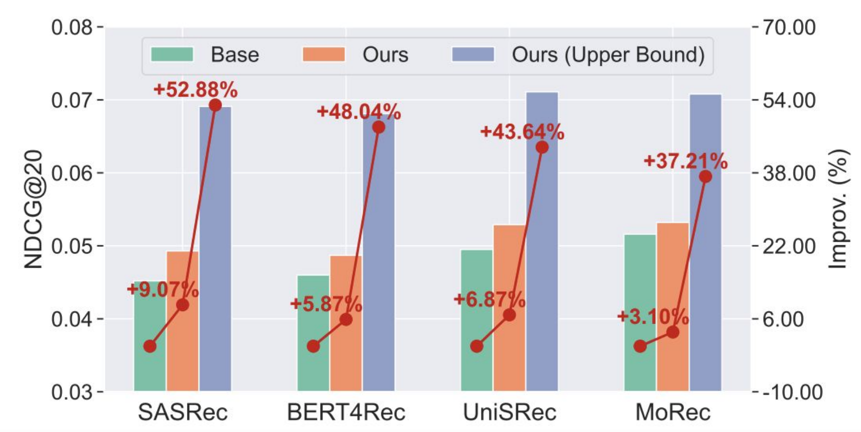
There have been explorations on model / data scaling of recommendation models

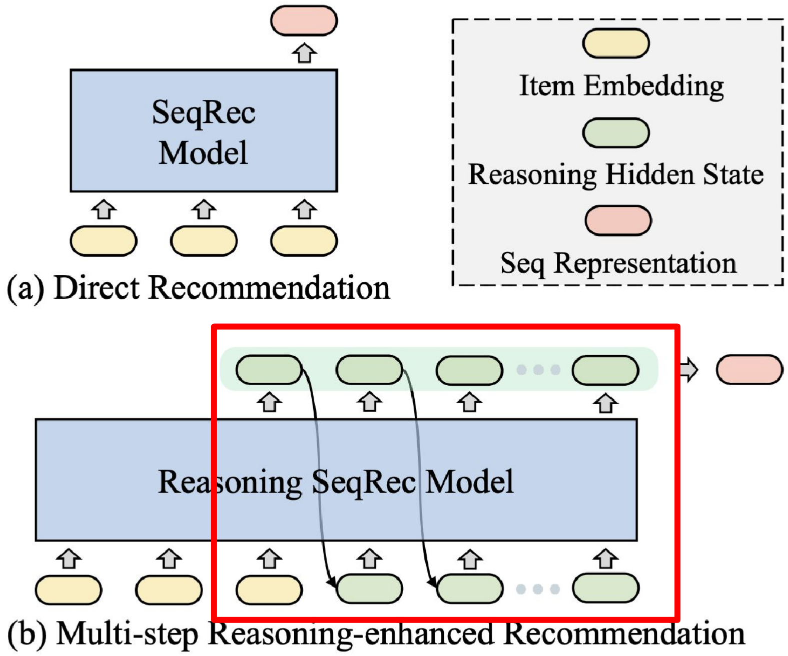


Test-time scaling is ~~still under~~ being actively exploration

Test-time Scaling & Reasoning

Reasoning over latent hidden states to scale up test-time computation.





Tang et al. Think Before Recommend: Unleashing the Latent Reasoning Power for Sequential Recommendation. arXiv:2503.22675.

Test-time Scaling & Reasoning

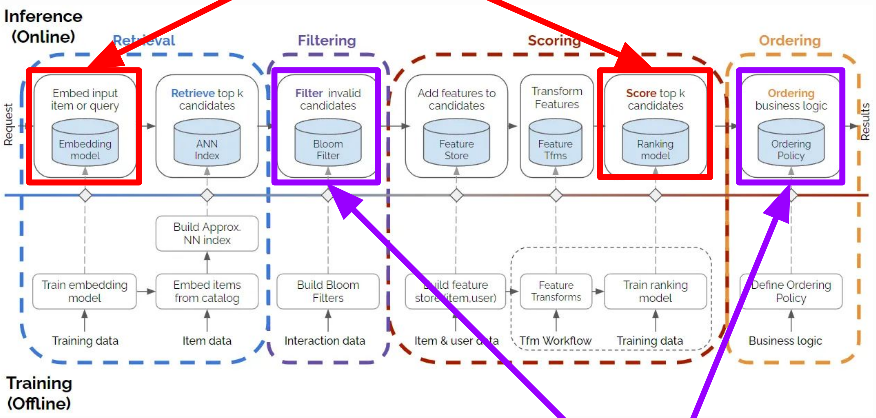
Reasoning over

text tokens

Liu et al. OneRec-Think: In-Text Reasoning for Generative Recommendation. arXiv:2510.11639.

Unified Retrieval & Ranking

models



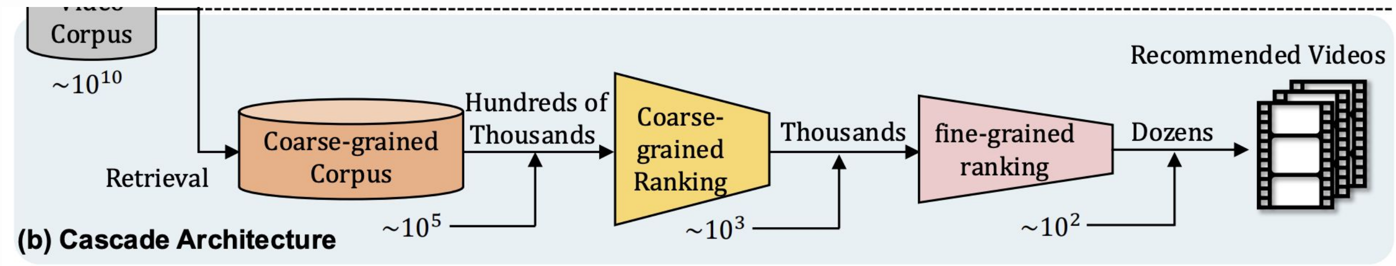
rules, strategies, heuristics

Complicated Architecture

- Difficult to be optimized in an **end-to-end** way
- **Latency** between / within different modules

Unified Retrieval & Ranking

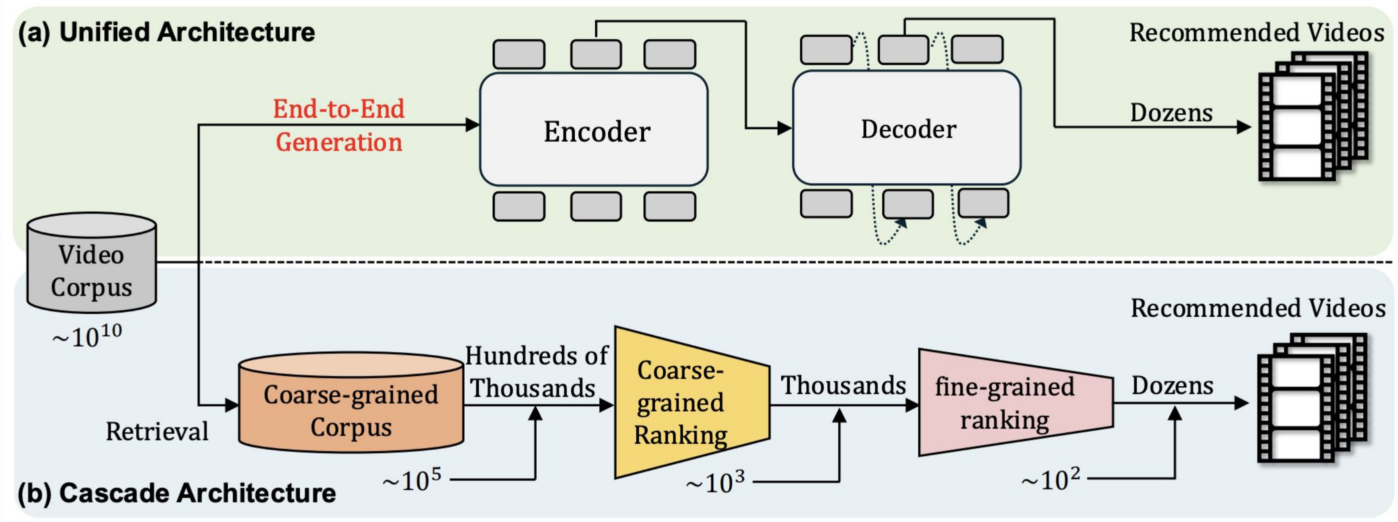
Is it possible to replace traditional cascade architecture



Deng et al. OneRec: Unifying Retrieve and Rank with Generative Recommender and Iterative Preference Alignment. arXiv:2502.18965.

Unified Retrieval & Ranking

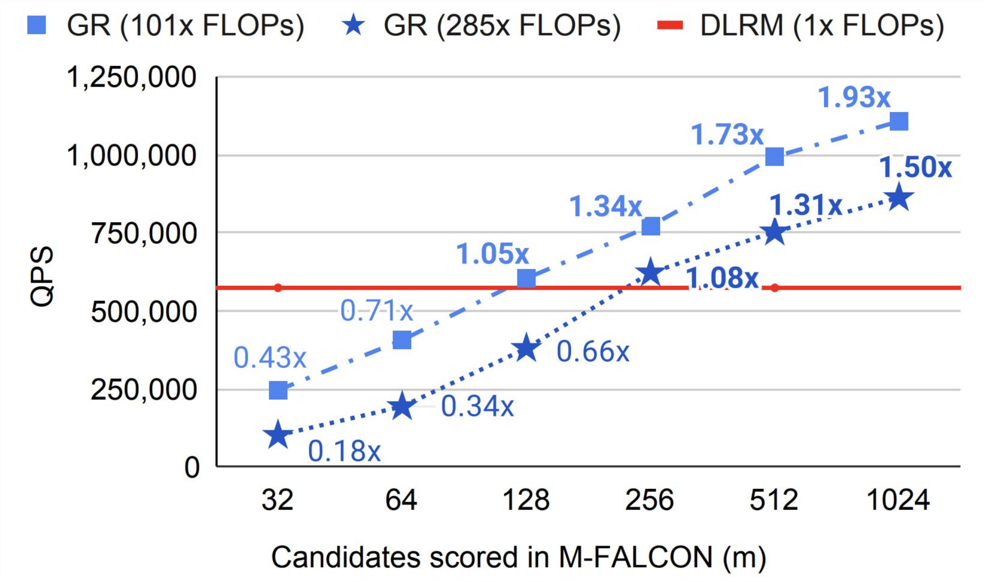
Is it possible to replace traditional cascade architecture with a **unified generative model**?

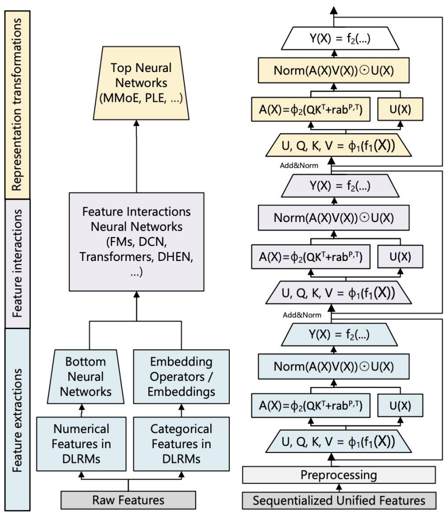


Deng et al. OneRec: Unifying Retrieve and Rank with Generative Recommender and Iterative Preference Alignment. arXiv:2502.18965.

Unified Retrieval & Ranking

Better throughout when ranking more candidates





DLRM GR

Q & A

Thank you for coming!

Please kindly refer to

large-genrec.github.io

for *slides*, *paper list*, ...